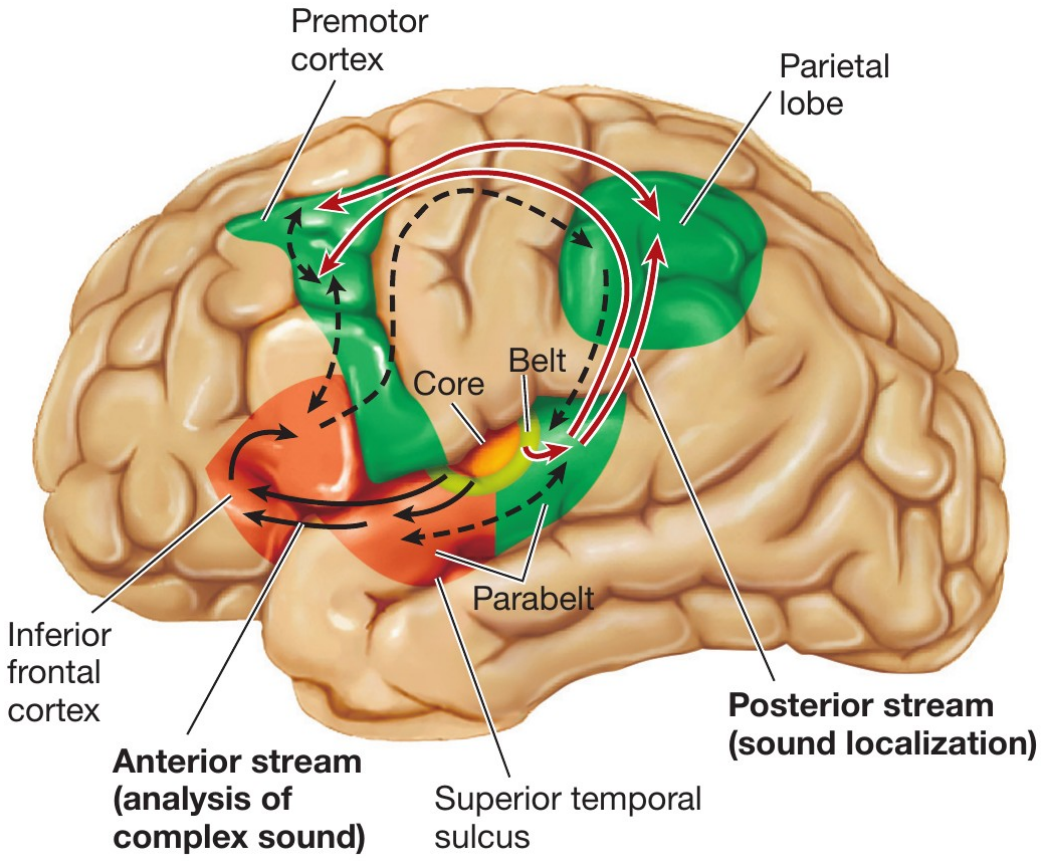


Carlson and Birkett (2022), Fig. 7.9, p. 200.

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Audition II: Central Auditory Processing

The Listening Brain: How We Find Meaning in Sound, Speech, and Music

Fall 2026

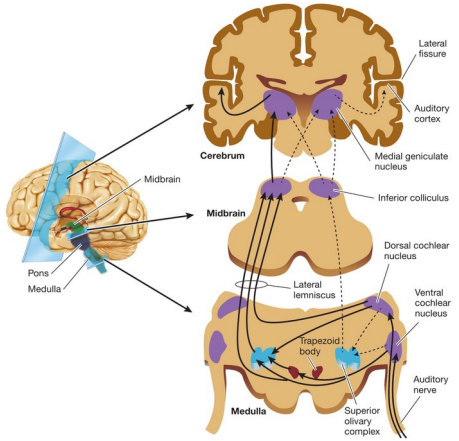
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Hearing depends on computations across the brain

Central hearing begins with neural signals that have already entered the brain. Brainstem, midbrain, thalamic, and cortical circuits transform those signals into estimates of what happened, where it happened, and whether it requires a response. Recognizing a familiar voice therefore requires additional operations beyond detecting that sound energy was present.

The same conversation can be understood while the speaker moves or background noise changes. Neurons must preserve useful features while combining them with context, attention, and memory. A circuit representing a sound's location performs a different computation from one identifying the speaker, even though both receive information about the same event.

The rotating anatomical reconstruction follows pathways connecting human auditory nuclei. Its colors identify reconstructed structures and connections, rather than showing thoughts or sound moving through tissue. Comparing the three-dimensional arrangement with the pathway diagram helps locate the small central nuclei that exchange signals before information reaches the auditory cortex.



Carlson and Birkett (2022), Fig. 7.7, p. 198.



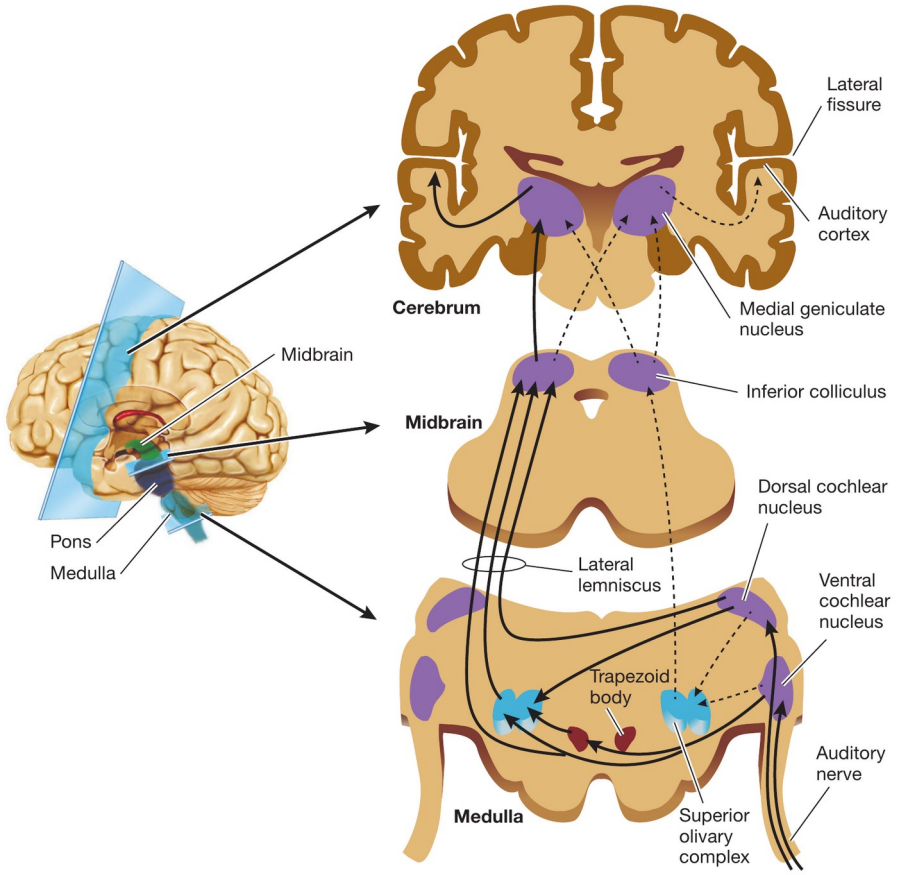
Sitek et al. (2019): Human auditory pathways, 3D reconstruction.

Central auditory pathways distribute information in parallel

The **cochlear nuclei** are central auditory structures in the brainstem that receive incoming auditory nerve axons. Their neurons distribute information to several targets rather than forwarding one unchanged message along a single chain. Different cell populations emphasize timing, changes in intensity, and patterns across sound frequencies for subsequent computations.

Connections reach the **superior olivary complex**, a group of brainstem nuclei that compares information associated with the two ears. Ascending pathways continue through the lateral lemniscus toward the inferior colliculus in the midbrain. These connections provide repeated opportunities to combine excitation and inhibition before a sound produces a conscious judgment.

The **medial geniculate nucleus** of the thalamus sends major projections to auditory cortex. Many central connections are bilateral, so each hemisphere receives information associated with both ears. Damage to one cortical region consequently differs from removing all input from one ear; the behavioral outcome depends on which central computation was interrupted.



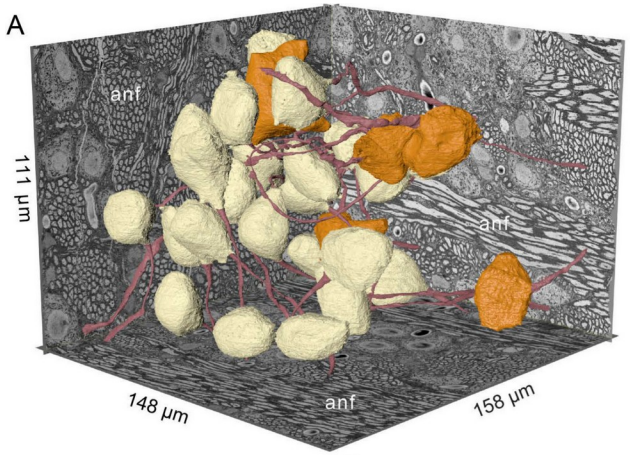
Carlson and Birkett (2022), Fig. 7.7, p. 198.

Large brainstem synapses preserve the timing of sound

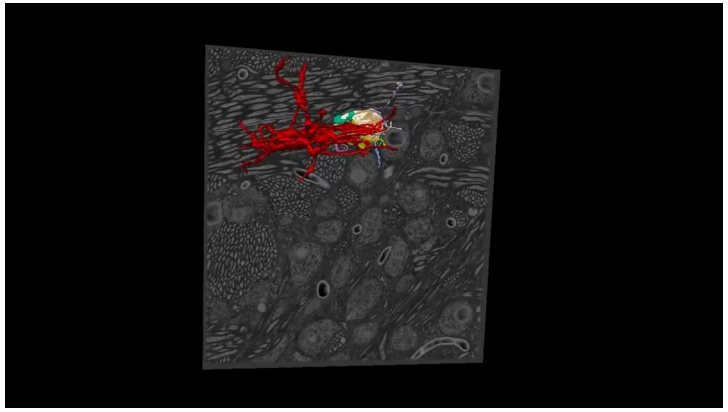
A **globular bushy cell** is a brainstem neuron that contributes to precisely timed auditory signaling. Large terminals called **endbulbs** contact its cell body. Their broad contact surfaces contain many transmitter-release sites, allowing an incoming action potential to produce a substantial excitatory response close to the neuron's spike-generating region.

Spirou and colleagues reconstructed mouse cochlear-nucleus tissue using serial electron microscopy. Successive images revealed cell membranes, axons, and synaptic contacts through a volume of brain tissue. Following a terminal across these images established which central neuron it contacted and how much of the cell surface the terminal occupied.

The source movie moves through the original tissue volume and its reconstructed cell. It identifies a physical arrangement of inputs; it does not directly record the neuron's firing during listening. The researchers used the measured geometry to constrain electrical models, allowing them to test how different combinations of those inputs could trigger spikes.



Spirou et al. (2023), Fig. 1, selected panels.



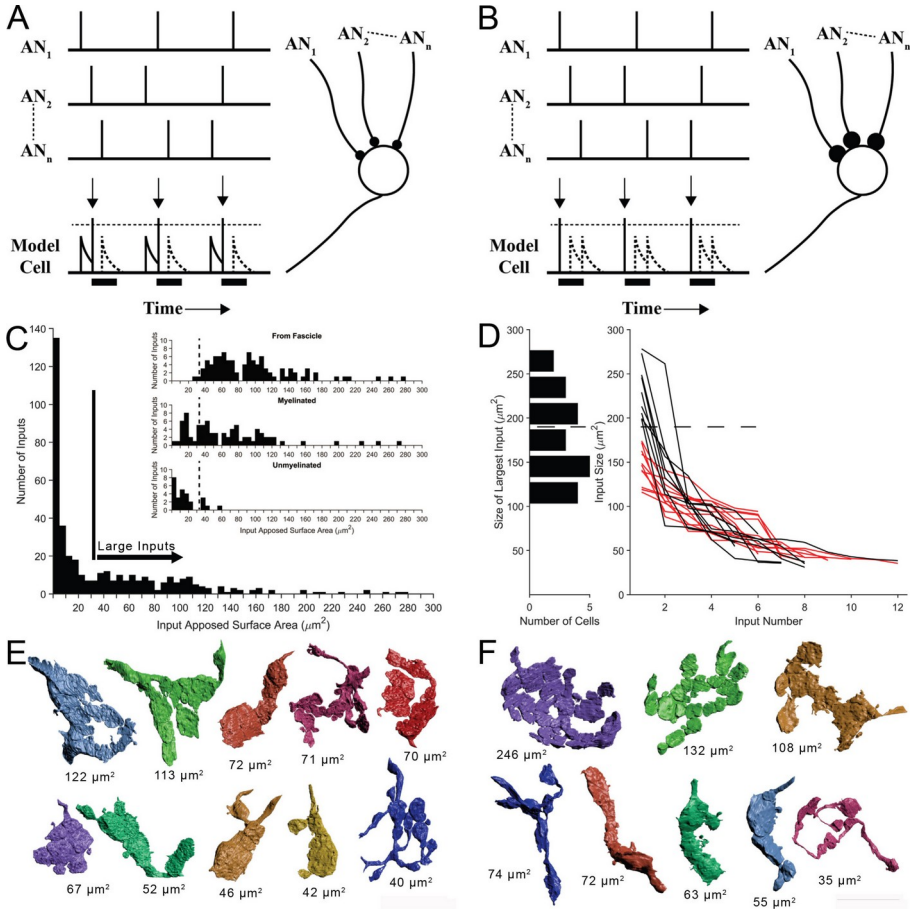
Spirou et al. (2023): Serial tissue volume and reconstructed bushy cell.

Fast excitation preserves precise timing

An **excitatory postsynaptic potential** is a depolarization produced when an input makes a neuron more likely to fire. For auditory timing comparisons, that depolarization must rise and decline quickly. A long-lasting response would allow signals arriving at substantially different moments to overlap, reducing the precision of a coincidence computation.

Rapid glutamate receptors and specialized membrane channels help some auditory brainstem neurons respond within short time windows. Potassium currents can shorten depolarization and restore the membrane toward its resting state. Strong contacts near the cell body also reduce the electrical distance between the activated synapse and the region where spikes begin.

These properties do different jobs. A large synaptic input increases the chance of reaching threshold, while a brief response restricts when another input can combine effectively with it. Changing either strength or duration can alter output timing, even if the presynaptic axon continues producing action potentials at exactly the same moments.



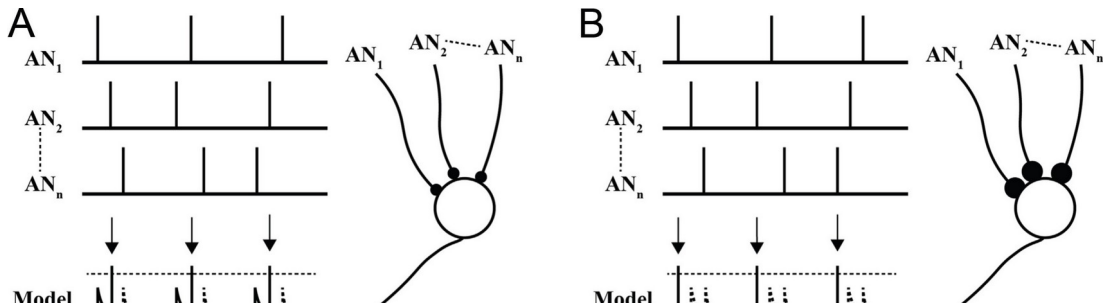
Spirou et al. (2023), Fig. 2.

A bushy cell combines a small set of powerful inputs

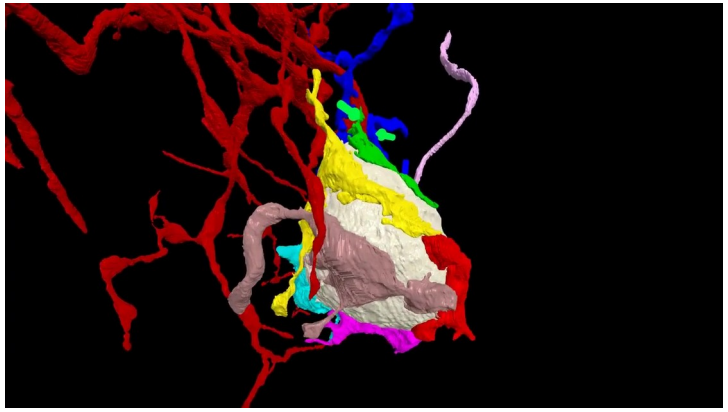
The number of terminals contacting a bushy cell determines which incoming signals can converge on it. However, counting terminals alone does not establish their electrical influence. A small contact and a large endbulb can differ substantially in their number of release sites and in the excitatory current they can deliver.

The reconstructed mouse cells received different combinations of large somatic inputs. Spirou and colleagues incorporated those measured contacts into electrical models. Some modeled cells required coincident activity across several subthreshold inputs, whereas others could be driven by one or two stronger inputs, producing different routes to an output action potential.

A **subthreshold input** depolarizes the neuron without producing a spike by itself. When several such responses overlap, their combined effect can cross threshold. A stronger individual input can cross threshold alone. These modeled alternatives explain how anatomical differences could change whether a central neuron emphasizes coincidence or follows a particularly influential incoming axon.



Spirou et al. (2023), Fig. 2, selected panels.



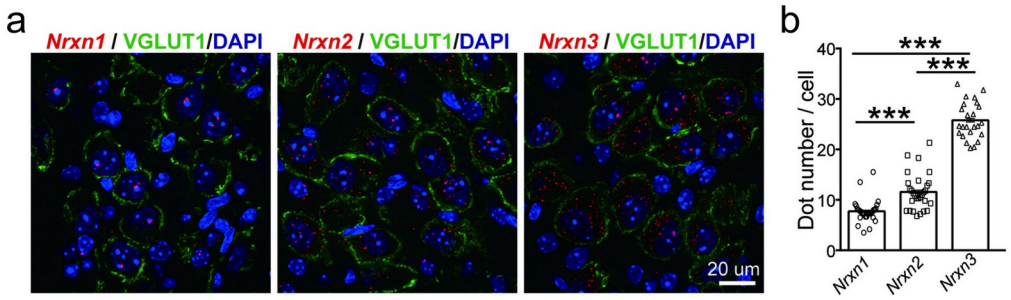
Spirou et al. (2023): Somatic inputs to a reconstructed bushy cell.

Inhibition changes auditory brainstem output

Glycine is an inhibitory transmitter used prominently in auditory brainstem circuits. At the lateral superior olive, glycinergic input from the medial nucleus of the trapezoid body opposes excitatory drive. The relative timing of these signals determines whether excitation reaches spike threshold before inhibition reduces the neuron's response.

Jiang and colleagues removed **neurexins**, proteins that organize presynaptic contacts, from selected mouse brainstem neurons. The manipulation weakened and changed the timing of glycinergic transmission onto lateral superior olive neurons. The investigators examined synaptic currents rather than inferring the effect solely from the presence of a labeled anatomical connection.

Calcium entering the presynaptic terminal normally helps trigger vesicles to release transmitter. The experiments indicated that removing neurexins disrupted the close coordination between calcium channels and release machinery. A less precisely coupled inhibitory input changes both how much inhibition arrives and how tightly that inhibition follows the triggering presynaptic event.



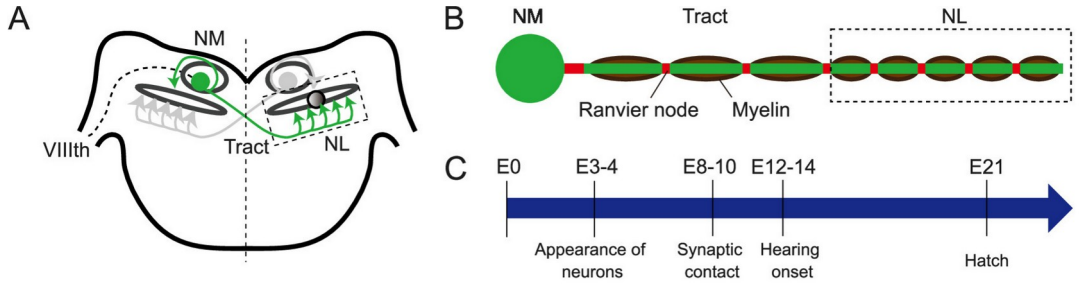
Jiang et al. (2024), Fig. 1, selected panels.

Myelin organizes conduction along sound-localization axons

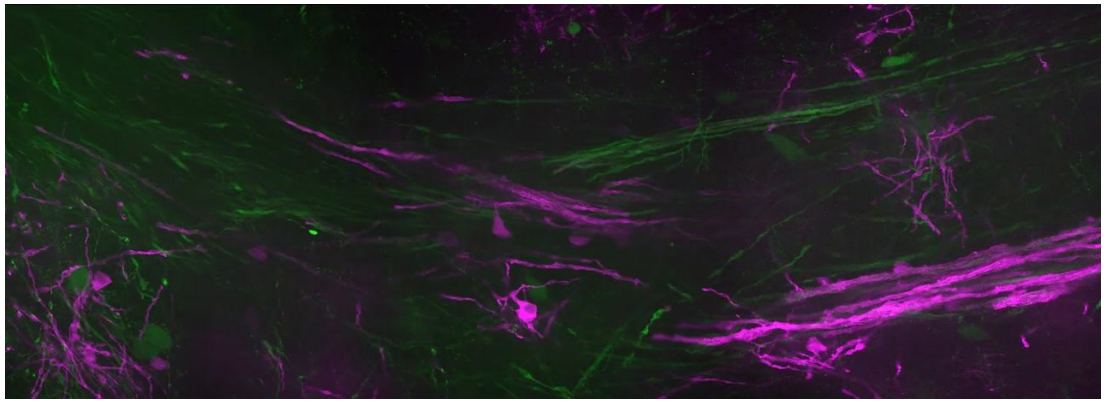
Myelin is an insulating membrane sheath produced by glial cells called **oligodendrocytes** in the central nervous system. Short gaps between sheaths, the **nodes of Ranvier**, contain membrane machinery that regenerates the action potential. Their arrangement influences how quickly an electrical signal travels along a central auditory axon.

Egawa and colleagues examined the chick brainstem pathway from nucleus magnocellularis to nucleus laminaris. These nuclei contribute to the comparison of sound timing across the two sides. Three-dimensional microscopy allowed the researchers to follow labeled axons alongside the oligodendrocytes that wrapped different portions of their length.

The source movie shows the tract region, where axons travel toward their targets. Green labels identify axons and magenta labels identify sparsely marked oligodendrocytes. Following the labeled processes reveals how one glial cell contributes sheaths to nearby axons, making the cellular arrangement underlying conduction timing visible within the actual tissue.



Egawa et al. (2025), Fig. 1, selected panels.



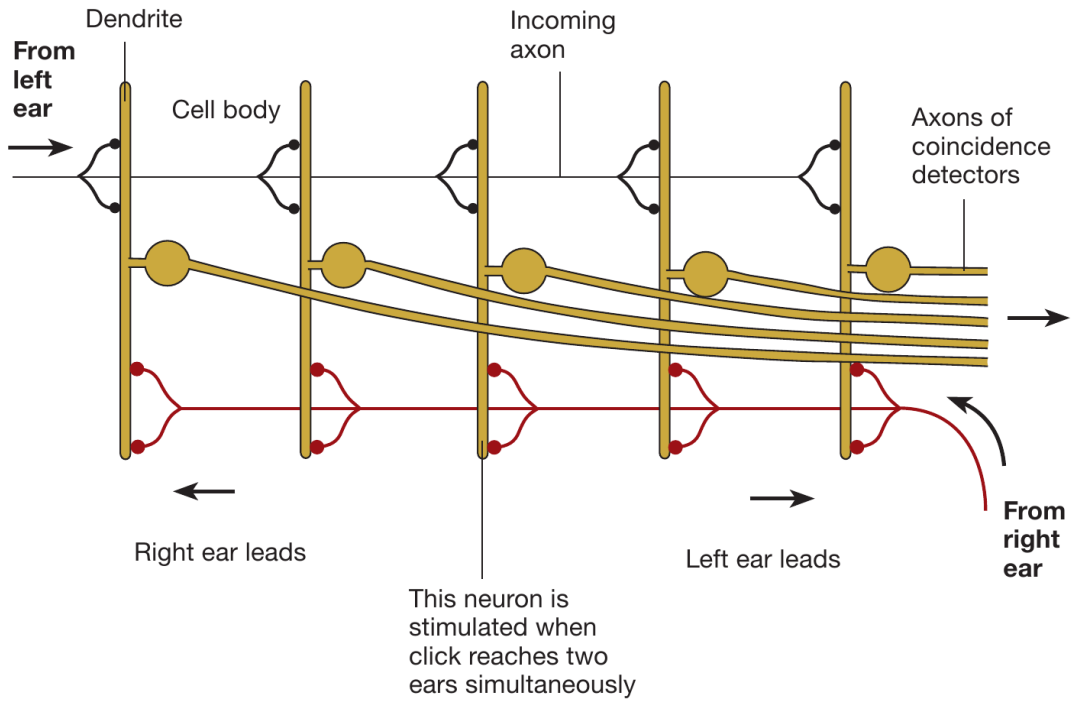
Egawa et al. (2025): Axons and oligodendrocytes in the tract region.

Coincidence detection compares the timing of two inputs

A **coincidence detector** responds especially strongly when two inputs arrive within a narrow time window. In auditory localization, inputs associated with the two ears can converge on such a neuron. Their excitatory responses combine most effectively when they overlap, converting a difference in arrival time into a difference in firing.

Conduction delays can align signals that began at different moments. An earlier signal traveling along a longer route can reach a neuron together with a later signal taking a shorter route. The coincidence concerns arrival at the receiving neuron, which differs from simultaneous arrival of sound at the ears.

Timing-sensitive circuits occur in birds and mammals, but their detailed organization differs. The owl's nucleus laminaris provides a well-studied example of coincidence-based spatial computation. A population of neurons with different timing preferences can distinguish alternative sound directions because each direction changes which inputs overlap most effectively at their synapses.



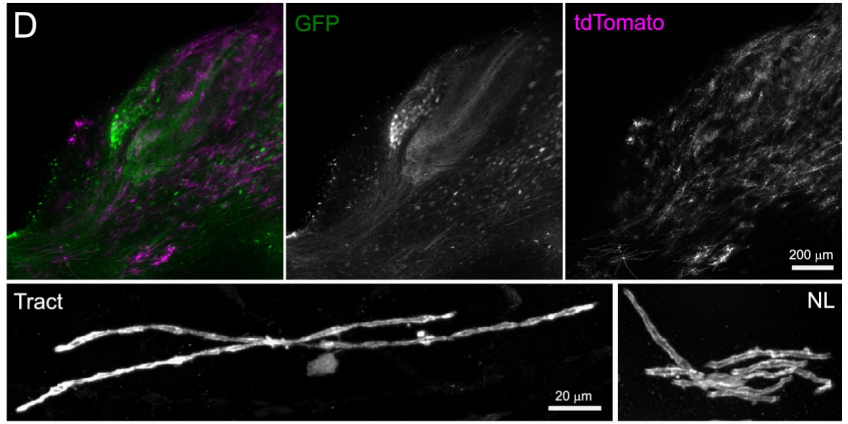
Carlson and Birkett (2022), Fig. 7.13, p. 204.

Myelinating cells differ across axon regions

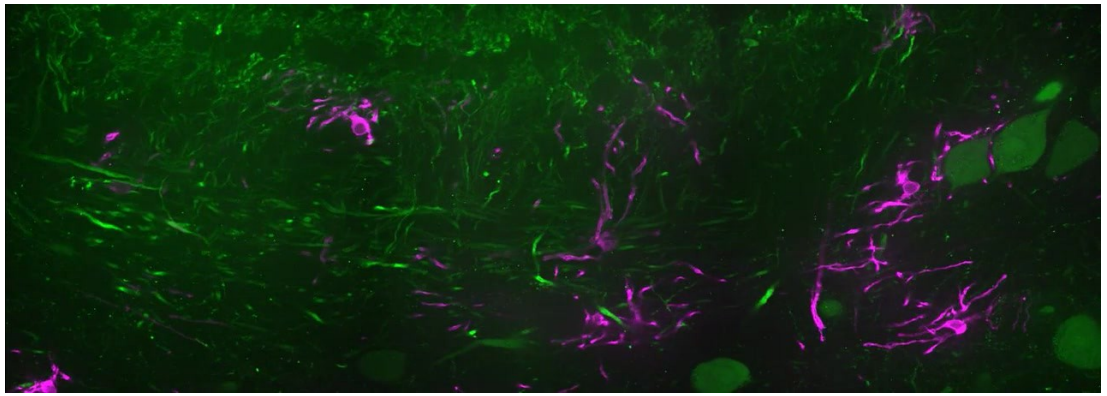
The same auditory axon can pass through a fiber tract and then enter the nucleus where its terminals contact receiving neurons. These regions contain different local cellular environments. Egawa and colleagues compared oligodendrocytes in the tract with those in nucleus laminaris while tracing how their processes contributed to myelin along individual axons.

The glial cells differed in morphology and regional density. Their sheaths also contributed to a nonuniform pattern of distances between nodes. The physical organization of the myelinating cells therefore provided a candidate explanation for why one axon could have different internodal spacing before and after entering its target region.

The second microscopy movie shows the nucleus-laminaris region rather than another view of the tract. Comparing the two preparations reveals the regional difference in cell organization. The study also interfered with transmitter release and found effects on myelination coverage, while the regional pattern of internodal length was comparatively preserved.



Egawa et al. (2025), Fig. 3, selected panels.



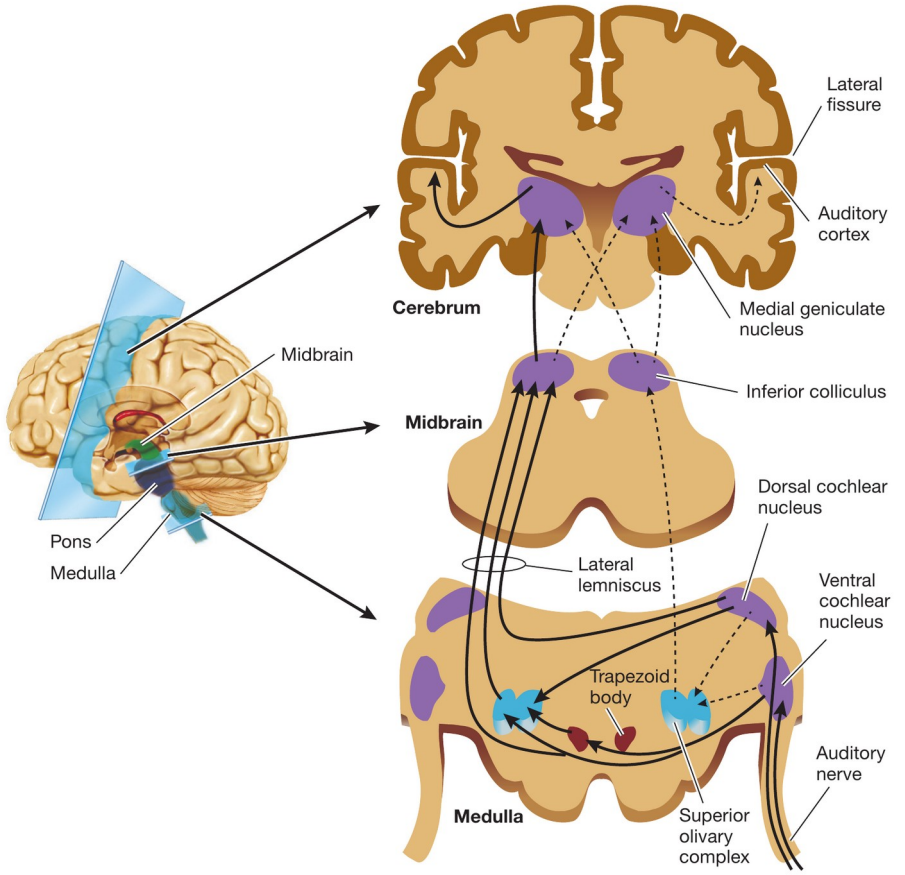
Egawa et al. (2025): Axons and oligodendrocytes near target nuclei.

Excitation and inhibition compare sound levels across ears

An **interaural level difference** occurs when a sound produces different signal strengths associated with the two ears. Lateral superior olive neurons receive excitatory and inhibitory inputs that carry information from opposite sides. Their output depends on the balance of these inputs rather than on one side's activity considered in isolation.

Stronger excitatory drive can increase firing, while stronger opposing inhibition can reduce it. Shifting that balance changes the response to a sound located on one side of the listener. A downstream population can compare the activity of neurons with different balances to estimate which sound directions are most consistent with the input.

Time and level comparisons supply complementary spatial information. The medial and lateral superior olive emphasize different aspects of those comparisons, with additional processing occurring in the midbrain and cortex. The result is a distributed estimate of sound direction that can be updated when the head moves or another sound enters the scene.



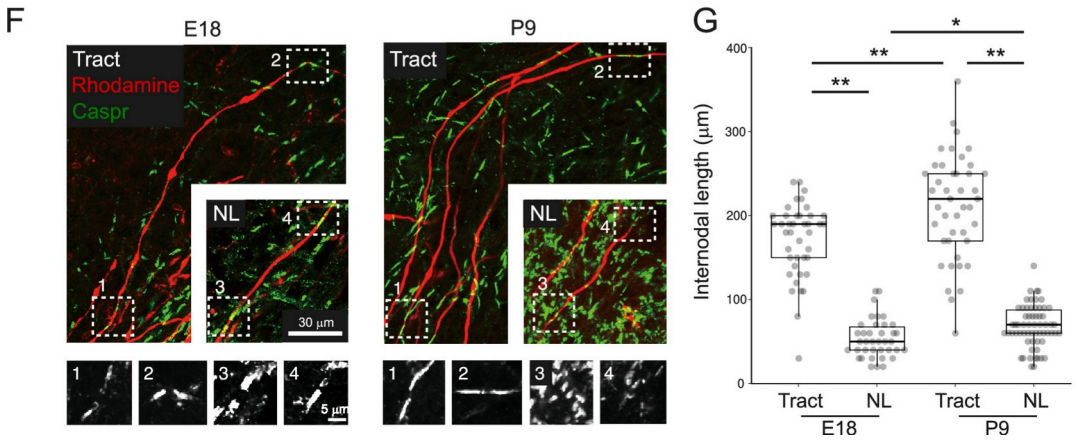
Carlson and Birkett (2022), Fig. 7.7, p. 198.

Node spacing can vary along one auditory axon

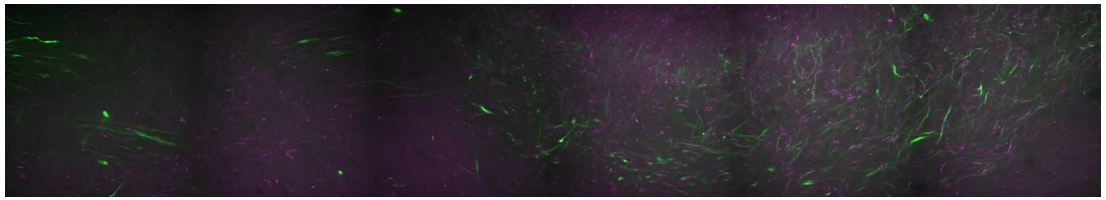
An **internode** is a myelinated stretch between adjacent nodes of Ranvier. Its length affects conduction alongside axon diameter, membrane properties, and the organization of the sheath. An auditory pathway can consequently adjust signal travel time through the arrangement of its axon and glial cells, without changing the sound being represented.

In the chick pathway, Egawa and colleagues observed regionally biased node spacing along individual axons. Detailed reconstruction allowed them to ask whether changes in axon structure explained that pattern. Their results instead emphasized regional differences in oligodendrocyte organization as a prominent contributor to the spacing of the myelin segments.

The third source movie marks axons and structures bordering their nodes through a stitched tissue volume. Tracing one axon across regions reveals that its nodal arrangement changes along its course. Such variation matters to a timing circuit because the receiving neuron compares the arrival of spikes, after those spikes have completed their axonal travel.



Egawa et al. (2025), Fig. 2, selected panels.



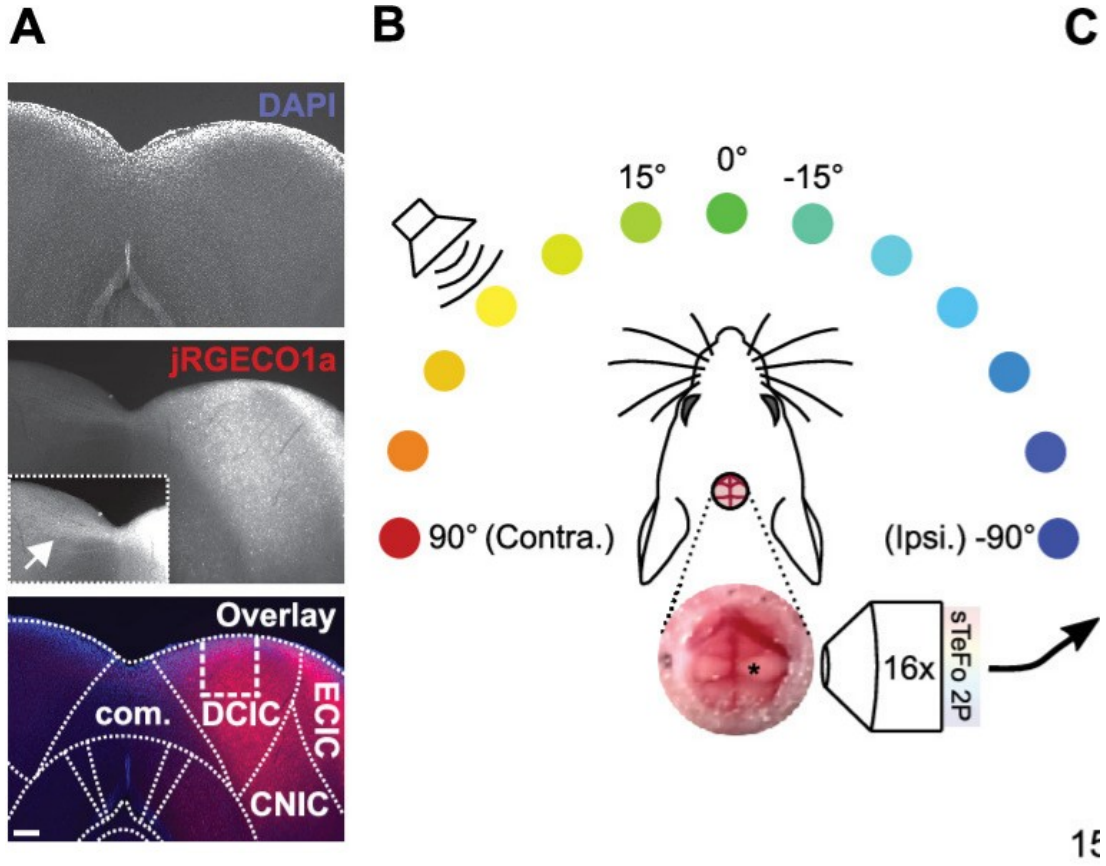
Egawa et al. (2025): The transition between tract and target region.

Populations encode sound direction

A neuron's response varies across repeated presentations of the same sound. This **trial-to-trial variability** means that one cell's firing cannot always identify the source location reliably. A population can still carry useful spatial information when different neurons change their responses in systematically different ways as the sound moves.

Boffi and colleagues recorded many neurons in the dorsal inferior colliculus of awake mice. They presented sounds from different horizontal directions and measured activity using calcium imaging and electrical recordings. Both approaches revealed noisy individual responses, but combinations of responses could identify sound direction on individual trials.

A **decoder** is an analysis that uses measured activity to predict an experimental variable, here the direction of the speaker. Successful decoding establishes that the recorded population contains usable spatial information. It does not require every neuron to prefer one unique location or every repetition to produce the same activity pattern.



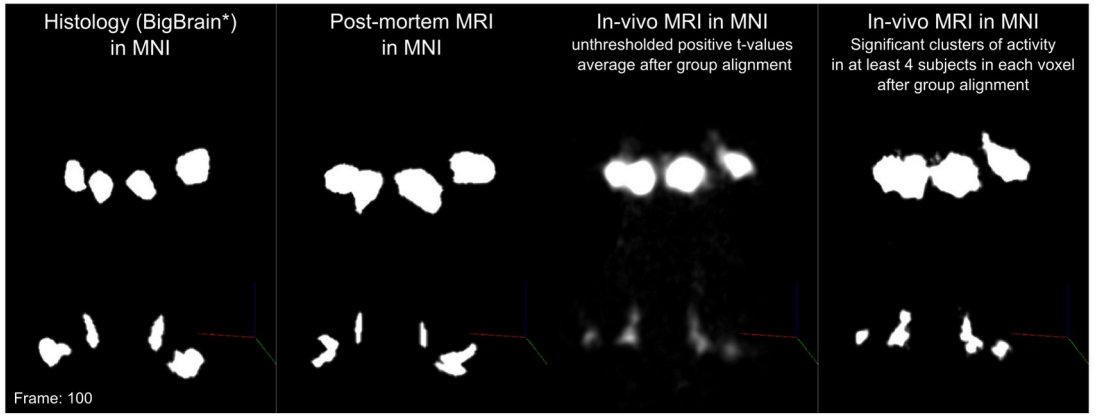
Boffi et al. (2024), Fig. 1, selected panels.

Mapping human nuclei in a shared brain space

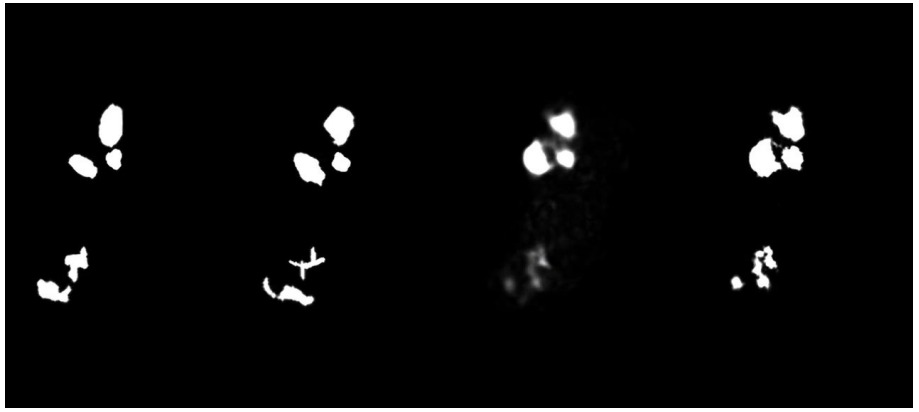
Human auditory brainstem nuclei are small and densely packed. Their apparent position can differ across individuals and imaging methods. Sitek and colleagues combined histological sections, postmortem magnetic resonance imaging, and living-human scans to construct complementary representations of these central structures and the pathways connecting them.

An **atlas** places labeled structures into a coordinate system that other measurements can use. Aligning an individual's brain with that system helps compare a functional response with known anatomy. The match remains a measured alignment between datasets; its colored boundaries are not direct photographs of every cell in that individual.

The source animation compares three-dimensional representations in a shared brain space. It helps locate the cochlear nuclei, superior olivary complex, inferior colliculi, and medial geniculate bodies relative to one another. Those landmarks allow a researcher to ask which central auditory station contributes to a response measured deep below the cortical surface.



Sitek et al. (2019), Fig. 9.



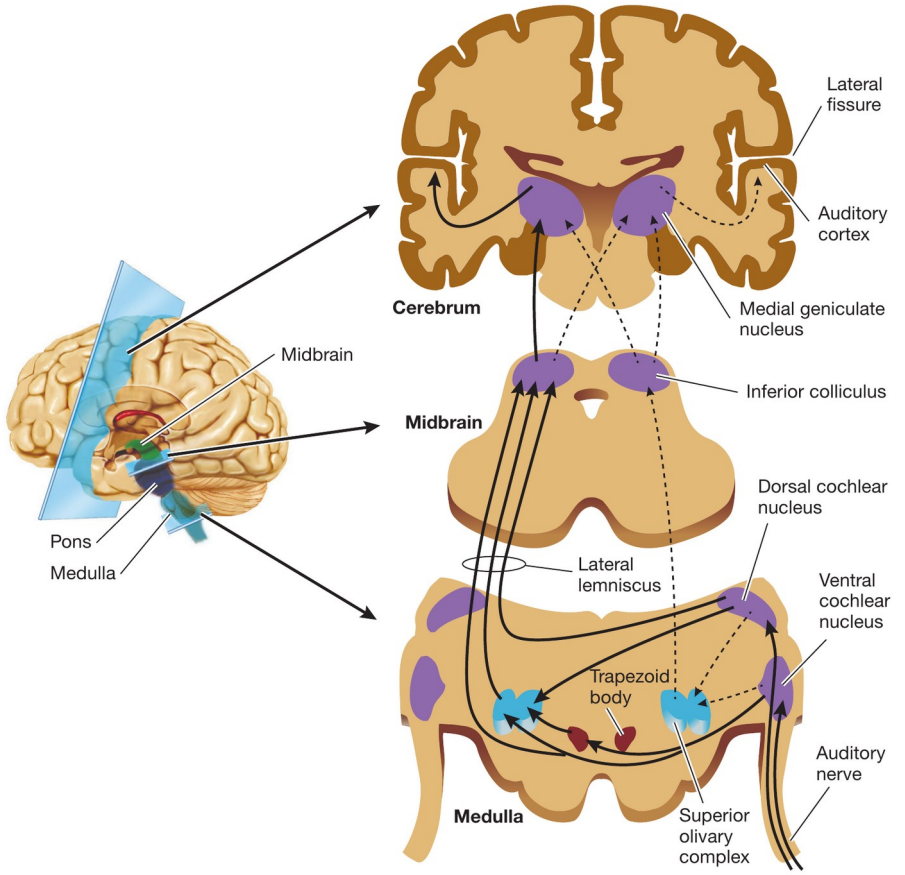
Sitek et al. (2019): Comparing histology and MRI in shared brain space.

The inferior colliculus integrates sound features

The **inferior colliculus** is a midbrain auditory structure receiving several ascending inputs and descending cortical connections. Its neurons combine information about frequency, timing, and spatial differences. Some regions also receive signals associated with movement and other senses, placing acoustic input within the changing conditions of an animal's behavior.

Combining inputs can change a neuron's response even when the incoming sound remains constant. Excitation from one pathway may reinforce an onset response, while inhibition from another can suppress activity during a different time interval. The same acoustic event can therefore produce different output patterns depending on the circuit state.

These signals contribute to several subsequent operations. Projections toward auditory thalamus supply information for cortical processing, while other connections help support rapid orientation and behavioral responses. A central auditory structure can participate in detection, localization, and coordination without representing each of those functions in one identical population code.



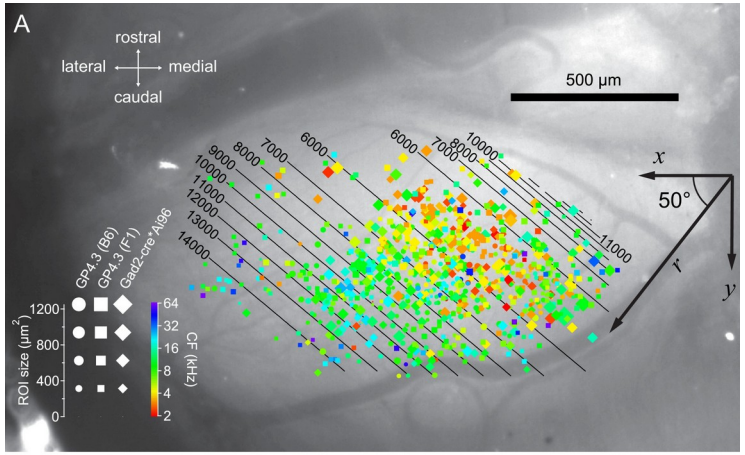
Carlson and Birkett (2022), Fig. 7.7, p. 198.

Frequency preferences organize the midbrain

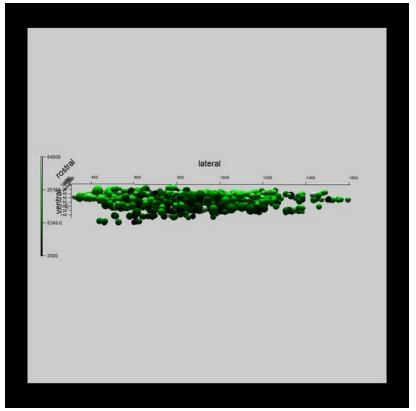
Tonotopy is an orderly relationship between preferred sound frequency and neural location. Wong and Borst imaged sound responses in the dorsal inferior colliculus of awake mice. They compared neurons that used the inhibitory transmitter GABA with other neurons, asking how response properties were arranged across the imaged tissue.

Nearby cells could differ in the time course of their sound responses. Nevertheless, frequency preference followed a broader spatial organization that appeared in both neuronal groups. A locally mixed arrangement of temporal responses therefore coexisted with a more systematic gradient of preferred frequencies across the larger recorded region.

The rotating source map places sound-responsive regions at their measured three-dimensional coordinates and colors them by characteristic frequency. It makes the larger frequency organization visible while retaining individual recording locations. Position in this map identifies sampled tissue, whereas the color records the frequency preference estimated from that tissue's response to the experimental sounds.



Wong and Borst (2019), Fig. 6, selected panels.



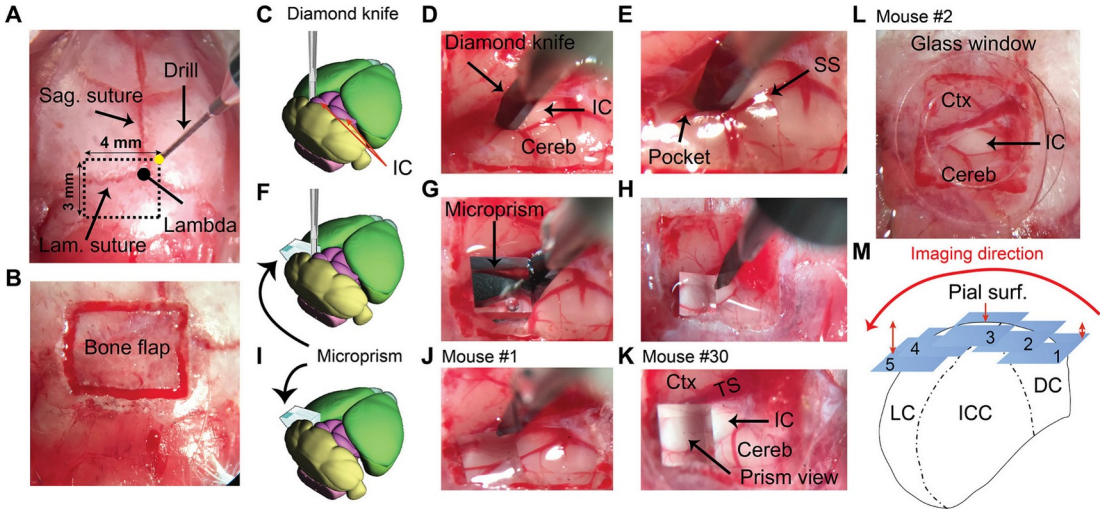
Wong and Borst (2019): Frequency organization in the inferior colliculus.

A microprism reveals deeper auditory activity

Some auditory midbrain regions are hidden along a deep fold, making them difficult to reach with ordinary surface imaging. Ibrahim and colleagues used a **microprism**, a small optical element that redirects the microscope's view. This allowed them to image the lateral cortex of the inferior colliculus from a different anatomical angle.

The researchers compared cellular regions called **modules** with the surrounding **matrix**. These structural compartments differed in their sound responses and in the way somatosensory stimulation affected those responses. Module neurons more often emphasized sound offset, while matrix neurons more often emphasized sound onset in the reported comparisons.

Combining sound and touch did not simply increase every neuron's activity. Responses in modules could be enhanced, while responses in matrix tissue could be suppressed. The anatomical compartments therefore helped separate different ways of combining sensory inputs, allowing a highly connected midbrain region to retain distinct response patterns within neighboring tissue.



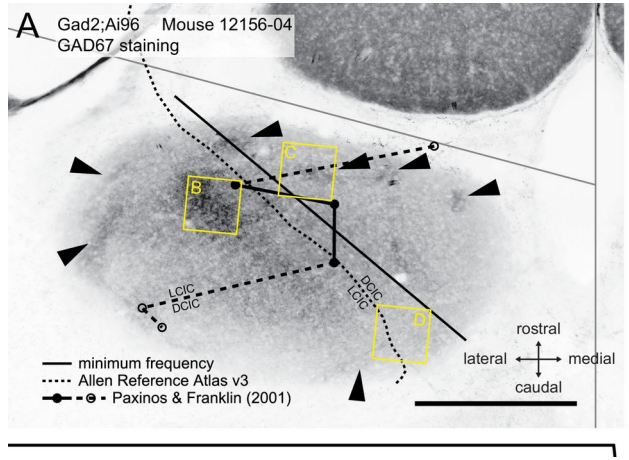
Ibrahim et al. (2025), Fig. 1.

Inhibitory modules organize midbrain inputs

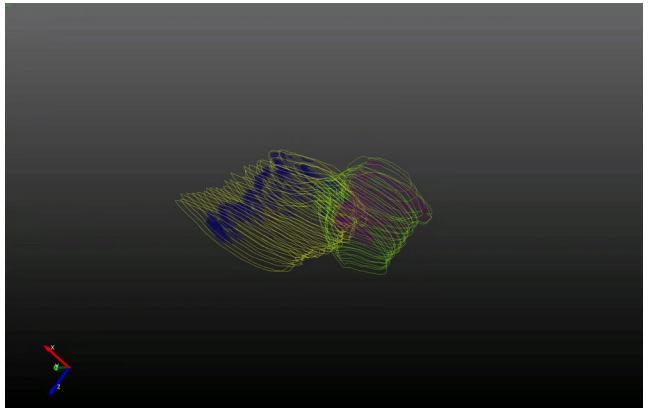
GABAergic neurons release GABA, a transmitter that usually reduces the activity of their receiving cells in the mature brain. The inferior colliculus contains organized inhibitory populations as well as other neurons. Their distribution provides an anatomical framework within which auditory and movement-related signals can interact during ongoing behavior.

Wong and Borst examined both sound responses and activity associated with spontaneous movement. They also reconstructed inhibitory modules from tissue sections. The combination allowed them to compare functional organization with anatomical landmarks instead of assuming that a boundary drawn from one kind of measurement must describe every property of the region.

The source movie reconstructs labeled modules across the left and right inferior colliculi. Each shape derives from the original serial sections, whose depth determines its position in the volume. The animation depicts the organization of inhibitory tissue; the paper's live recordings separately supply evidence about sound tuning and movement-associated cellular activity.



Wong and Borst (2019), Fig. 9, selected panels.



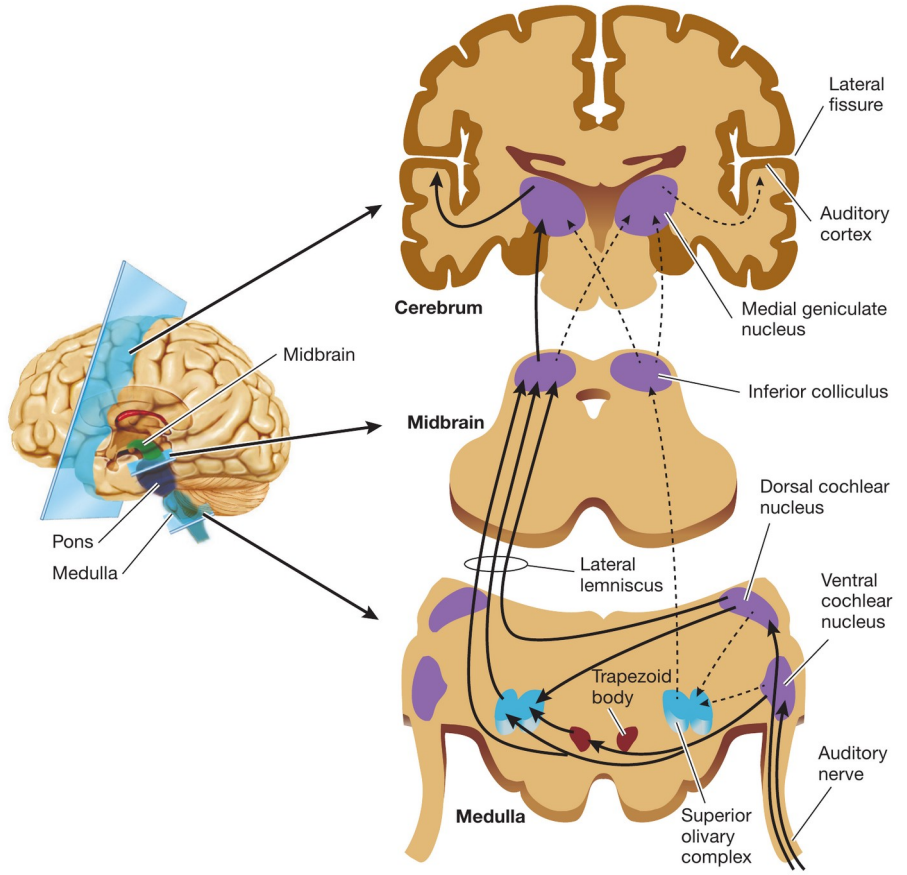
Wong and Borst (2019): Reconstructed inhibitory modules.

Auditory thalamus connects midbrain and cortex

The **auditory thalamus** includes the medial geniculate body, which receives major input from the inferior colliculus. Its subdivisions participate in different central pathways. Some projections strongly target primary auditory fields, while others reach additional cortical regions, distributing information across circuits that differ in their response properties and connections.

Thalamic neurons are influenced by both ascending input and feedback from cortex. The message reaching a cortical cell therefore depends on interactions within the pathway rather than on one isolated incoming spike train. Excitation, inhibition, and membrane state determine which inputs generate a thalamic response and when that response reaches cortex.

The pathway diagram shows a major route from midbrain to thalamus and temporal cortex. Other central connections coexist with this route, including projections that return toward earlier stations. Learning how these structures are connected makes it possible to distinguish a change generated in cortex from one already present in the signal arriving there.



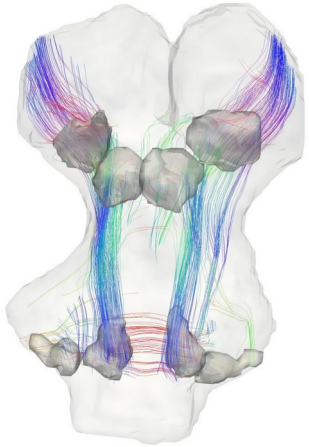
Carlson and Birkett (2022), Fig. 7.7, p. 198.

High-field imaging traces human auditory pathways in vivo

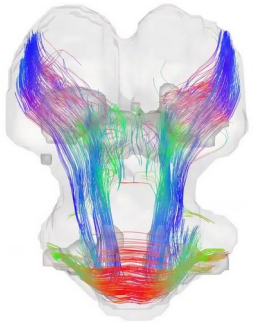
Diffusion MRI measures how water movement varies across tissue. Researchers use those measurements to estimate likely orientations of white-matter pathways. Sitek and colleagues combined this information with anatomical labels of the human auditory nuclei, generating three-dimensional representations of possible connections between structures that are difficult to inspect in a living brain.

The resulting **tractography streamlines** follow estimated directions through successive image locations. They provide evidence about the spatial course of a pathway, but an individual streamline is not a directly photographed axon. Histology and postmortem measurements helped compare the living-human estimates with anatomy sampled at finer spatial resolution.

The animation rotates a living-human reconstruction so that overlapping structures can be separated visually. Connections that look adjacent in one two-dimensional slice can follow different courses through depth. Viewing the volume from several angles helps identify which central nuclei lie along a reconstructed route and where different pathways approach one another.



Sitek et al. (2019), Fig. 6, selected panels.



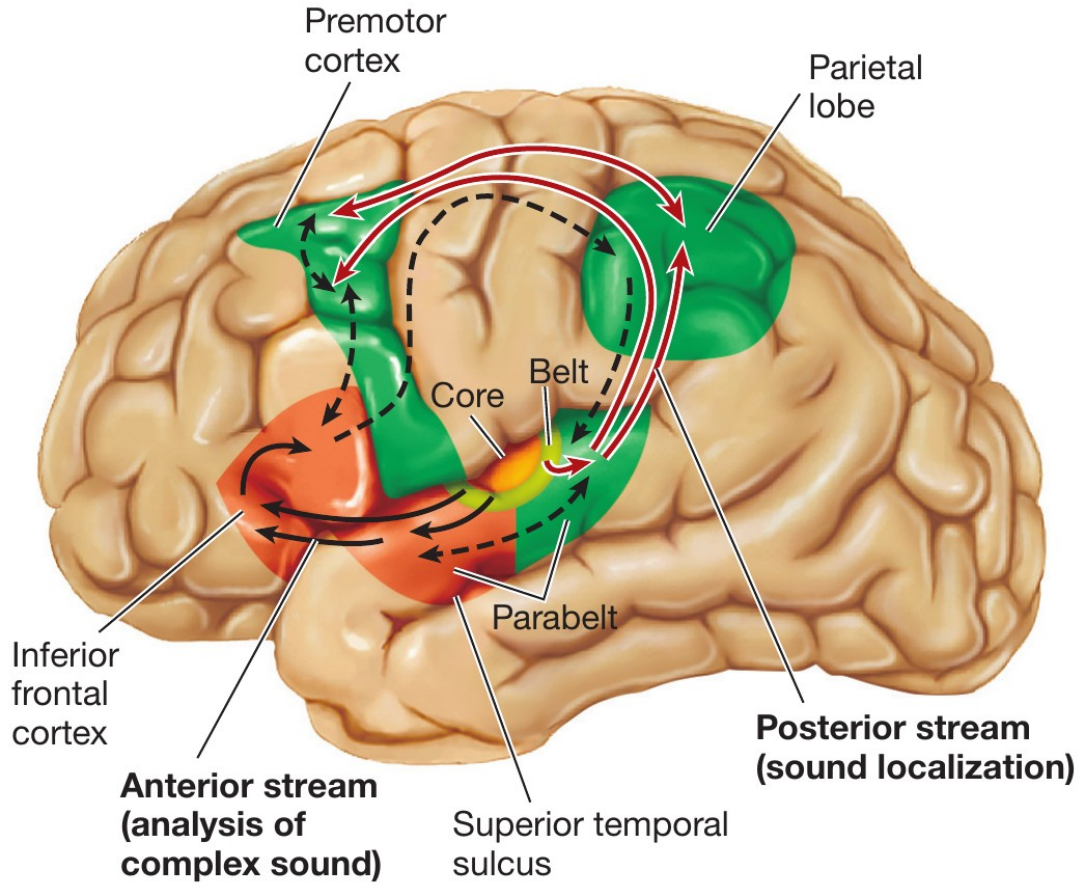
Sitek et al. (2019): Human auditory tractography.

Core, belt, and parabelt transform sound signals

Primary auditory cortex lies on the superior temporal surface and receives major input from auditory thalamus. In primates, a central **core** is surrounded by **belt** and **parabelt** regions. These regions form interconnected networks that progressively combine acoustic features with information about objects, actions, and the listener's current task.

Several fields contain organized frequency preferences, so cortical tonotopy consists of multiple related maps. A position within one map does not identify a complete word, instrument, or melody. Natural sounds activate patterns across neurons whose responses also depend on timing, combinations of frequencies, and other features of the stimulus.

Connections extend anteriorly toward regions involved in recognizing complex sounds and posteriorly toward spatial and sensorimotor networks. These routes pass through several surrounding cortical regions. Their functions overlap and interact, allowing the same voice to be identified, located, and used to guide a turn toward the person who is speaking.



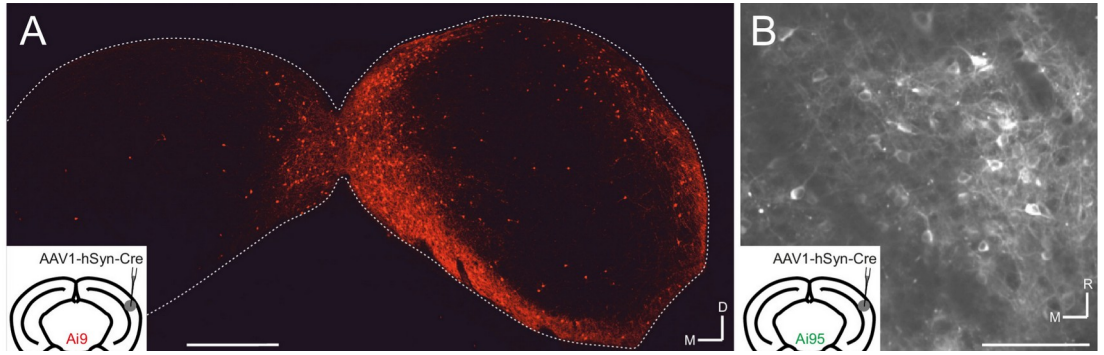
Carlson and Birkett (2022), Fig. 7.9, p. 200.

Midbrain activity predicts a mouse's sound-detection choice

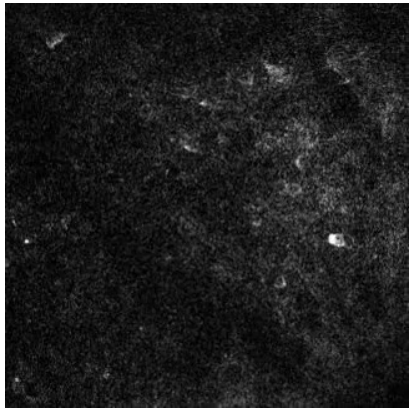
Lee and colleagues trained mice to report whether a sound was present. They imaged neurons in a shell region of the inferior colliculus that receives cortical input. Responses varied with more than the physical stimulus: activity also contained information about the animal's behavior during the sound-detection task.

The researchers tested whether the measured population could predict the mouse's response. They then examined animals with bilateral auditory cortical lesions. Behavioral information remained available in the recorded midbrain activity, showing that this information did not require an intact auditory cortex to appear within the studied collicular population.

The original recording shows calcium fluorescence during the task. A bright change reports a change in the indicator signal, which follows cellular calcium rather than displaying individual spikes directly. The analysis compared these signals with stimulus and behavioral events, separating a response to the sound from activity associated with the animal's decision and action.



Lee et al. (2024), Fig. 4, selected panels.

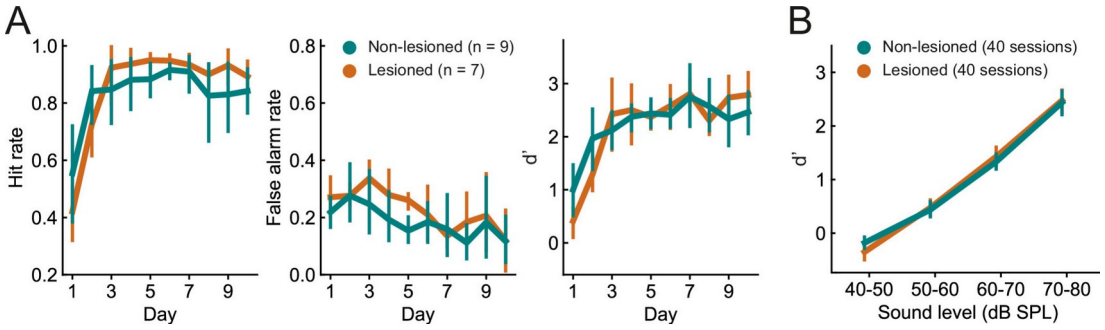

Lee et al. (2024): Calcium responses in mouse inferior colliculus.

Detecting a sound differs from identifying it

Detection asks whether a sound occurred, whereas **discrimination** asks how it differed from another sound. Recognizing a voice, understanding a sentence, and locating a quiet target place different demands on the central system. A circuit sufficient for one task may not preserve the information or connections required for another.

Auditory cortical damage consequently need not erase every response to sound. Subcortical circuits can still carry acoustic and behavioral information, as the mouse detection experiment demonstrated. More complex judgments may depend on cortical networks that compare sound features with learned categories, recent context, or information arriving from other sensory systems.

A lesion result must therefore be interpreted relative to the behavior that was tested. Reporting that a mouse can detect a trained signal does not establish normal recognition of every natural sound. The relevant biological distinction concerns which remaining pathways can transform the available information into the specific action required by that task.



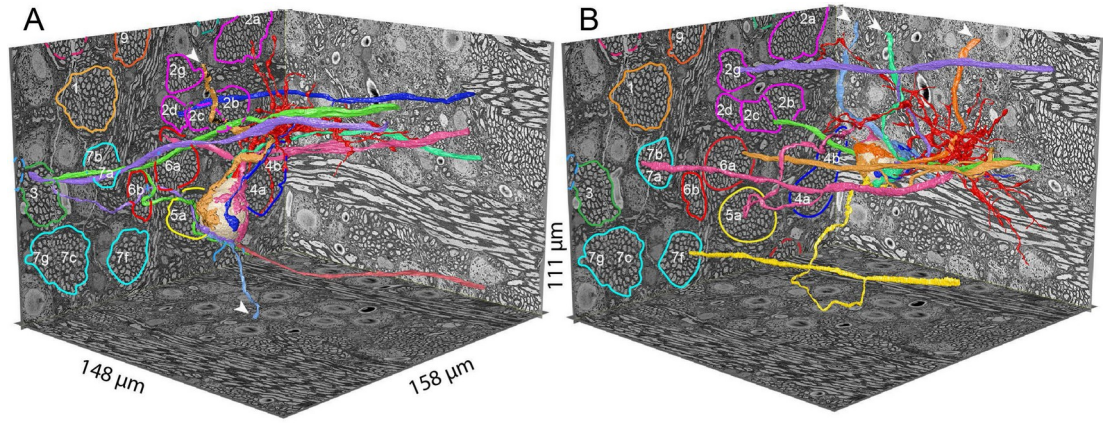
Lee et al. (2024), Fig. 3.

Several auditory nerve endings converge on one central cell

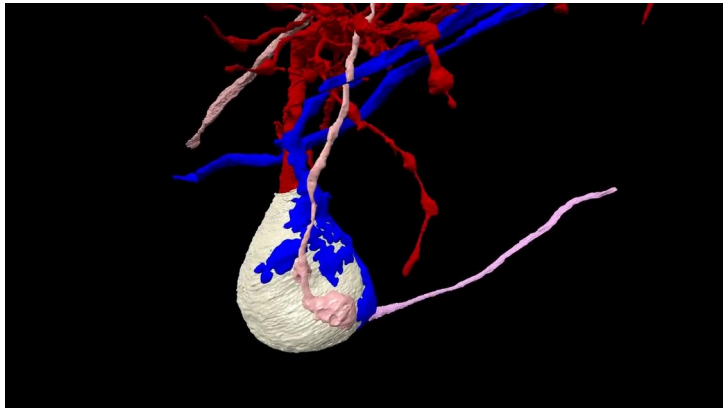
A large auditory nerve axon can branch before making terminal contacts within the cochlear nucleus. Those branches distribute its activity to central receiving cells. Reconstructing the axon together with its terminals reveals which contacts share a presynaptic origin, information that cannot be obtained by counting disconnected terminal profiles in individual tissue images.

Spirou and colleagues followed large somatic contacts backward through the original electron-microscopy volume. The branches differed in their length and arrangement, and several terminals converged on the same globular bushy cell. The reconstruction therefore specified both the receiving cell's input set and the anatomical paths supplying those inputs.

The movie separates a reconstructed neuron, its large terminals, and their parent axons by color. These colors identify structures traced through tissue rather than different measured sound frequencies. Following a branch to its contact explains how one incoming spike can recruit multiple release sites close to the receiving neuron's cell body.



Spirou et al. (2023), Fig. 3, selected panels.



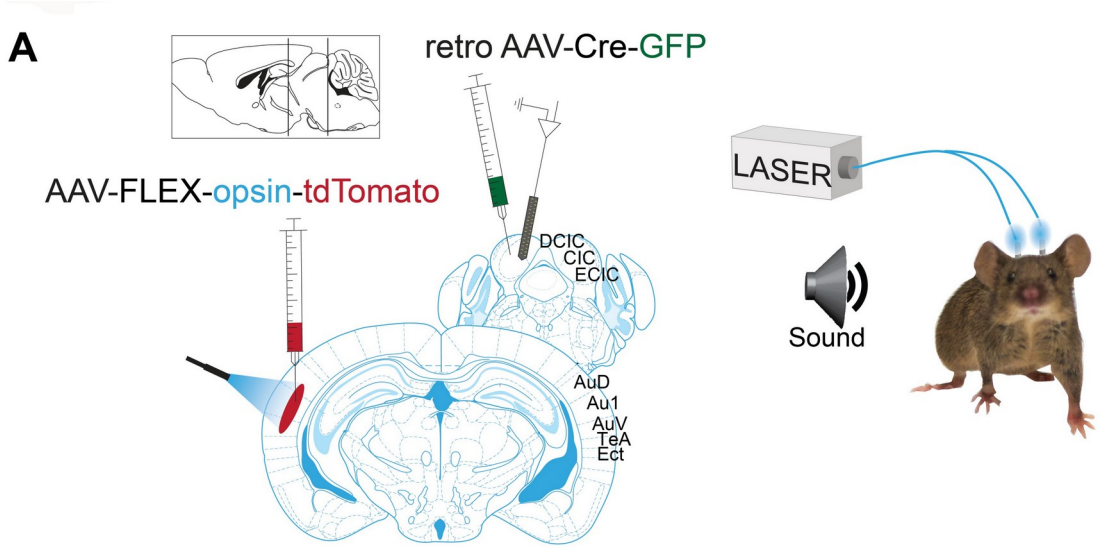
Spirou et al. (2023): Converging auditory nerve terminals.

Auditory cortex sends signals back to the inferior colliculus

Corticofugal feedback consists of projections that travel from cortex toward earlier processing stations. In the auditory system, cortical neurons send such projections to the inferior colliculus. A sound response in the midbrain can therefore reflect both ascending acoustic information and activity returning from a higher level of the central pathway.

Blackwell and colleagues introduced light-sensitive proteins into mouse corticocollicular neurons and manipulated their activity. Activating the projection changed inferior-colliculus responses, including their frequency selectivity. Suppressing the projection under the tested conditions did not produce a corresponding broad effect, indicating that its contribution depended on how the pathway was being recruited.

A descending excitatory projection can influence both excitatory and inhibitory target circuits. Its net effect need not be a uniform increase in firing across the inferior colliculus. Altering which cells are recruited changes the balance of those local interactions, providing a route through which cortical state can reshape earlier sound responses.



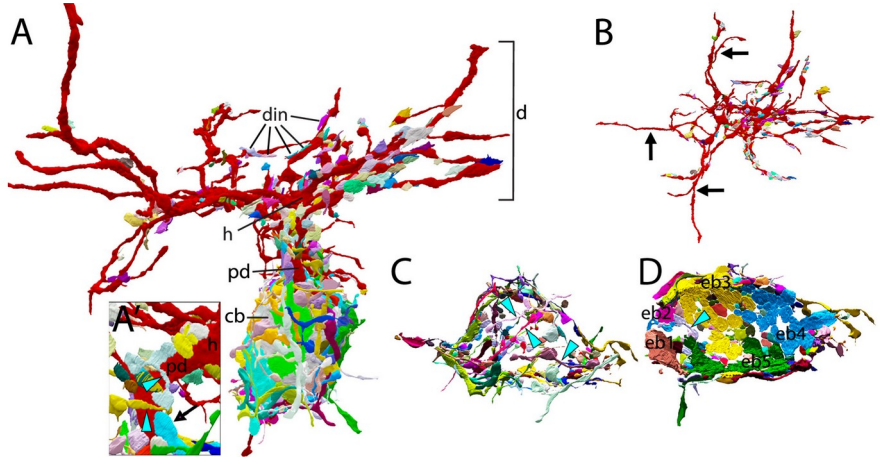
Blackwell et al. (2020), Fig. 1, selected panels.

Dendritic inputs add another route into a brainstem neuron

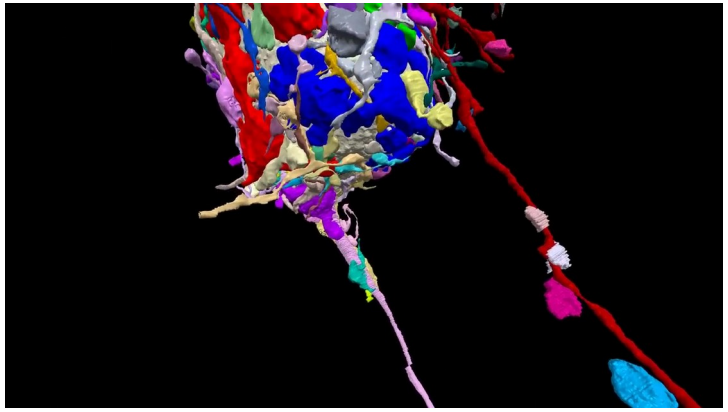
A neuron's **dendrites** receive inputs at locations electrically separated from its cell body. The size and shape of those branches influence how a synaptic current spreads toward the spike-generating region. A contact on a thin distant branch can therefore contribute differently from a large terminal located directly on the soma.

The mouse brainstem reconstructions included dendritic inputs as well as large somatic endbulbs. Spirou and colleagues measured those compartments and incorporated their geometry into electrical models. This allowed the influence of dendritic arrangement to be examined alongside soma size, input strength, and the structure of the axon initial segment.

The source movie follows one reconstructed cell and its identified synaptic inputs through the tissue-derived volume. Some dendritic structures lacked the expected innervation, showing why visible branch shape alone cannot specify the complete connection pattern. Reconstructing actual contacts supplies the additional information needed to test how that branch could contribute to central auditory integration.



Spirou et al. (2023), Fig. 8, selected panels.



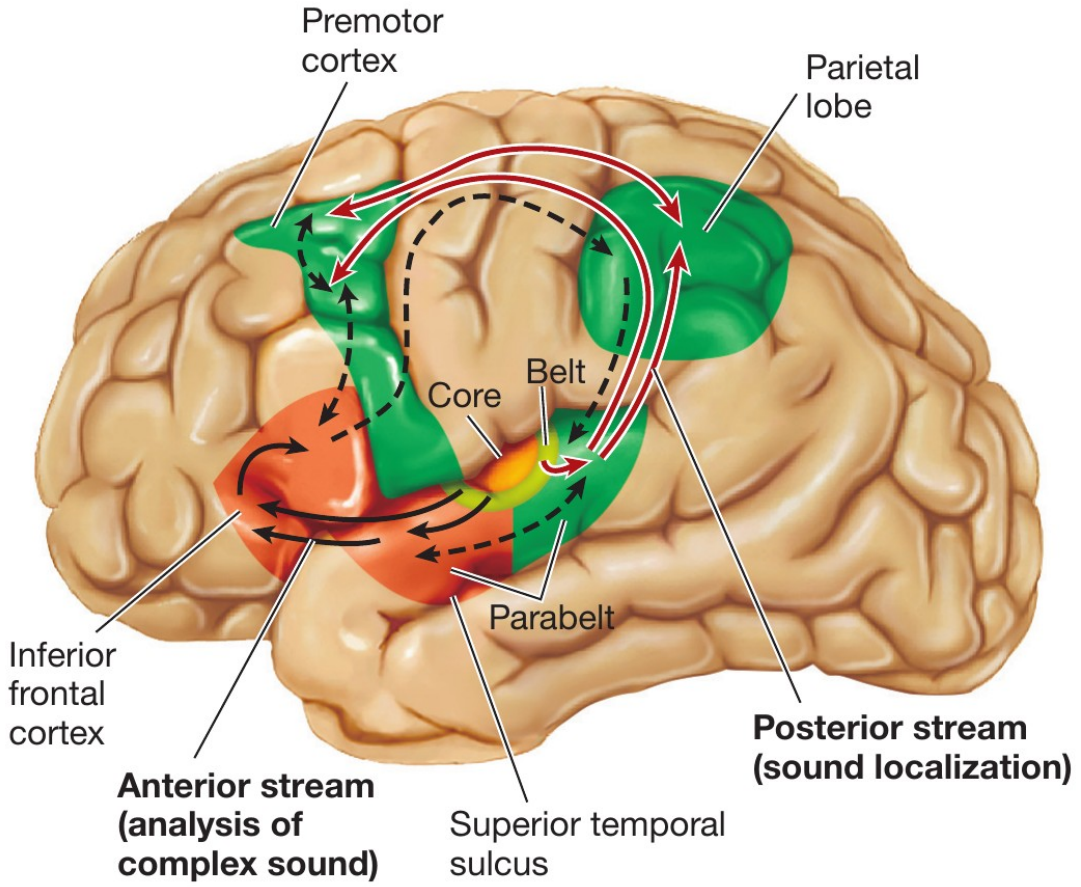
Spirou et al. (2023): Reconstructed dendrites and synaptic contacts.

Auditory streams emphasize identity and location

The **anterior auditory stream** contributes to identifying complex sounds, while a **posterior stream** contributes to spatial and sensorimotor operations. These routes connect auditory regions with temporal, parietal, and frontal cortex. Their different connections allow information from the same sound to contribute to recognition, localization, and selection of an appropriate action.

Recognizing a barking dog requires matching its changing acoustic pattern with learned sound categories. Turning toward it additionally requires estimating its location relative to the body. Identifying the sound does not supply that spatial transformation automatically, and locating it does not require identifying every feature of the individual animal producing it.

The streams exchange information rather than operating as sealed channels. A familiar voice can direct attention toward a particular location, and spatial separation can help identify that voice among competitors. Damage can nevertheless affect identity and location judgments differently, revealing that their successful integration depends on distinguishable operations within the central auditory network.



Carlson and Birkett (2022), Fig. 7.9, p. 200.

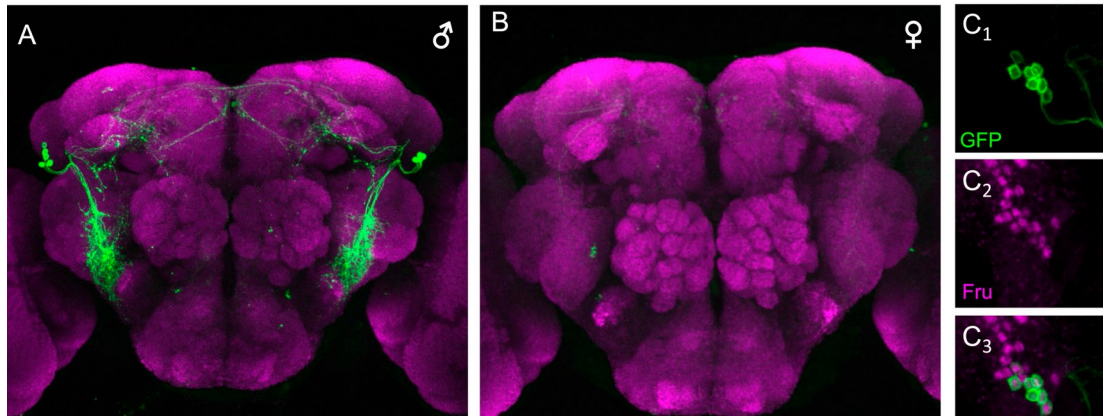
Activating song-responsive cells evokes courtship

Male fruit flies produce courtship songs whose timing influences other flies' behavior. Zhou and colleagues identified central neurons that respond to these signals. They examined a population called **vPN1**, which receives information from earlier auditory neurons and sends projections into brain regions involved in processing the social significance of sound.

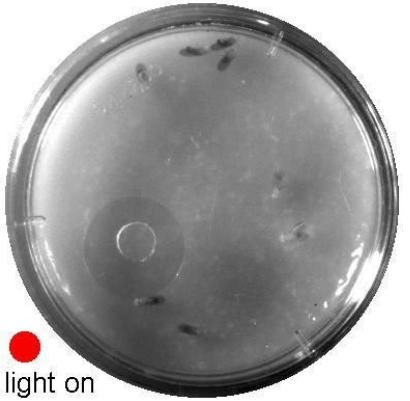
The researchers introduced **CsChrimson**, a light-sensitive protein, into selected neurons. Red illumination could then increase those cells' activity without presenting the usual song. Activating vPN1 recruited **courtship chaining**, in which male flies orient toward and follow other flies, connecting activity in an identified central population with a recognizable behavioral response.

The movie shows that behavioral manipulation in the original experimental animals. Light supplies an artificial trigger for the targeted circuit; the experiment does not show that the flies were hearing an external singer. Comparing the recruited action with song-evoked behavior tests which central cells can help transform an auditory signal into social approach.

R72E10-GAL4AD ∩ VT9665-GAL4DBD



Zhou et al. (2015), Fig. 2, selected panels.



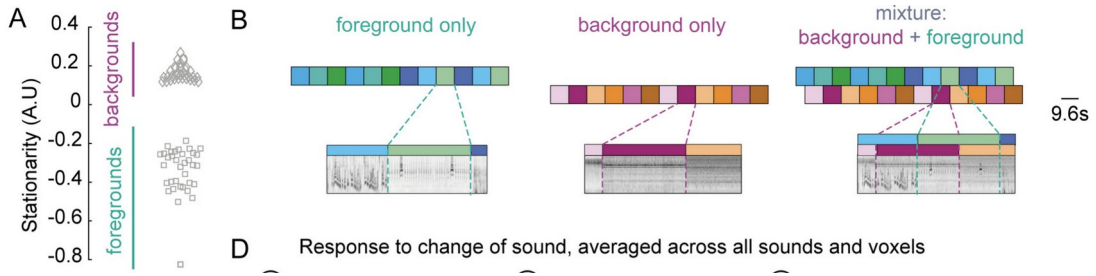
Zhou et al. (2015): vPN1 activation during fly courtship.

Higher regions separate natural sound mixtures

A foreground sound can be partly masked by background noise that overlaps its frequencies. Landemard and colleagues presented ferrets with natural sounds alone and in mixtures. They measured activity across auditory cortical regions using **functional ultrasound**, which detects changes in cerebral blood volume associated with local neural activity.

Responses in primary auditory cortex reflected contributions from both foreground and background. In higher auditory regions, responses became less sensitive to which background accompanied the foreground. This **background invariance** preserves information about a potential sound source while reducing the influence of surrounding acoustic activity that changes independently of it.

A spectrotemporal model could explain much of the ferret hierarchy through tuning to relatively simple acoustic features. The same model did not fully explain the corresponding organization in human non-primary auditory cortex. The cross-species comparison separates a mechanism recoverable from local acoustic filtering from additional processing needed to account for the human response patterns.



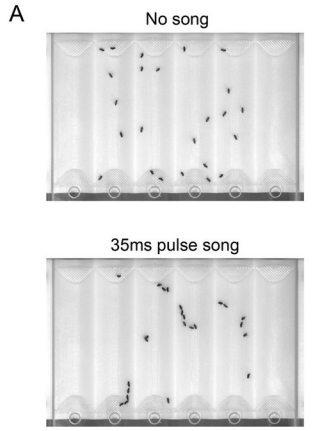
Landemard et al. (2025), Fig. 1, selected panels.

A downstream population recruits social action

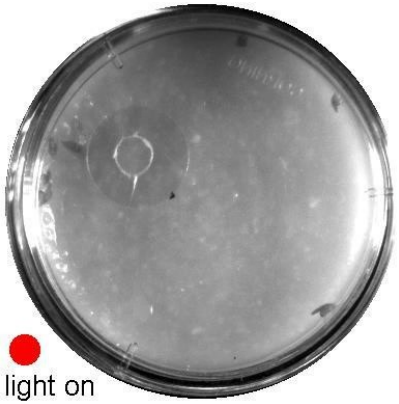
The fly central pathway continues from vPN1 toward a group called **pC1**. Zhou and colleagues measured how these neurons responded to courtship pulse songs with different intervals between pulses. Their response preferences corresponded more closely to the timing preferences of the behavioral response than did those of the preceding population.

Silencing either central population disrupted song-induced male chaining. Conversely, directly activating pC1 with a light-sensitive protein recruited chaining behavior. Removing activity tests whether the population contributes to the normal response, while adding activity tests whether recruiting that population can help produce the action under the experimental conditions.

A song contains several acoustic dimensions, but its behavioral significance depends on which features receiving neurons retain or emphasize. Different tuning at successive stages can transform a broad response to pulses into a pattern more closely aligned with social behavior. That transformation occurs within the central pathway rather than being supplied by the external sound alone.



Zhou et al. (2015), Fig. 3, selected panels.



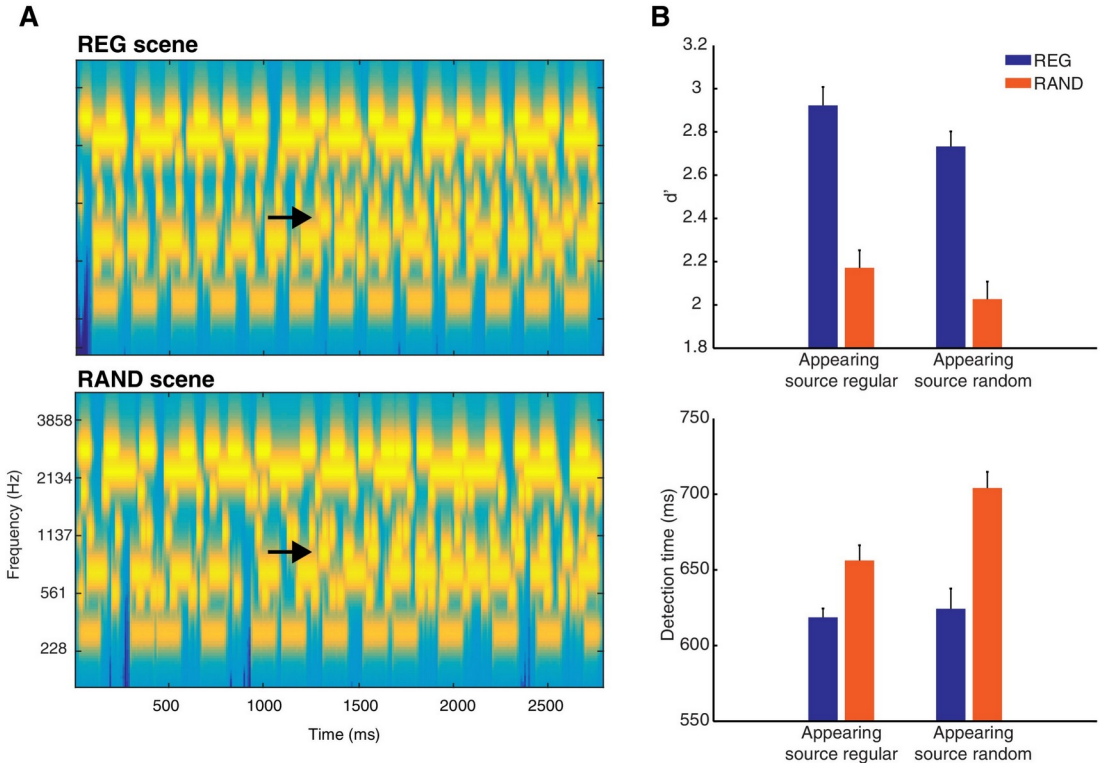
Zhou et al. (2015): pC1 activation during fly courtship.

Shared timing helps the brain group sounds into objects

Auditory scene analysis separates a mixture into candidate sound sources. Components beginning together or changing together can be grouped because they may share an origin. A voice, for example, produces coordinated changes across many frequencies, while an independently moving background source usually follows a different time course within the same interval.

Regularity provides another cue. Repeated footfalls or evenly spaced tone pips create a pattern against which a newly arriving event can be compared. The brain can track these regularities across several concurrent streams, retaining an estimate of each stream instead of treating every brief sound as an unrelated isolated event.

Sohoglu and Chait tested scenes containing regular or randomly timed tone-pip sources. Listeners detected an added source more quickly and accurately when the background sources were regular. The advantage arose from temporal organization, showing how a predictable background can make a new auditory event easier to separate from the ongoing mixture.



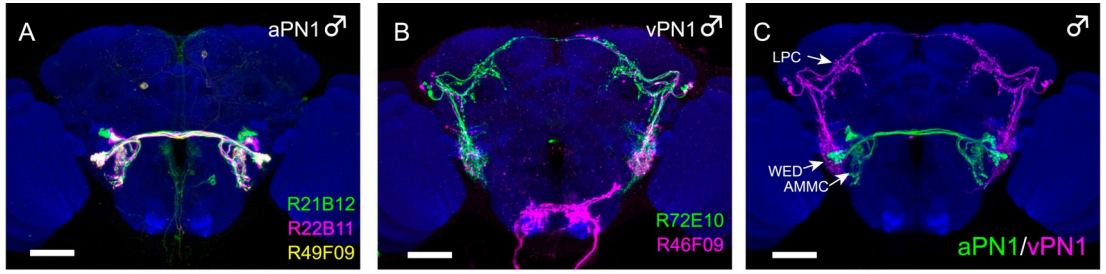
Sohoglu and Chait (2016), Fig. 1.

Aligning successive auditory populations

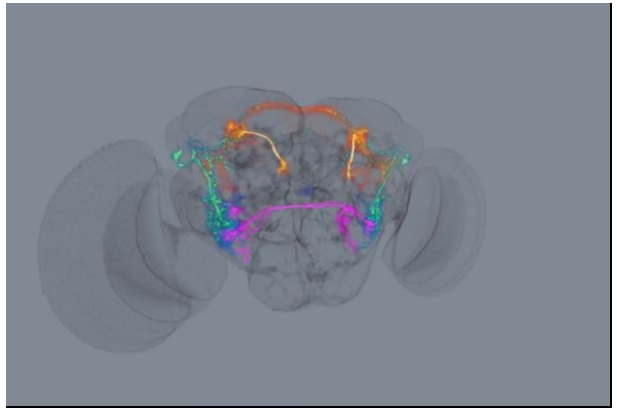
Identifying a central auditory pathway requires tracing how successive neuronal populations occupy and connect different brain regions. In the fly song study, investigators labeled earlier auditory neurons, vPN1 neurons, and pC1 neurons. Their reconstructed positions provided an anatomical framework for the physiological and behavioral manipulations applied to those populations.

Co-registration aligns images from different preparations into a shared anatomical space. The original three-dimensional movie uses that alignment to display the populations together. Their colors identify labeled cell classes and reconstructed projections, allowing a viewer to follow where one population's processes approach the territory occupied by the next central population.

Anatomical overlap becomes more informative when combined with measured responses and selective interference. In this study, calcium responses identified song-sensitive activity, while silencing and activation tested contributions to chaining behavior. Together, those measurements specify successive components through which acoustic timing can be transformed into a centrally organized social response.



Zhou et al. (2015), Fig. 1, selected panels.



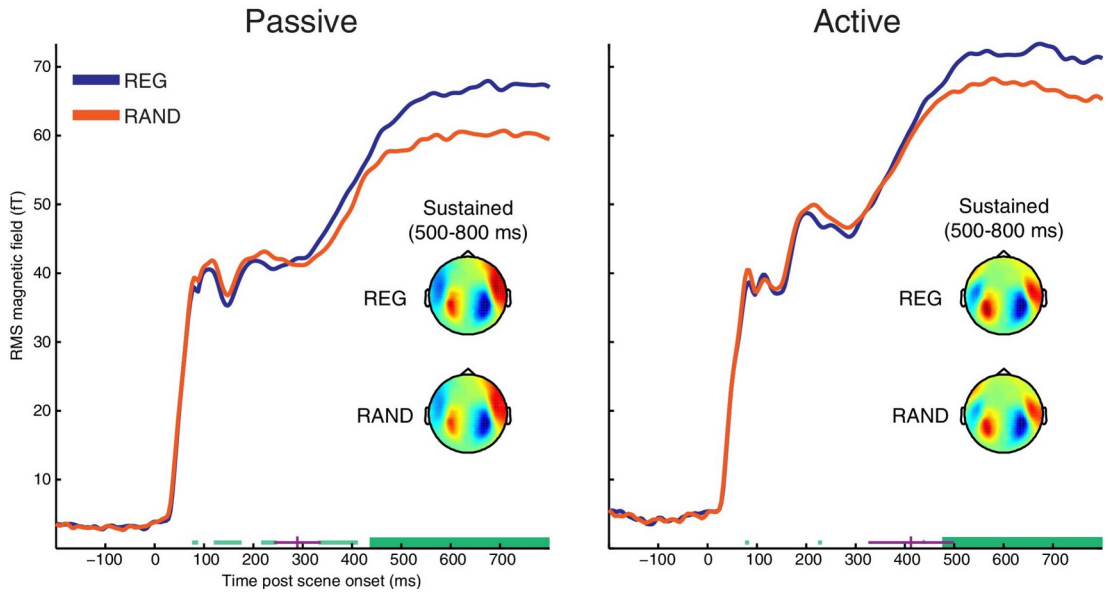
Zhou et al. (2015): Registered central song-responsive populations.

Regularity helps a sound emerge from noise

Sohoglu and Chait constructed auditory scenes from several simultaneous tone-pip streams. Within a stream, timing could repeat regularly or vary randomly. A new stream sometimes appeared during the scene, creating a detection problem in which the listener had to distinguish a newly arriving source from several sounds that were already present.

Regular scenes produced different sustained brain responses from random scenes. **Magnetoencephalography**, or MEG, measured changing magnetic fields associated with coordinated neural activity. Analyses implicated auditory and parietal regions, indicating that the brain registered temporal organization across the ongoing scene rather than responding only when the new source appeared.

The new source also evoked different responses depending on background regularity and whether listeners actively attended to the task. Predictable ongoing input changed both behavior and the response to a later event. This provides a concrete route through which an estimate built from recent sound history can influence what is detected next.



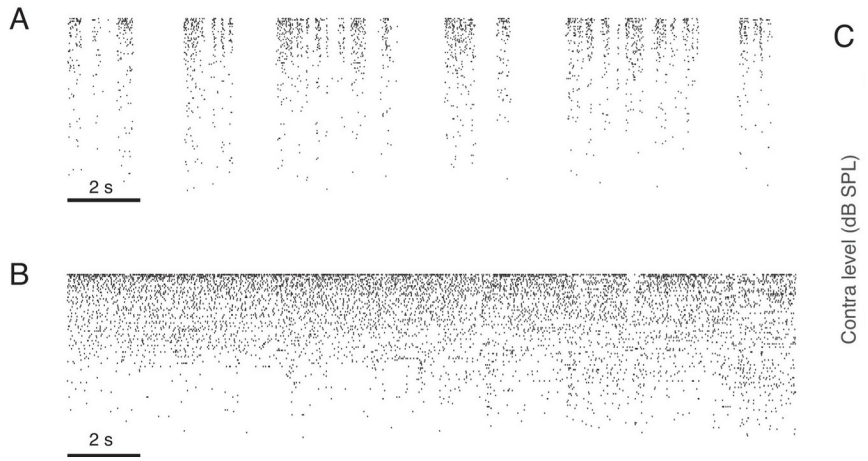
Sohoglu and Chait (2016), Fig. 2.

Cortical state changes the code for sound direction

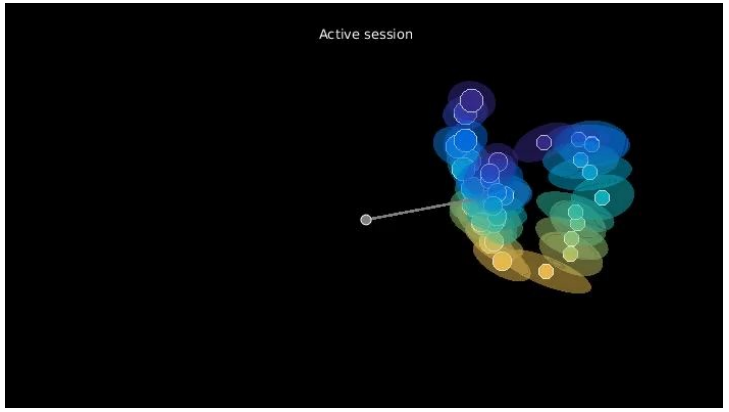
Cortical state describes patterns of ongoing activity that change how a population responds to incoming signals. In rat auditory cortex, investigators compared more active and less active states while presenting sounds that differed in overall level and in the level difference associated with the two ears. Both variables affected the population response.

In the more active state, population activity supported a representation of sound lateralization that was less dependent on overall sound level. The analysis asked how changes associated with the stimulus were arranged relative to spontaneous population fluctuations. Those relationships changed with state, altering how clearly a decoder could separate alternative sound locations.

The source movie projects recorded population activity into a three-dimensional analytical space. Its axes describe combinations of neuronal responses, not physical directions through the brain. Comparing the projected patterns shows how an unchanged acoustic distinction can become easier or harder to recover when ongoing neural fluctuations occupy different response directions.



Kobak et al. (2019), Fig. 1, selected panels.



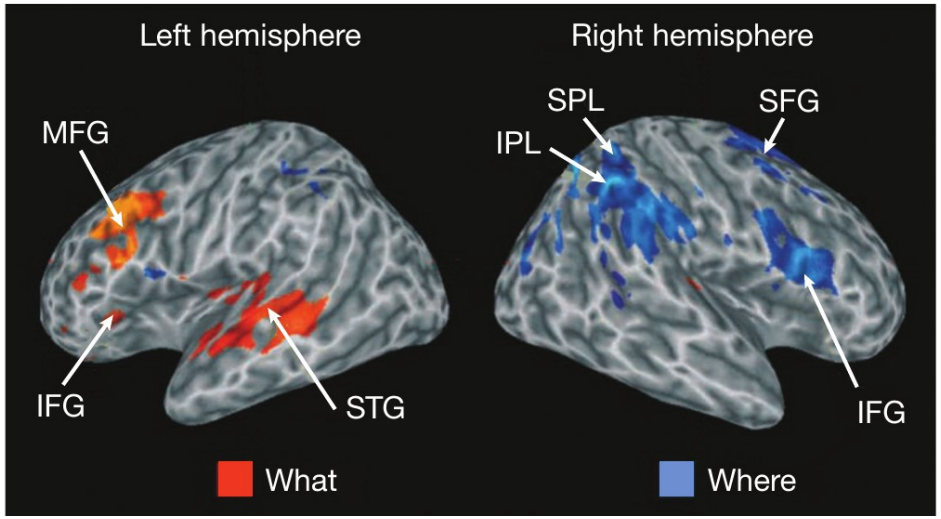
Kobak et al. (2019): Cortical population activity across sound conditions.

Attention changes which sound stream guides a response

Selective auditory attention prioritizes information from one sound source while other sources continue to reach the nervous system. A listener following one conversation must preserve the selected voice's changes and reduce interference from competing voices. The task concerns which central representation controls recognition and behavior, rather than which sound alone entered the room.

Selection can use location, voice identity, temporal structure, and information from other senses. Frontal and parietal networks interact with auditory regions as the listening goal changes. This interaction can alter the influence of particular neural populations without requiring the physical sound mixture to become quieter or less complicated in the environment.

The same stimulus can also be judged in different ways. A listener may compare where sounds occurred on one trial and which sources produced them on another. These instructions recruit different central operations because the information needed for a spatial decision differs from the information needed to recognize a sound category.



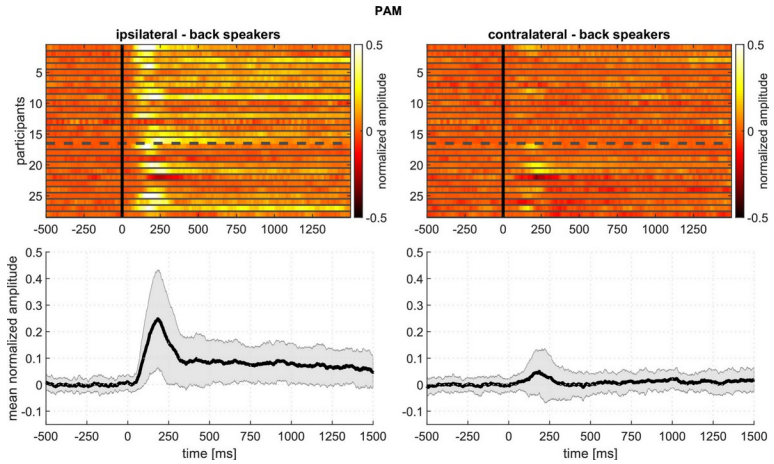
Carlson and Birkett (2022), Fig. 7.17, p. 206.

Listening direction recruits ear-muscle activity

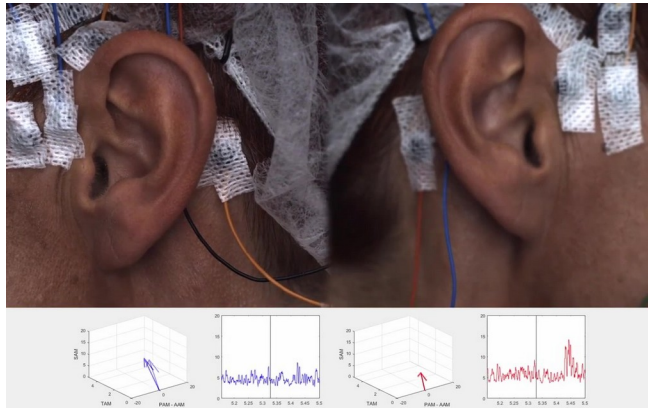
Strauss and colleagues asked whether the direction of attention appears in subtle motor activity during listening. Participants attended to a story from one side while ignoring a competing story on the opposite side. Surface electrodes recorded electrical activity from muscles around the ears, providing a readout of an otherwise inconspicuous orienting response.

Muscle activity was stronger on the attended side, with patterns that differed from those produced by an unexpected sound. The source movie shows a participant with particularly visible sustained movement during the voluntary-attention condition. The phenomenon concerns a motor output accompanying centrally directed listening, rather than the transformation of sound by auditory receptors.

Most people do not visibly aim their ears at a speaker as many other mammals do. The measured ear-muscle activity was stronger on the side toward which participants directed their attention. Small electrical muscle responses occurred even while the participant remained largely still, providing a motor readout of the attended direction.



Strauss et al. (2020), Fig. 3.



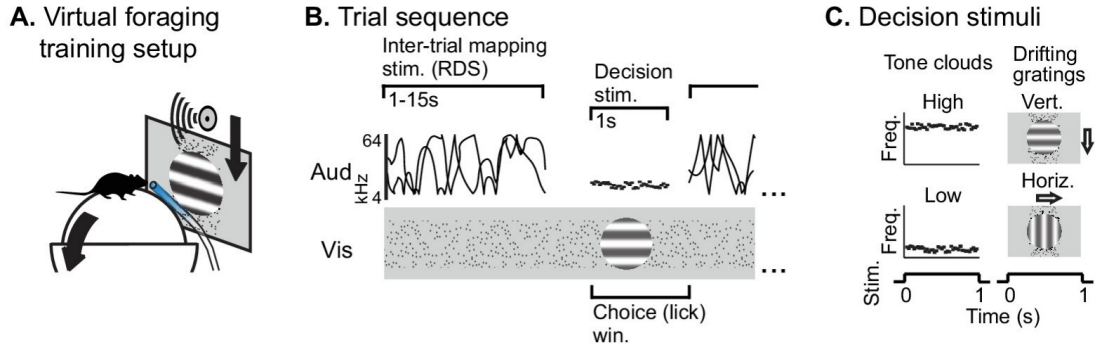
Strauss et al. (2020): Ear-muscle responses during voluntary attention.

Task goals change auditory cortical responses

Attention does not always increase the total firing produced by a sound. A population can become more useful if activity unrelated to the required distinction is reduced. The important question is whether the response pattern preserves information needed for the task, which differs from asking whether every recorded neuron becomes more active.

In a mouse audiovisual switching task, Morrill and colleagues compared activity when animals attended to sound with activity when they attended to visual information. Auditory attention reduced ongoing and stimulus-evoked firing in particular cortical populations. Sound information remained recoverable from the resulting activity, despite the reduction in the total number of recorded spikes.

The effect was especially prominent in deeper cortical layers and in neurons without strong spectrotemporal tuning. Reducing this activity could limit background interference while preserving the responses useful for discrimination. The same sensory input can therefore be represented more efficiently when a task changes which components of the cortical response are emphasized.



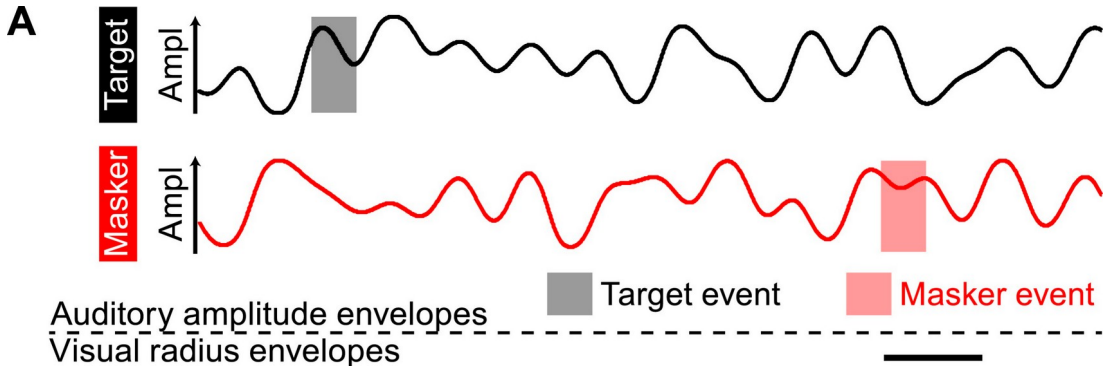
Morrill et al. (2022), Fig. 1, selected panels.

Visual timing can strengthen auditory selection

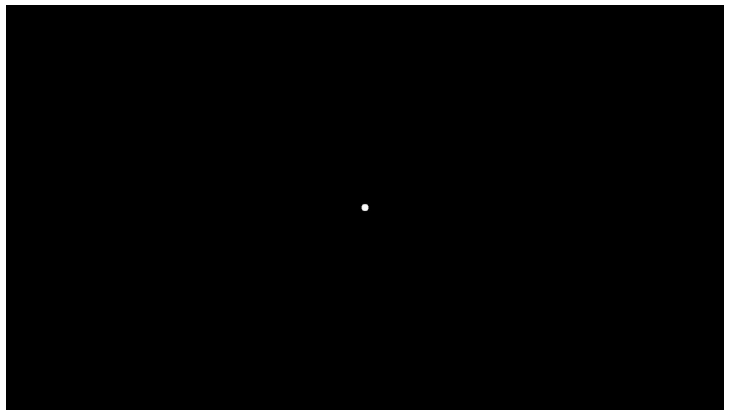
Maddox and colleagues presented competing sound streams whose loudness envelopes varied independently. A visible disc also changed size over time. When the disc's expansion and contraction followed the target stream's envelope, listeners performed better on judgments about that auditory stream, even though the visual information did not directly specify the answer.

An **amplitude envelope** describes how sound level changes across time. Matching the envelope to a visual change creates temporal coherence across the senses. This supplies a cue that the changing visual object and one acoustic stream belong together, helping that stream remain distinct from another source mixed into the same listening interval.

The original stimulus movie presents the target-matched condition. Its relevance comes from the shared timing, not from a picture naming the sound or a mouth articulating a word. The experiment therefore isolates a contribution of audiovisual temporal organization that can assist selection even when no linguistic information is available in the image.



Maddox et al. (2015), Fig. 1, selected panels.



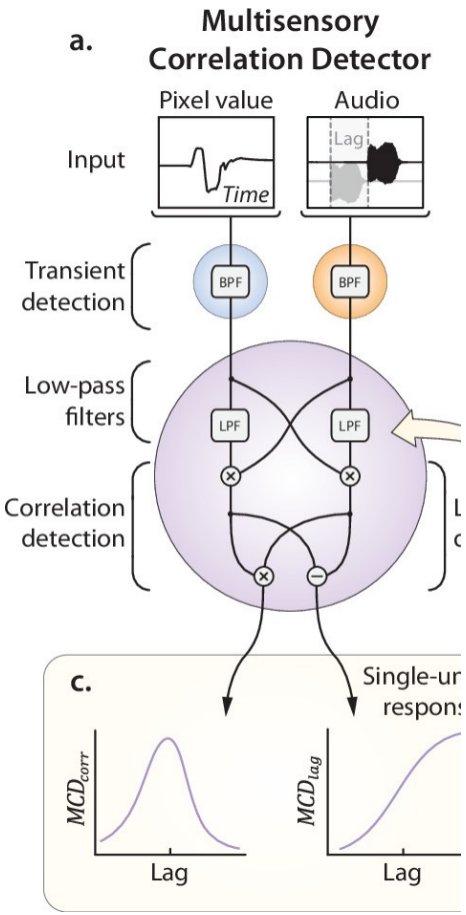
Maddox et al. (2015): Visual timing matched to the target stream.

Do sights and sounds belong together?

A visible event and a sound may have one cause or separate causes. Their timing helps the brain make that distinction. A closing mouth, a struck drum, or a bouncing object produces coordinated changes across the senses, while an unrelated background sound is less likely to follow the same pattern through time.

Audiovisual integration combines information when signals are treated as belonging together. The useful timing window is not infinitesimally narrow because physical transmission and neural processing introduce delays. Within that window, visual information can change the interpretation of an auditory event, including its apparent location or the speech category that it resembles.

A model of correlation-sensitive units tests one possible computation underlying these effects. It compares changes in image luminance with changes in sound level across time. Those units are simulated components of an explanatory model; their output can be compared with human perceptual measurements without treating the simulation itself as a recording from the brain.



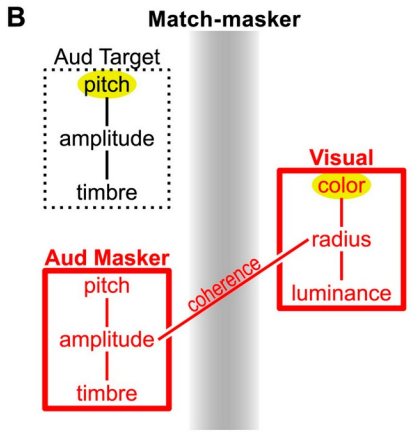
Parise (2025), Fig. 1, selected panels.

Visual coherence can favor a competing stream

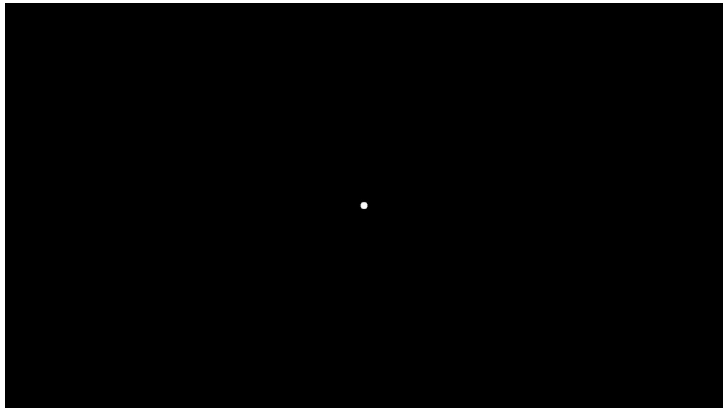
A synchronized visual object can match a competing sound rather than the one a listener is trying to follow. The attention experiment included this **masker-matched** condition. The visible disc followed the irrelevant auditory stream's envelope, while the target stream followed a different independent pattern during the same stimulus interval.

This manipulation separated the presence of a visual stimulus from the usefulness of its timing for selecting the target. Simply seeing a changing object did not reproduce the advantage obtained when its changes matched the attended sound. The information supplied by vision depended on which acoustic stream shared its temporal structure.

The movie preserves the original competing streams and visual changes. Comparing its organization with the target-matched example explains why seeing something move need not improve every listening task. Temporal coherence provides a cue about a candidate audiovisual object, and that object can be helpful or distracting depending on which stream the listener needs to select.



Maddox et al. (2015), Fig. 3, selected panels.



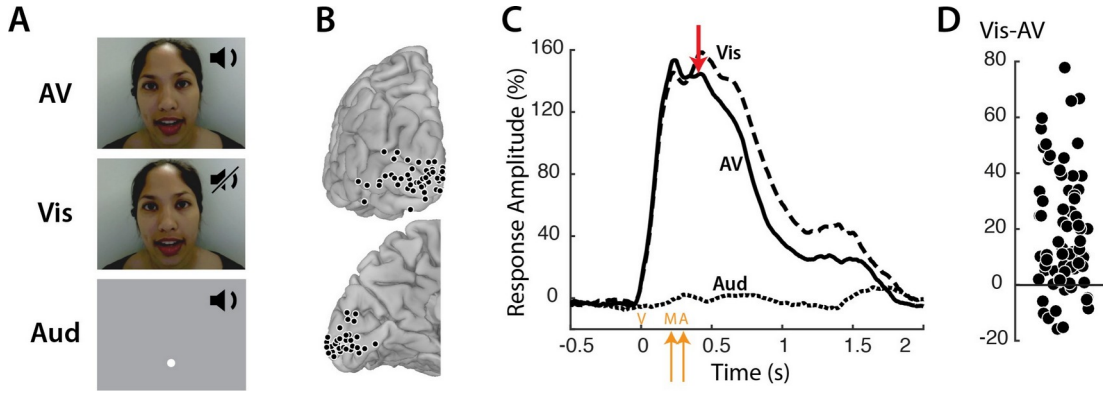
Maddox et al. (2015): Visual timing matched to the competing stream.

Frontal cortex helps select a talker's mouth during speech

During speech, the mouth provides visual information about when sounds occur and how they are articulated. Ozker and colleagues recorded directly from cortical electrodes while participants viewed speech under conditions with different auditory information. They examined visual responses corresponding to the talker's mouth rather than treating the entire face as one undifferentiated stimulus.

When auditory speech was absent, responses increased especially in visual regions representing the mouth. Trial-by-trial activity relationships also showed stronger coupling between frontal cortex and these mouth-related visual regions than between frontal cortex and other measured regions. This pattern was consistent with selective recruitment of visual speech information when it became particularly useful.

A mouth's shape does not specify every spoken sound uniquely, but it can constrain alternatives and provide timing. Selecting that information can help a listener combine a partial acoustic signal with the visible articulatory pattern. The measured cortical relationships concern how relevant visual evidence is selected, rather than a single area independently recognizing an entire conversation.



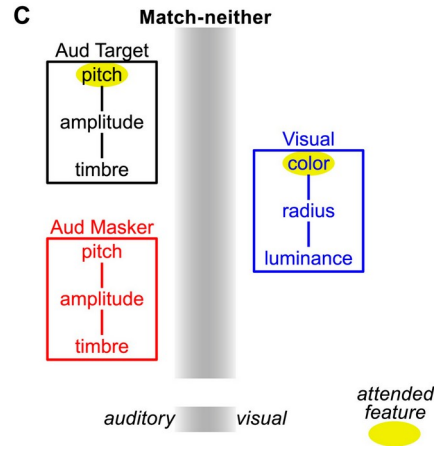
Ozker et al. (2018), Fig. 1.

What happens when visual timing does not match?

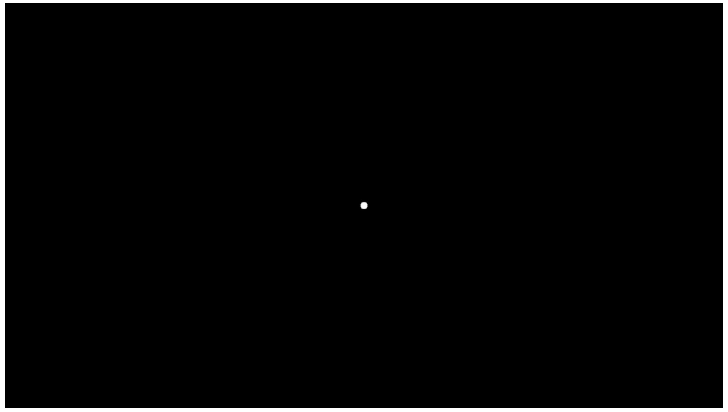
The audiovisual attention study also included a condition in which the visible disc matched neither sound stream. Its size changed, and both sounds continued, but their independent temporal patterns supplied no consistent visual cue identifying the auditory target. This condition separated general visual stimulation from useful cross-modal coherence during the listening task.

Comparing target-matched, masker-matched, and unmatched conditions tested which shared pattern improved performance. The target-matched advantage could not be explained solely by adding another changing stimulus to the display. The relation between the streams determined whether the image helped bind and select the auditory information relevant to the listener's judgment.

These distinctions also apply to a crowded conversation. A visible gesture or mouth movement is informative when its timing belongs to the person whose voice is being followed. Another person's equally visible movement may identify a different sound source, while an unrelated flickering object supplies little consistent evidence about either voice's ongoing acoustic pattern.



Maddox et al. (2015), Fig. 3, selected panels.



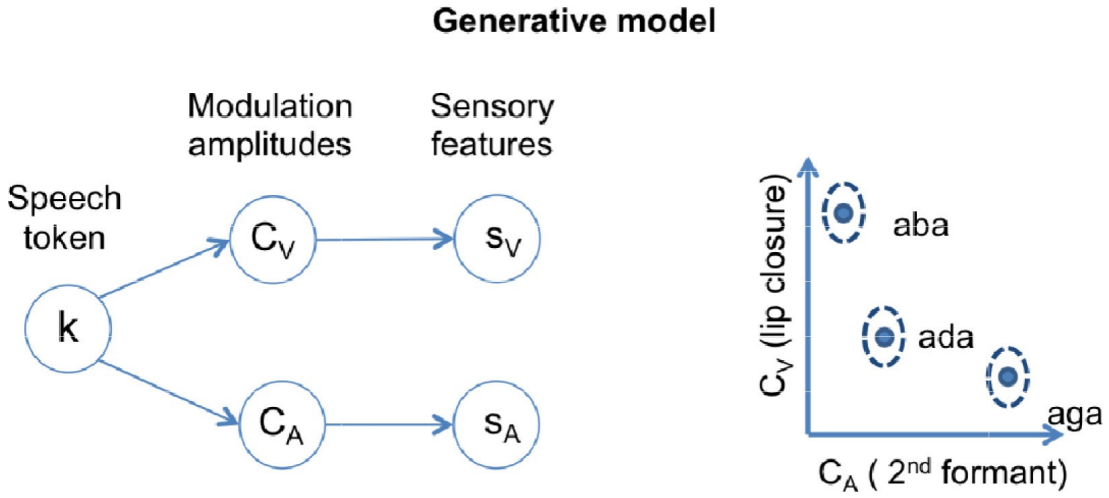
Maddox et al. (2015): Visual timing unmatched to either stream.

Experience changes speech categories

A **speech category** groups acoustically variable examples that listeners treat as the same kind of sound. Different speakers produce different vocal frequencies and articulation patterns, yet a listener can often recognize the same syllable across them. Central processing combines the current acoustic evidence with learned regularities about how those examples vary.

Recent experience can change this interpretation. A speaker consistently producing an unusual acoustic cue can shift which category is most plausible for a later ambiguous example. That adjustment must remain flexible enough to accommodate a new speaker while preserving longer-term knowledge acquired across many previous conversations and listening situations.

Olasagasti and Giraud modeled this problem using representations updated at different timescales. A rapidly changing representation followed the immediate environment, while a more stable representation preserved older category information. Their comparison with speech-recalibration behavior tested how the interaction between these memories could produce rapid adjustment without permanently replacing the learned category.



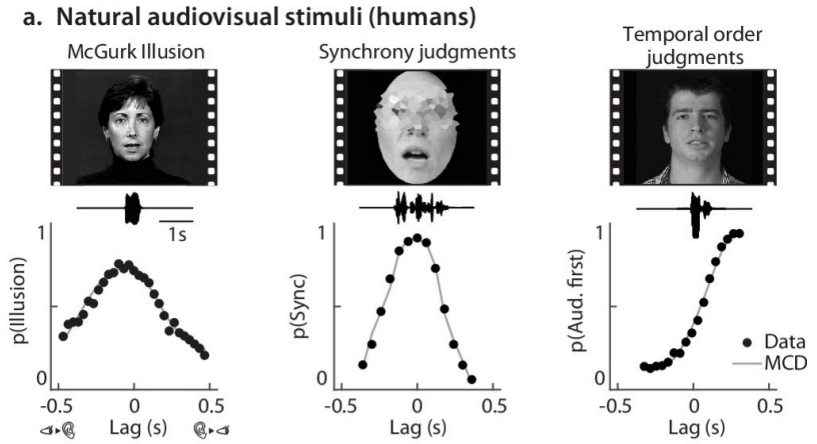
Olasagasti and Giraud (2020), Fig. 1.

A moving mouth can change the syllable you hear

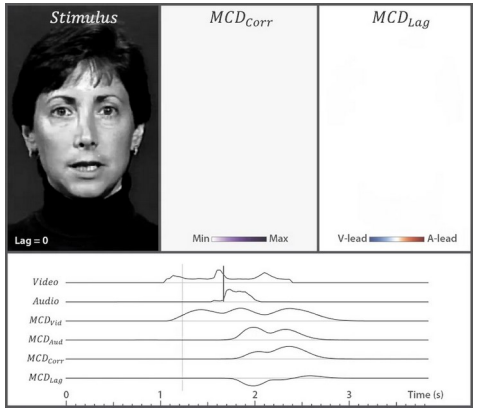
In the **McGurk illusion**, mismatched lip movements can change the speech sound an observer reports hearing. The original demonstration pairs an auditory recording of /pa/ with a video of the speaker articulating /ka/. The acoustic track remains the same, while the visible articulatory evidence changes the combined percept for many observers.

The brain normally uses the coordinated sound and movement of a real speaker. When those cues disagree in a controlled movie, their interaction becomes easier to detect. A changed report therefore does not require a changed auditory waveform; it can arise from the central interpretation of evidence arriving through two sensory routes.

The demonstration belongs to a study testing a computational account of audiovisual correlation. Its model evaluates how visual and acoustic changes align through time, including when the tracks are deliberately shifted. Comparing different alignments tests the temporal conditions under which the cues are likely to be combined into one perceived speech event.



Parise (2025), Fig. 2, selected panels.



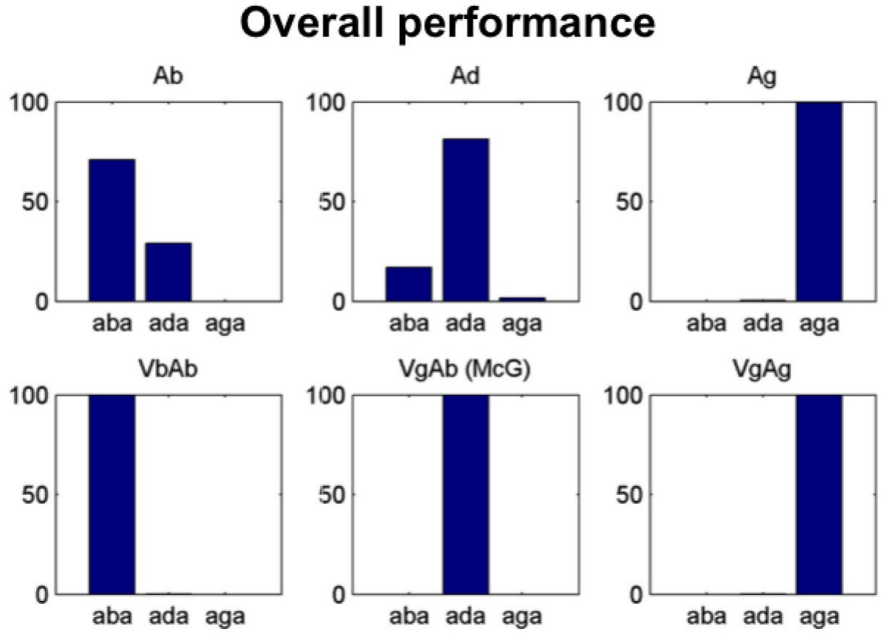
Parise (2025): Original audiovisual speech simulation.

Speech perception recalibrates across timescales

A short exposure to mismatched audiovisual speech can influence the interpretation of a later ambiguous sound. **Recalibration** describes this adjustment of the working relationship between acoustic cues and speech categories. The change concerns how the central system interprets the signal, rather than a physical alteration of the speaker's original recorded waveform.

Olasagasti and Giraud compared models that adjusted one category representation with a model retaining representations at multiple timescales. The latter could accommodate fast changes after McGurk exposure while also preserving information from longer-term speech experience. The model explains behavior by specifying how new evidence changes the current estimate of a category.

Different timescales solve different parts of the problem. Rapid updating allows a listener to adapt to a particular encounter, while slower memory prevents a brief unusual example from erasing an established distinction. Their interaction makes the current category estimate sensitive to recent evidence without requiring every ambiguous syllable to be learned again from the beginning.



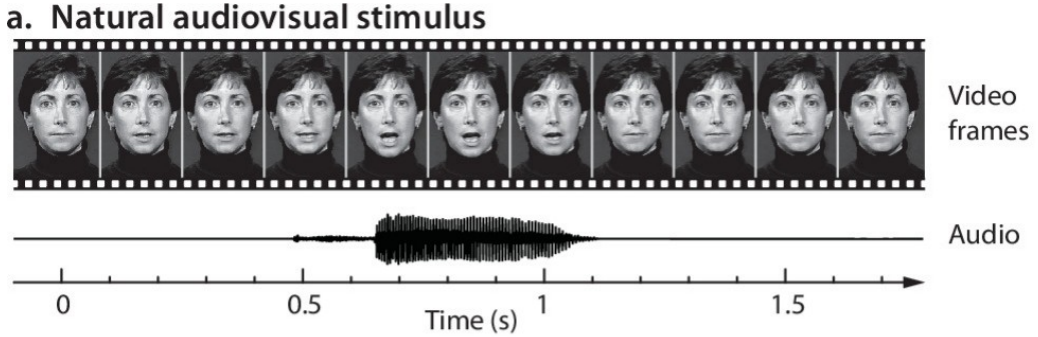
Olasagasti and Giraud (2020), Fig. 2.

Visual events can shift a voice's apparent location

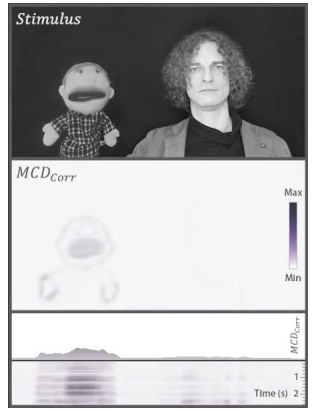
A ventriloquist coordinates a voice with movement of a dummy while minimizing visible speech movements of the human performer. The sound can then seem to originate from the visually active character. This **ventriloquist effect** concerns the estimated location or source of a sound, adding a spatial example of central audiovisual interpretation.

The source movie includes the original performance example and output from a computational model. The overlaid activity represents simulated correlation-sensitive units, not electrodes recording from a listener. Its purpose is to test whether shared acoustic and visual timing supplies information that could favor one visible object as the apparent source of the voice.

Changing the relationship between the tracks changes which image locations correlate with sound fluctuations. A plausible audiovisual source can consequently emerge from coordinated events rather than from the sound waveform alone. The broader model was compared with perceptual experiments, allowing its predictions about these correlations to be evaluated against measured behavior.



Parise (2025), Fig. 3, selected panels.



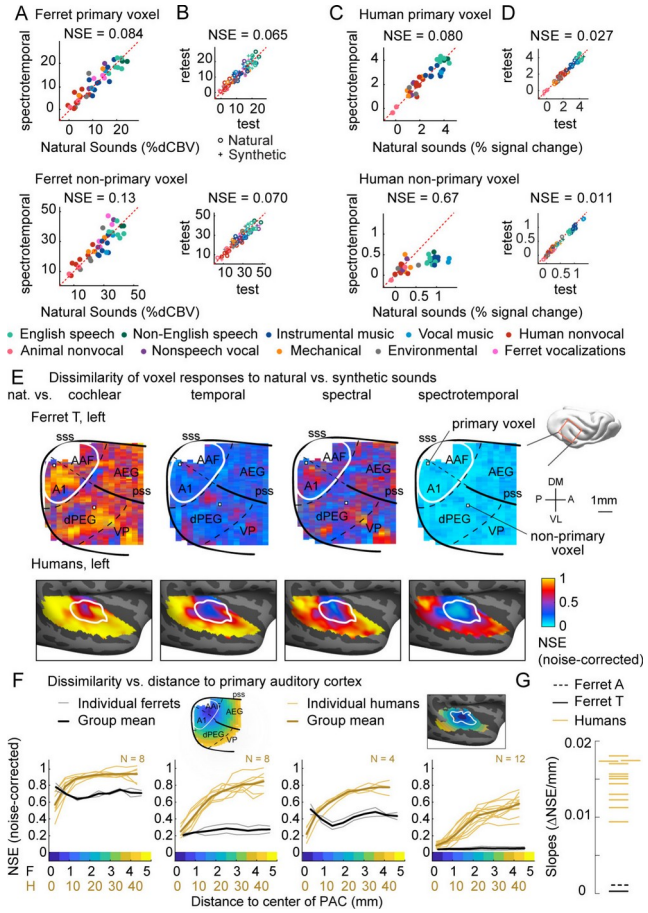
Parise (2025): Original spatial audiovisual simulation.

Recognizing a voice requires more than measuring its pitch

A **voice** contains information about the speaker as well as the spoken message. Pitch contributes, but so do timing, spectral structure, articulation, and changes across a phrase. A listener can recognize one person while that person speaks at several pitches, or understand the same sentence spoken by different individuals.

Central auditory populations must therefore combine features across time instead of assigning identity to one frequency. Human and ferret studies of natural sounds show responses to combinations of acoustic properties, while higher-order human responses can distinguish categories not captured fully by simpler acoustic models. These representations retain more than an instantaneous frequency measurement.

The listener's task determines which aspect becomes useful. Voice identity helps follow a person through a mixture, linguistic content supplies the message, and prosody contributes information about emphasis or emotion. These features coexist within the signal, but recognizing one does not guarantee that every other feature has been identified accurately.



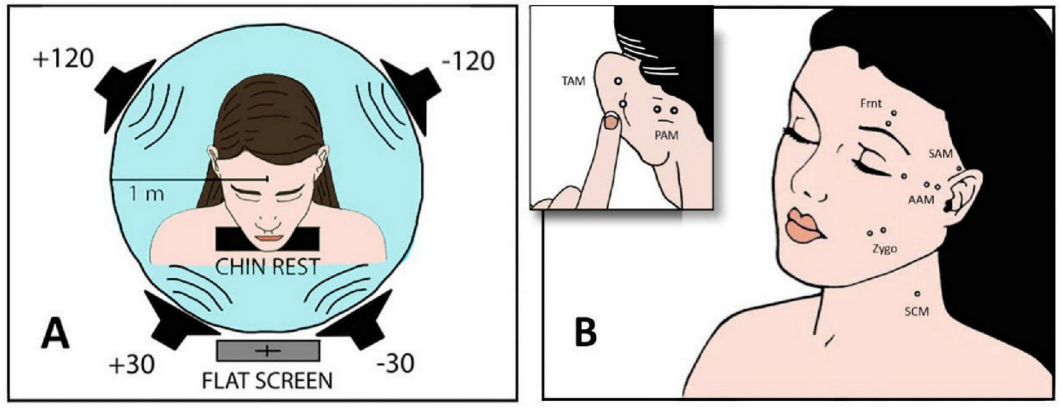
Landemard et al. (2021), Fig. 2.

A surprising sound recruits an orienting response

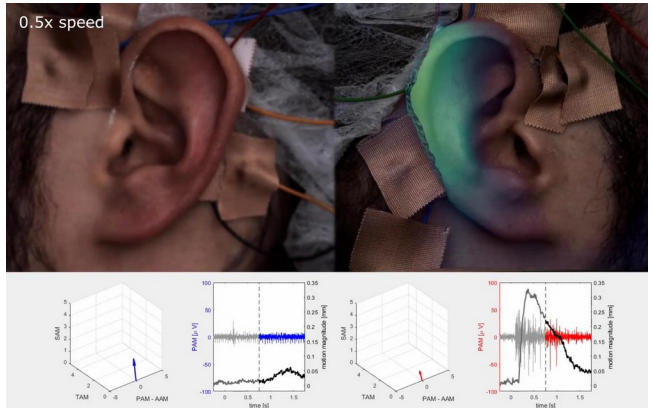
An unexpected sound can redirect attention before a person decides to follow it. Strauss and colleagues presented novel sounds from different speaker locations while participants silently read. Electrical recordings from muscles around the ears showed side-dependent responses, providing a measurable component of the orienting reaction to the new auditory event.

This reflexive condition differed from voluntarily attending to one of two competing stories. The two tasks produced different patterns across the measured muscles, including differences in the direction of subtle ear movement. An externally triggered interruption and a sustained listening goal therefore did not generate one identical pattern of peripheral motor output.

The original movie shows a response to a novel sound from a posterior speaker. Digitally enlarged movement makes the small displacement visible, while the accompanying displays retain the source recording and analysis. The effect illustrates a centrally organized orienting response; it does not mean that a person consciously chose to move the ear toward the sound.



Strauss et al. (2020), Fig. 1.



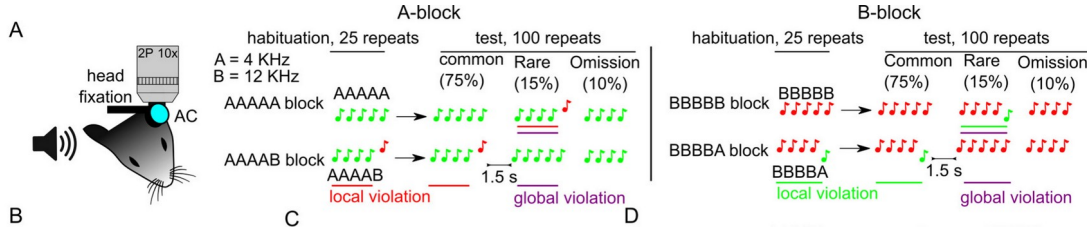
Strauss et al. (2020): Ear-muscle responses during reflexive attention.

Predictions span short and longer sound sequences

An **auditory prediction** is an expectation about an upcoming event based on preceding input and learned regularities. A repeated tone creates a local expectation, while a repeated arrangement of several tones creates a longer sequence expectation. A sound can fit the immediately preceding event yet violate the larger pattern in which it occurs.

Jamali and colleagues examined this distinction in mouse auditory cortex. Different neuronal populations responded to violations at different sequence levels. Responses to unexpected events involved both adaptation-related changes and additional surprise responses, showing why an increased response to a rare sound cannot automatically be assigned to one universal prediction mechanism.

An expected sound can also be omitted. In the awake animals, particular inhibitory interneurons responded when a locally expected ending was missing. That response carried information about sequence structure even though no new sound arrived at that moment, allowing an internally maintained expectation to become visible in the measured central activity.



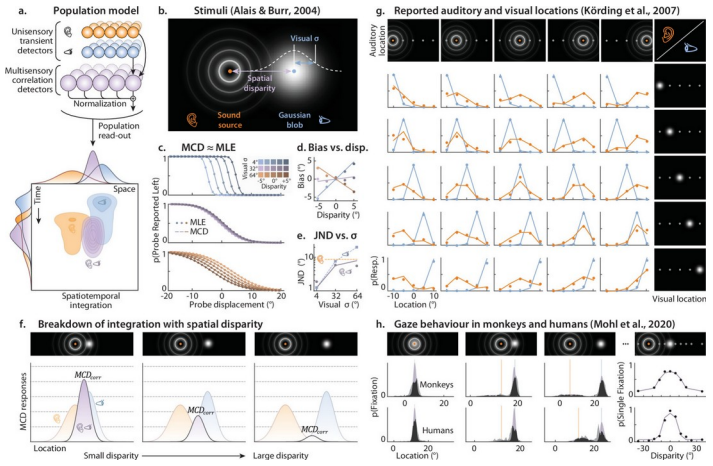
Jamali et al. (2024), Fig. 1, selected panels.

Comparing audiovisual predictions with human gaze

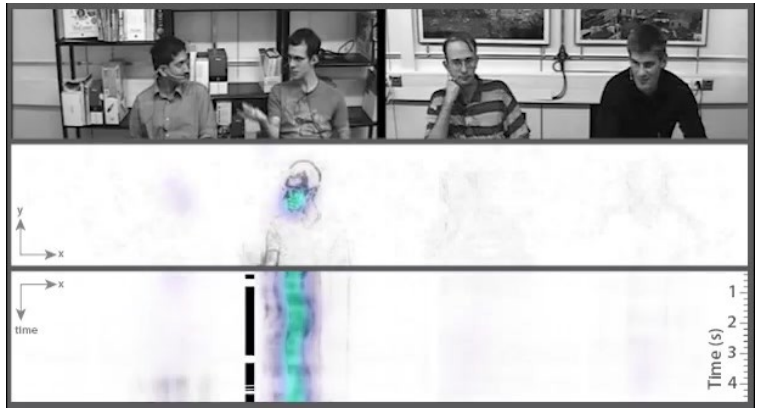
A **saliency map** estimates which locations in a scene are likely to attract attention. The audiovisual correlation study generated such maps by combining changes in the image with changes in sound. Its simulated units operated directly on the movie and soundtrack, allowing predictions to follow the continuous time course of a natural scene.

The source example overlays recorded human gaze positions on the film alongside the model output. Gaze points are measured behavior, whereas the model's changing activity is a proposed computation. Comparing their spatial and temporal patterns tests whether moments of shared audiovisual change help identify the objects that observers tend to inspect.

This distinction separates an explanation from the observations it is meant to explain. If one visible object repeatedly changes when the sound changes, it becomes a stronger candidate source than nearby objects with unrelated timing. That association can direct looking toward a speaker or event even before its full identity has been recognized.



Parise (2025), Fig. 4, selected panels.



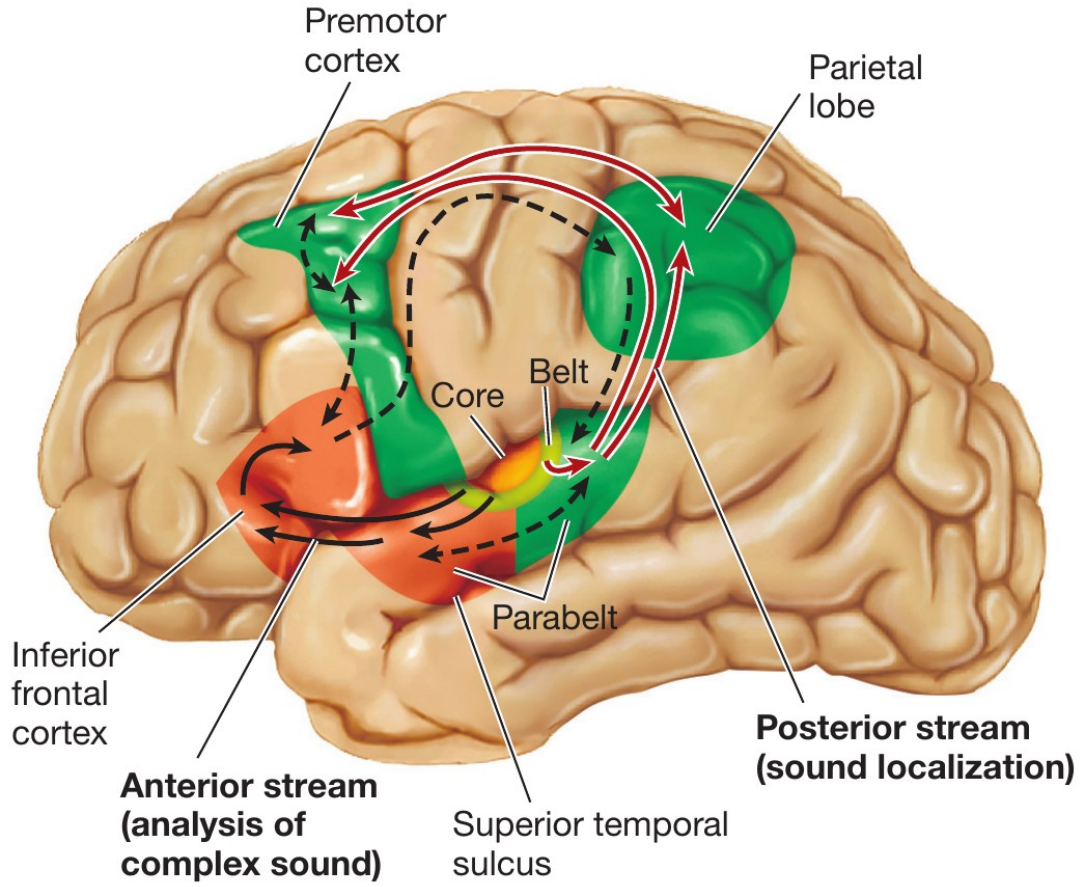
Parise (2025): Audiovisual model and recorded gaze.

Pitch depends on relationships among sound components

Pitch is the perceived highness or lowness of a sound, which differs from the identity of the source producing it. A voice and a violin can produce notes with similar pitch while remaining easy to distinguish. Central auditory circuits must therefore extract pitch-related structure without discarding the additional information identifying the source.

Many pitched sounds contain a **harmonic series**, with components at multiples of a common fundamental frequency. When the fundamental component is removed, the remaining pattern can still evoke a pitch corresponding to it. The perceived pitch is consequently not always a direct report of the lowest frequency physically present in the sound.

For example, components at 200, 300, and 400 hertz share a spacing compatible with a 100-hertz fundamental. Their combined relationships provide evidence about that periodic structure even without a 100-hertz component. Higher auditory processing can use such relationships to recognize a note or voice across changes in the mixture reaching the brain.



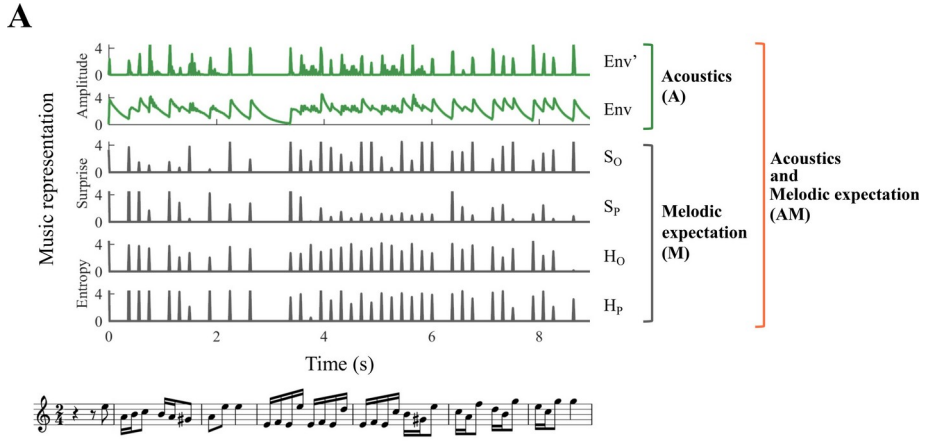
Carlson and Birkett (2022), Fig. 7.9, p. 200.

Recording cortical responses during a melody

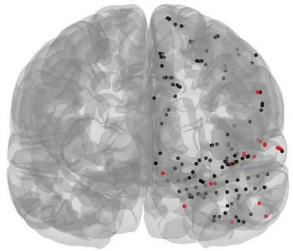
Electrocorticography, or ECoG, records electrical activity from electrodes placed on the cortical surface for clinical purposes. Di Liberto and colleagues analyzed such recordings alongside noninvasive measurements while participants listened to Bach melodies. This allowed them to compare ongoing musical events with responses measured close to human auditory cortical tissue.

A melody contains both acoustic changes and changes in expectation. A note can be loud or sharp in its onset, but it can also be surprising because of the notes that preceded it. The researchers modeled these components separately so that an expectancy-related response was not simply equated with every large acoustic change.

The first electrode-placement movie shows one participant's recording sites in three dimensions. Highlighted sites responded to the music, establishing where the measured signals originated. Their location helps interpret later response analyses: an expectation-related signal must be assigned to the recorded cortical region rather than assumed to represent an undifferentiated response of the entire brain.



Di Liberto et al. (2020), Fig. 1, selected panels.



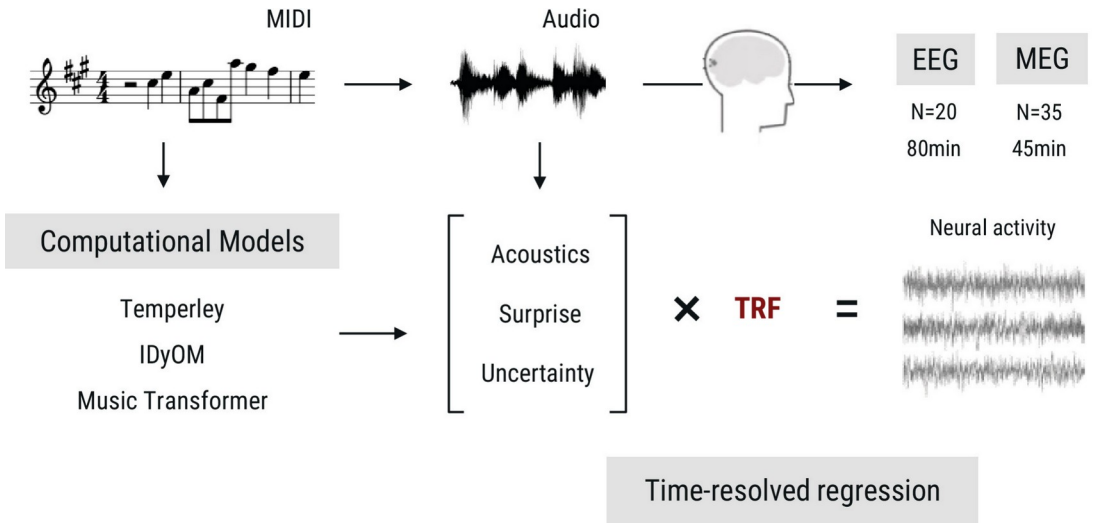
Di Liberto et al. (2020): Participant 1: cortical recording sites.

Experience shapes musical expectation

An unexpected note depends on the musical context that makes some continuations more likely than others. Kern and colleagues compared several computational accounts of these expectations while recording human brain activity during naturalistic music listening. Some models relied on simple local rules, whereas others incorporated statistical regularities learned from a larger musical collection.

Brain responses tracked note-level surprise even after acoustic and adaptation-related effects were considered. Models using longer-term musical experience predicted the measured surprise responses better than simple rules alone. However, the immediate context that contributed most strongly consisted of a relatively short sequence of preceding notes, rather than the entire composition at once.

Long-term learning and short-term context therefore performed different jobs in the account. Previous experience supplied probabilities about musical continuations, while the recent phrase identified which probabilities were relevant at that moment. A listener can consequently encounter a locally surprising note using a prediction system shaped by many earlier musical experiences.



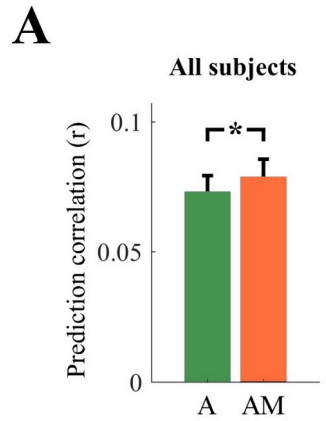
Kern et al. (2022), Fig. 1.

Unexpected timing changes cortical responses

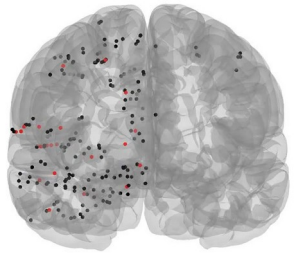
Note-onset expectation concerns when a note is likely to begin. A passage can establish a timing pattern even when its exact next pitch remains uncertain. Di Liberto and colleagues estimated onset expectations from the preceding melody and examined whether those estimates explained cortical activity beyond the effect of the note's physical onset.

The analysis related time-varying musical features to time-varying electrical responses. It identified contributions associated with the predictability of note timing in human temporal cortex. The result concerned measured responses during listening, with a computational model specifying which temporal events were expected within the musical material presented to the participant.

The second source movie locates recording sites in another participant. Those sites differ anatomically from the first participant's clinical electrode coverage. Combining their measured responses with the musical sequence allows the investigation to follow a timing-related effect across available cortical recordings, while retaining information about where each individual signal was actually sampled.



Di Liberto et al. (2020), Fig. 2, selected panels.



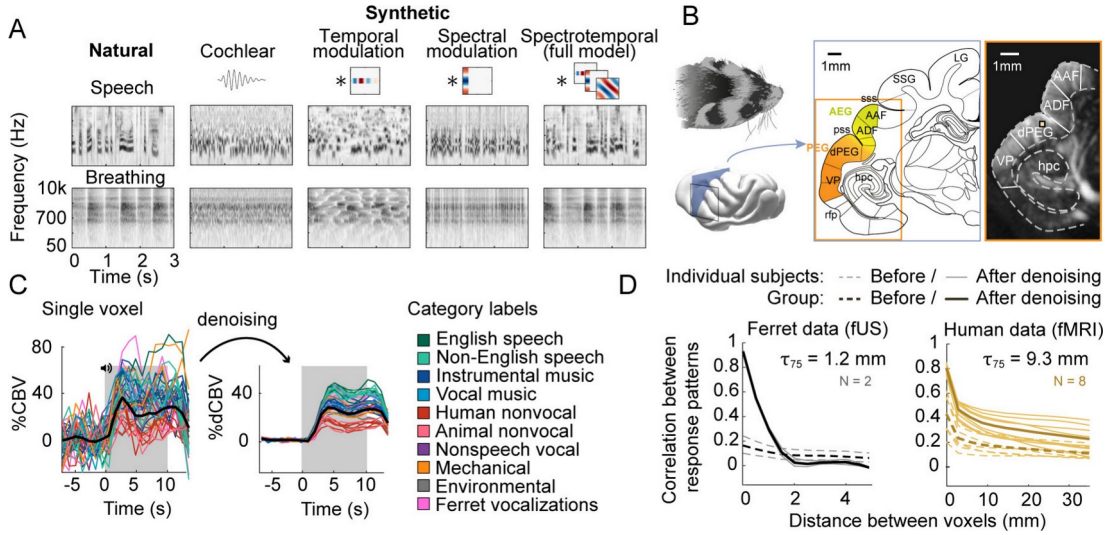
Di Liberto et al. (2020): Participant 2: cortical recording sites.

Timbre helps identify a source across changing notes

Timbre distinguishes sources that produce similar pitch and loudness. A piano and a flute differ in how their sound begins, changes, and ends, as well as in the relative strengths of their frequency components. Recognizing an instrument requires a central representation that can combine these features across a changing musical phrase.

The **attack** is the beginning of a note, while its later evolution and decay provide additional source information. A struck piano string produces a different time course from a sustained breath-driven tone. A listener can use that pattern even when both instruments play the same nominal note in the same musical context.

Natural-sound studies compare these changing patterns with responses across auditory cortex. A representation that preserves a source across several notes must tolerate changes in pitch while retaining other diagnostic features. This separates identifying an instrument from identifying one instantaneous frequency, and explains why the identity of a melody's performer is not carried by pitch alone.



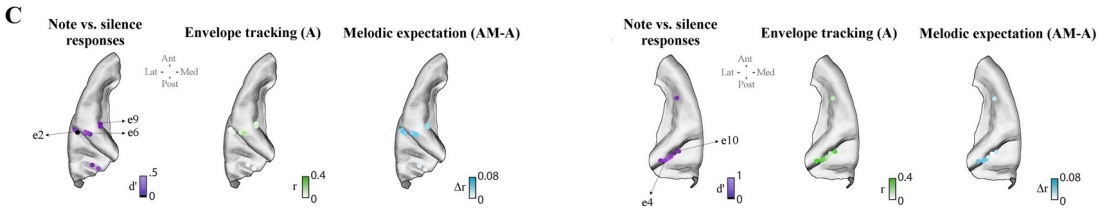
Landemard et al. (2021), Fig. 1.

Pitch surprise differs from onset surprise

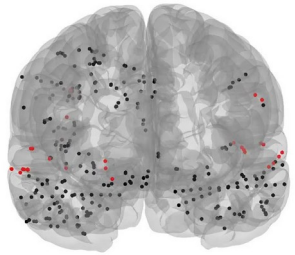
Pitch expectation concerns which note is likely to occur next. **Onset expectation** concerns when that note will occur. A familiar rhythm can contain an unexpected pitch, while a familiar sequence of pitches can be played with an unusual delay. These changes separate two dimensions of musical prediction that often occur together in ordinary listening.

Di Liberto and colleagues included estimates of both dimensions when modeling cortical responses to melodies. The analysis found distinct contributions from pitch and onset expectations, including responses in Heschl's gyrus and planum temporale. This placed expectation-related information within temporal auditory regions rather than assigning all musical prediction exclusively to a distant frontal area.

The third electrode-placement movie shows the recording coverage of the remaining clinical participant. The combined analysis relates responses from these measured sites to the melody's unfolding structure. The electrical signal follows the listener's response to the presented music; the model-derived surprise values specify features against which that measured signal was compared.



Di Liberto et al. (2020), Fig. 3, selected panels.



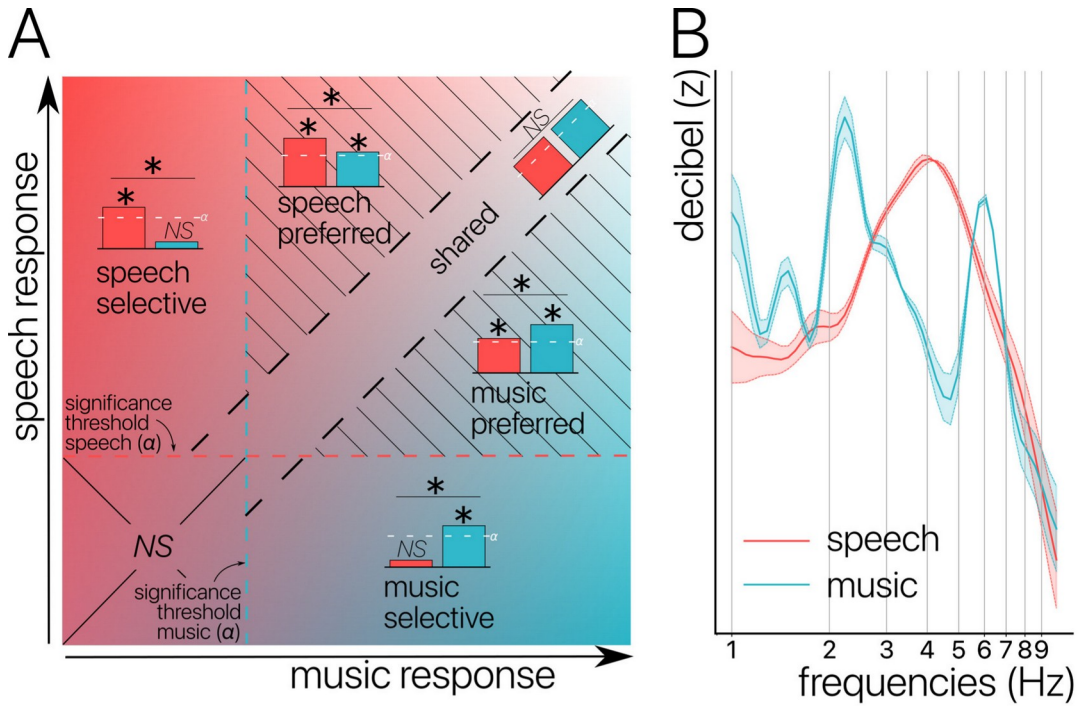
Di Liberto et al. (2020): Participant 3: cortical recording sites.

Speech and music recruit interacting networks

Speech and music share acoustic features but also contain different kinds of organization. Te Rietmolen and colleagues examined cortical responses to both categories using naturalistic listening data. Their analyses considered frequency-specific neural activity across distributed regions, allowing shared processing to be examined alongside distinctions between the two stimulus classes.

The measured networks overlapped without becoming identical. Some response patterns were shared across speech and music, while other patterns differed according to the category and frequency range of neural activity being analyzed. This organization is more specific than assigning language to one hemisphere and all music to the other.

An overlapping network can reuse computations such as tracking temporal changes while responding differently to a spoken sentence and a melody. The distinction depends on patterns across connected populations and time, not merely whether one isolated region becomes active. Comparing the categories under the same analytical framework identifies both their common demands and their separable response structure.



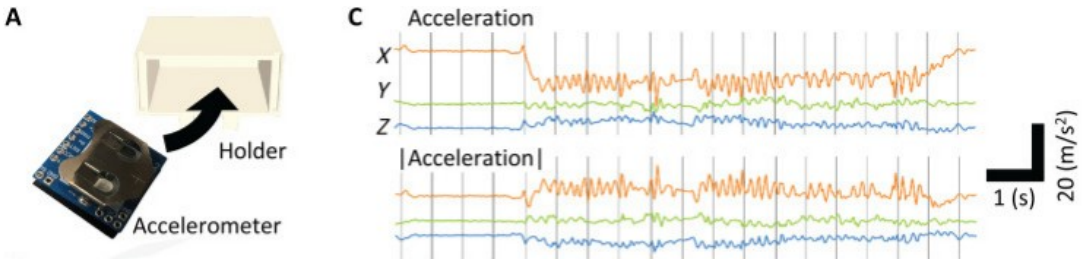
te Rietmolen et al. (2024), Fig. 1, selected panels.

Small rat head movements can align with musical beats

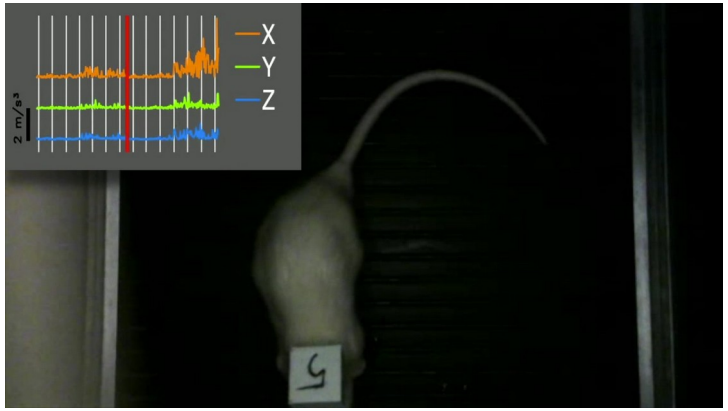
Ito and colleagues measured small head movements while rats listened to music. An accelerometer attached to the head captured movement too subtle to judge reliably by casual observation alone. Relating those measurements to the musical beat tested whether movement was distributed uniformly through time or tended to occur at particular phases of the music.

The rats showed beat-related movement, with a stronger relationship near the original musical tempo used in the study. The investigators also measured responses in auditory cortex. This combined approach examined both an observable movement and central activity associated with the temporal organization of the sounds, rather than inferring neural synchronization from the movement alone.

The original movie includes rat movement during music presentation and an author-supplied slow-motion segment. Its altered playback speed changes the pitch of the soundtrack in that segment. The quantitative timing analyses used the recorded experimental signals, so the slowed display should be interpreted as a way to inspect movement rather than as the original listening condition.



Ito et al. (2022), Fig. 1, selected panels.



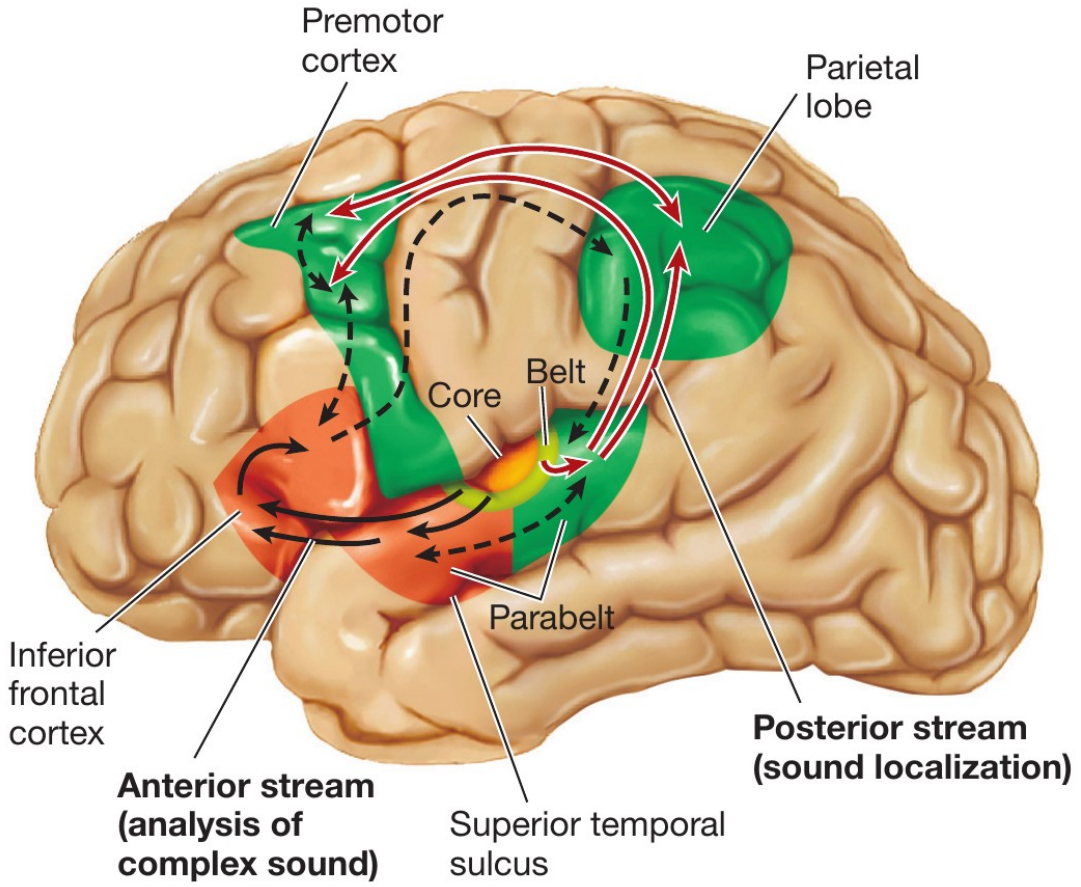
Ito et al. (2022): Rat movements during original music playback.

Beat perception recruits prediction and movement networks

A **beat** is a recurring temporal pulse that a listener can use to organize movement and predict upcoming events. It may remain perceptible when some expected notes are absent. Following a beat therefore involves an internally maintained timing estimate, rather than requiring a new external sound to trigger every moment of the rhythm.

Auditory regions interact with movement-related structures, including supplementary motor areas, basal ganglia, and cerebellar circuits. These systems contribute to different aspects of timing and movement coordination. Their interaction can help a listener prepare for an expected beat while using later sounds to correct the phase or pace of the ongoing estimate.

An omitted event provides a useful distinction. If timing depends only on reactions to each sound, there is no external trigger during the gap. Maintaining the pulse through that interval requires the system to carry its timing state forward, then compare the returning sound with the moment predicted from the preceding pattern.



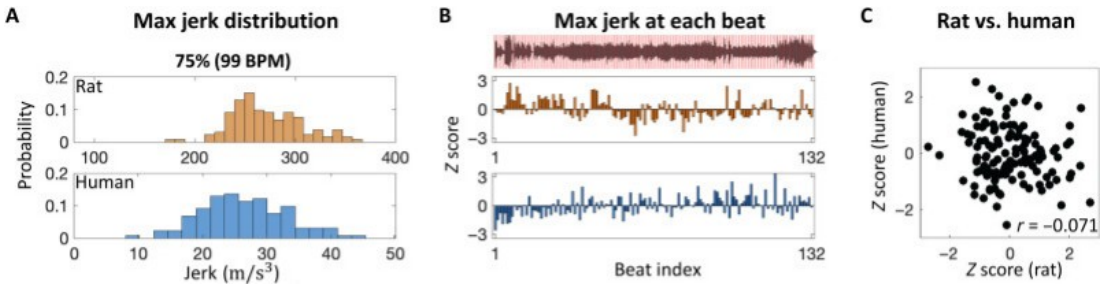
Carlson and Birkett (2022), Fig. 7.9, p. 200.

Changing tempo tests synchronization

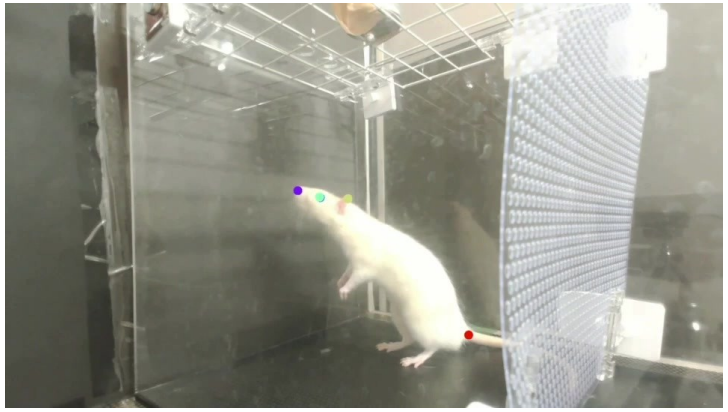
The rat study changed playback tempo to ask which musical timescale produced the strongest beat relationship. Speeding or slowing the sequence changed the interval between its beats. The preference did not simply scale with the faster movements usually associated with a small animal, suggesting that the relevant timing depended on more than body size.

The strongest effects occurred around tempos commonly used in human beat synchronization. Auditory cortical responses also depended on temporal structure. These measurements motivated a neural-dynamics explanation in which the recent history of sound changes the response to the next event, giving some intervals a stronger representation than others.

The second source movie shows a rat in a bipedal stance during excerpts from Mozart, Lady Gaga, and Queen. It provides examples of beat-related movement across different musical material. The tempo comparison itself comes from the study's controlled stimulus conditions and movement analyses, which separate the illustrative performance examples from the measured timing preference.



Ito et al. (2022), Fig. 2, selected panels.



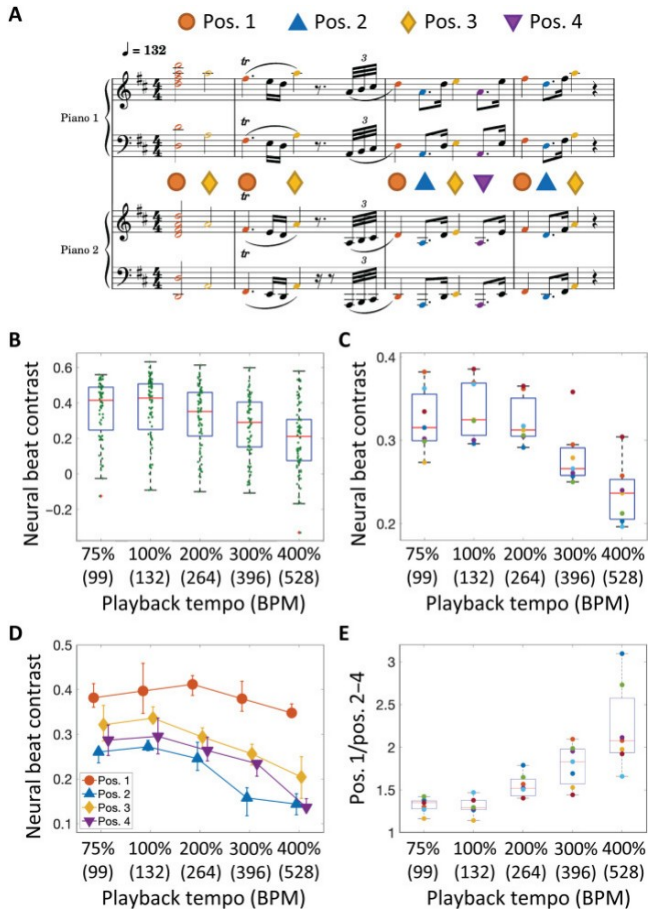
Ito et al. (2022): Rat movement during several musical excerpts.

Adaptation can give auditory networks a preferred timescale

Adaptation changes a neuron's response according to its recent stimulation. If a sound reduces responsiveness briefly, a second event arriving soon afterward may evoke less activity than one arriving after recovery. The response to an event therefore depends on both its current acoustic features and the time elapsed since earlier events.

Ito and colleagues modeled how this recent-history effect could shape beat-related responses. Repeated events at different intervals interact differently with the modeled recovery dynamics. The model reproduced important features of the observed timing preference, providing a candidate explanation for why auditory cortical responses emphasized a particular range of musical tempos.

The mechanism concerns a neural timescale rather than a command to move at one fixed speed. A movement circuit can receive a stronger beat-related signal when auditory responses distinguish recurring pulse positions from intervening events. The behavioral recordings and neural measurements test the resulting pattern, while the adaptation model specifies a possible process that generates it.



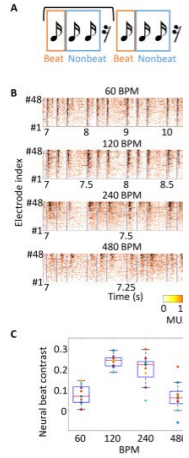
Ito et al. (2022), Fig. 3, selected panels.

Comparing rat and human movement against beats

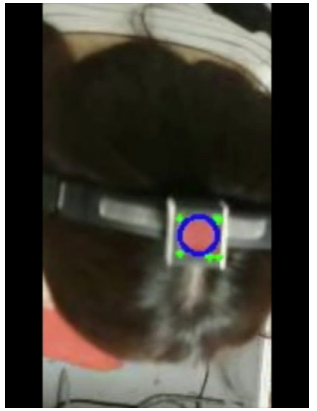
Human participants also listened to music while their head movements were recorded in the beat-synchronization study. Comparing movements against the same musical pulse made it possible to examine similarities and differences without relying on the visual impression that a person or animal seemed to be dancing. Each movement had to be related to a defined time point.

Different people could place the upper and lower parts of a head movement at different phases of the beat. The investigators therefore considered several movement measures, including acceleration and changes in acceleration. A repeated visible motion is not one indivisible event; its timing depends on which part of the movement is being measured.

The third source movie shows human movement during the Mozart excerpt used in the experiment. Its displayed trajectories correspond to the original participants and their recorded motions. Relating those trajectories to the beat distinguishes shared timing from differences in how individual listeners translate the same pulse into a physical pattern of movement.



Ito et al. (2022), Fig. 4, selected panels.



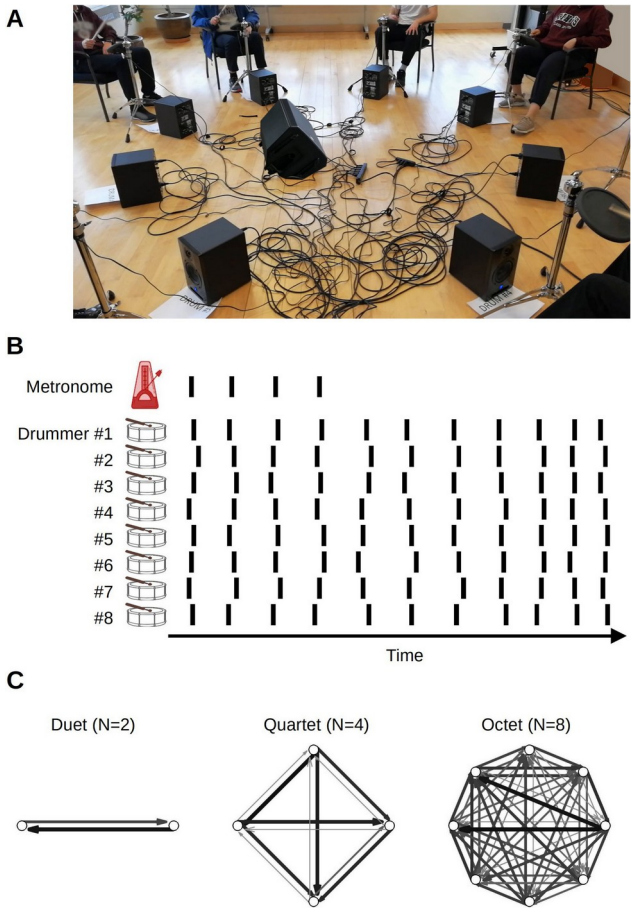
Ito et al. (2022): Human head movement during music playback.

Beat, rhythm, and meter describe different patterns

Rhythm describes the pattern of event durations and intervals, whereas a beat supplies a recurring temporal reference. A drummer can maintain a regular pulse while placing other notes between the pulse positions. The listener must represent both the recurring reference and the more varied arrangement of events played around it.

A **meter** groups beats into recurring patterns of relative emphasis. Learning such groupings allows a listener to anticipate positions within a phrase rather than responding to every note as equally likely. Musical experience can influence those expectations because different traditions organize timing and accent in different familiar ways.

When people perform together, each musician hears timing produced by the others while generating their own events. Small corrections can stabilize a group pulse, while overcorrection can create variability. Research on group drumming measures these coupled changes, treating ensemble timing as an interaction among listeners and performers rather than a collection of independent internal clocks.



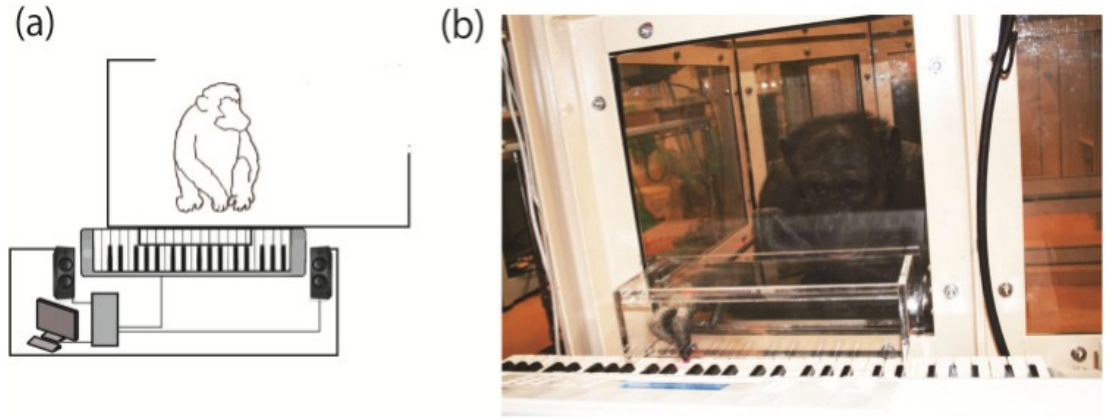
Dotov et al. (2022), Fig. 1.

A chimpanzee can adjust tapping during an auditory rhythm

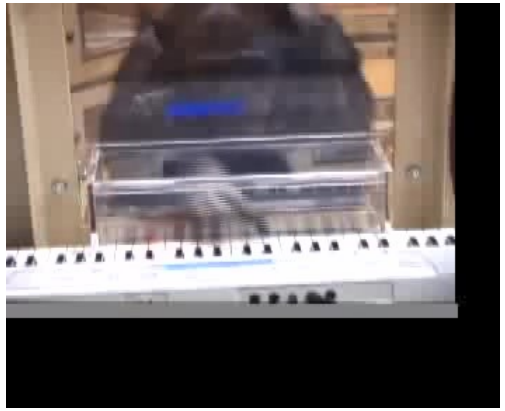
Hattori and colleagues trained chimpanzees to tap illuminated keys on an electronic keyboard. The timing task did not require the animals to follow an auditory beat. Adding an isochronous sound, with equal intervals between events, allowed the researchers to test whether an irrelevant external rhythm nevertheless influenced the timing of the ongoing tapping.

One chimpanzee spontaneously aligned tapping with the auditory rhythm under particular conditions. The effect did not appear uniformly across all individuals and tempos. Its importance lies in the measured adjustment of an already ongoing action, rather than in assuming that any repeated keyboard pressing demonstrates the full flexibility of human musical synchronization.

The first source movie shows the original keyboard task and the animal's responses during sound presentation. The auditory events were not instructions to make a particular musical performance. Comparing tap times with the sound sequence tested whether the external pulse attracted the timing of an action that the chimpanzee had learned for another purpose.



Hattori et al. (2013), Fig. 1.



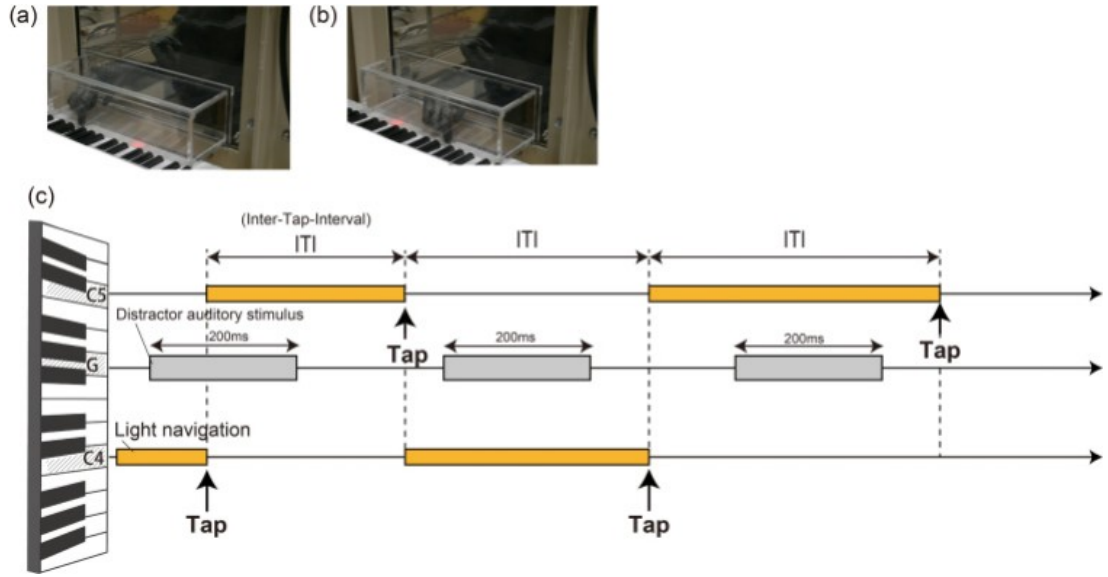
Hattori et al. (2013): Chimpanzee tapping during the original task.

Does a chimpanzee adjust its tapping tempo?

A repeated movement can reflect a preferred spontaneous tempo rather than adjustment to an external rhythm. The chimpanzee tapping study therefore compared conditions with different auditory intervals. If tapping remains unchanged while the external tempo shifts, apparent agreement in one condition may result simply from a coincidence between the two preferred rates.

The investigators examined how sound presentation affected tapping relative to baseline performance and how taps aligned with the auditory events. The synchronization observed in one individual was most apparent under particular timing conditions. This pattern identifies a capacity for auditory influence while also specifying the conditions under which that influence was expressed.

Both movement rate and event alignment are necessary parts of the comparison. Two processes can have the same average interval while occurring at different phases, just as two clocks can tick equally often without ticking together. Measuring the individual tap times establishes whether repeated actions remain aligned with the external sound rather than merely sharing its average pace.



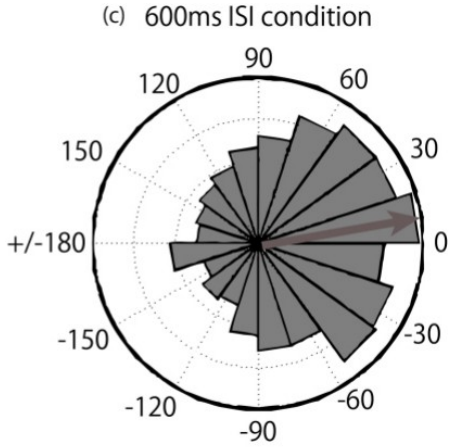
Hattori et al. (2013), Fig. 2.

The phase of each tap reveals its relation to the next sound

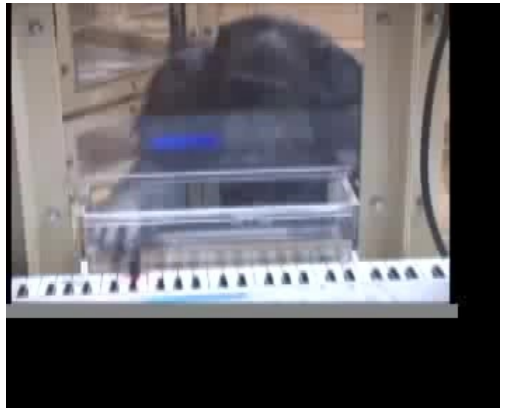
Phase specifies where one event falls within a repeating cycle. A tap just before a tone, a tap at the tone, and a tap halfway between tones occupy different phases even if they recur at the same rate. Comparing these positions makes the timing relationship between an action and a sound explicit.

The second chimpanzee movie supplies another original example from the keyboard experiment. Its key presses can be related to successive auditory events rather than judged only by how regular the movement looks. The study's analyses asked whether those relations clustered systematically instead of drifting freely across different points in the cycle.

A stable phase relationship requires the system to preserve or repeatedly correct alignment. Small differences between the movement's preferred period and the sound's period would otherwise accumulate. Tracking phase across successive events therefore provides information about auditory influence on movement that is lost when performance is summarized only by the total number of taps.



Hattori et al. (2013), Fig. 3, selected panels.



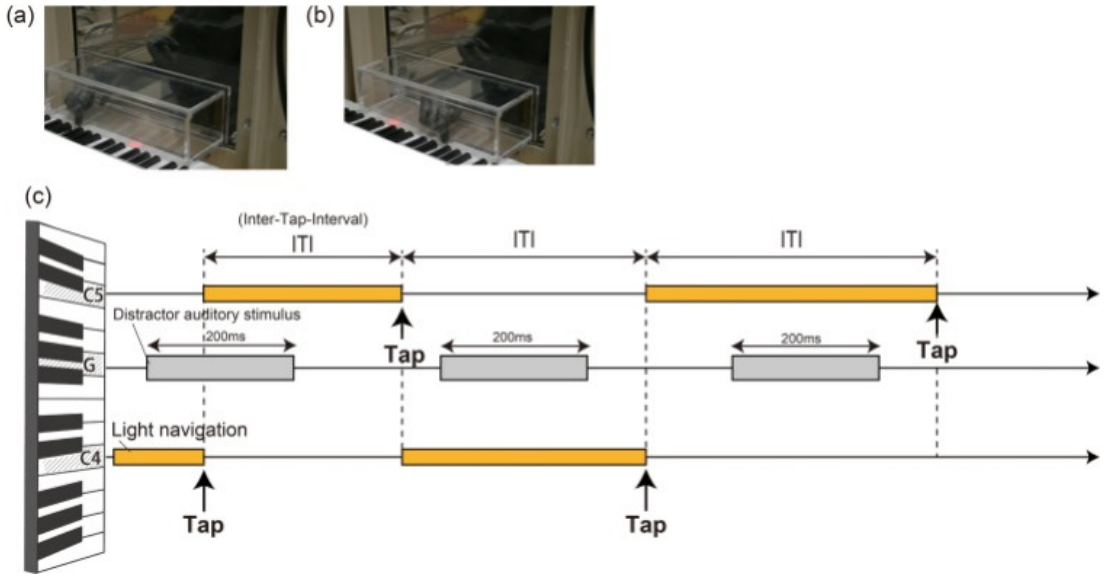
Hattori et al. (2013): A second tapping trial from the original task.

Synchronizing differs from reacting to each beat

A reaction follows an event after the time needed to detect it and initiate an action. **Synchronization** requires an ongoing timing relationship across a sequence, often with preparation before the next event arrives. Those operations differ even when both produce a movement close to the time at which a sound occurs.

Varying the sound interval helps separate the alternatives. A fixed sensory-to-movement delay can place a reaction near each sound, but it does not by itself establish that the next event was predicted. Alignment that persists through a missing sound or changes systematically with tempo provides different evidence about how an internal timing process contributes.

Animal rhythm studies therefore examine complete distributions of movement times and compare them with appropriate baseline conditions. The question is which aspects of human-like synchronization appear in that preparation, such as period adjustment or phase attraction. Defining those components allows a partial capacity to be described accurately without reducing all rhythmic behavior to one category.



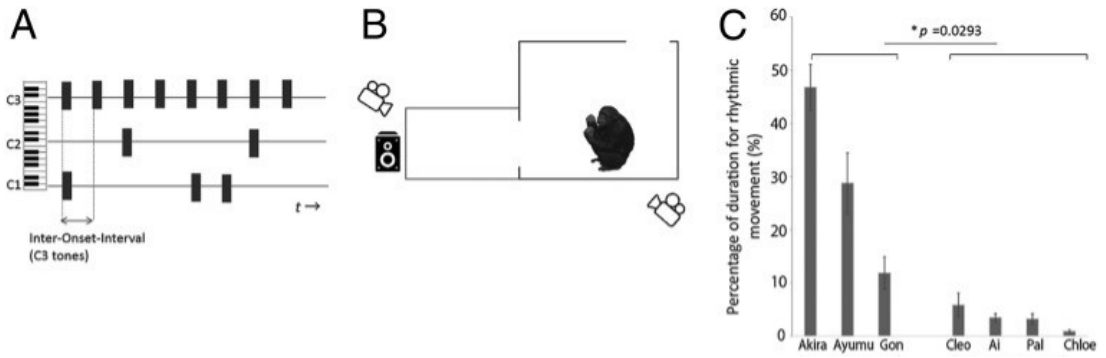
Hattori et al. (2013), Fig. 2.

Sound can evoke rhythmic swaying in chimpanzees

Hattori and Tomonaga presented auditory patterns to chimpanzees and recorded their spontaneous behavior. Sound evoked rhythmic swaying and other repeated movements without requiring a learned dance routine. The study examined whether acoustic events could recruit organized bodily motion, extending the comparison beyond animals trained to press a key at a particular moment.

The responses varied across individuals, with larger effects among the males in the sampled group. The investigators measured the duration and periodicity of movement and compared behavior across acoustic conditions. These outcomes describe observable responses to sound rather than establishing the subjective emotions or musical experience of the animals producing them.

The first source movie shows rhythmic swaying in an upright posture. Its repeated body displacement can be tracked against the ongoing auditory sequence. An animal moving in this way provides a test of sound-movement coupling even when its motions do not reproduce the precise predictive timing characteristic of a practiced human musician.



Hattori and Tomonaga (2019), Fig. 1, selected panels.



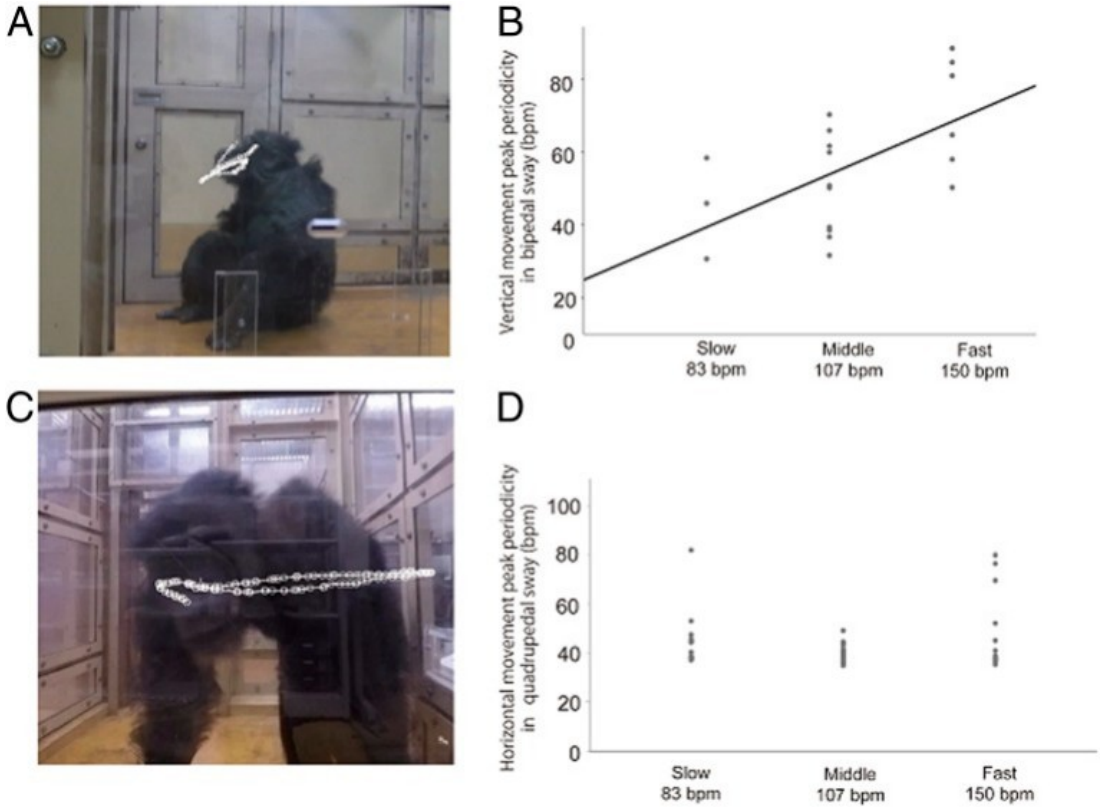
Hattori and Tomonaga (2019): Upright swaying during sound presentation.

Movement can occur without tracking a regular beat

The chimpanzee swaying study compared regularly timed sounds with irregular sequences. Both kinds could evoke rhythmic movement. Regularity was therefore not necessary for every observed swaying response, separating the general tendency to move during sound from the narrower ability to synchronize precisely with an externally specified musical beat.

Changing stimulus tempo affected movement periodicity in the tested chimpanzee when it stood upright. The relationship differed with posture, showing that the physical organization of the movement contributed to the measured timing pattern. A repeated action cannot be interpreted independently of whether the animal is standing, supported on four limbs, or using only part of the body.

These comparisons identify separate components of the response: whether movement begins, how long it continues, and how its period relates to the sound. Different acoustic conditions can affect those components differently. Measuring them separately provides a more informative account than assigning every sound-evoked movement the same label of dancing or beat synchronization.



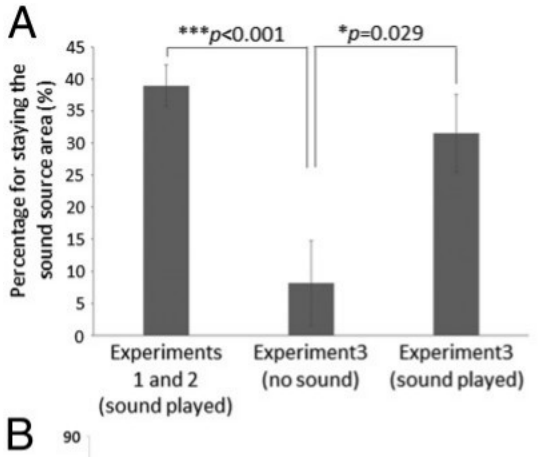
Hattori and Tomonaga (2019), Fig. 2.

Individuals differ in sound-induced rhythmic movement

The second source movie shows rhythmic swaying in a quadrupedal posture. Compared with upright swaying, this movement involves a different distribution of body support and a different path of head displacement. The change in posture therefore changes the physical signal that an observer would track when quantifying the animal's response to sound.

The study compared individual animals rather than assuming that one visible example represented every chimpanzee. Some showed stronger or more sustained sound-induced motion than others. These differences occurred within a group sharing the experimental setting, making individual variability part of the observed behavioral pattern rather than merely a difference between species.

Tracking motion over time distinguishes the period of the repeated action from its total duration. A chimpanzee can sway for longer without changing the rate of individual cycles, or alter cycle timing without increasing the time spent moving. These are different ways in which a sound can influence an organized central motor response.



Hattori and Tomonaga (2019), Fig. 3, selected panels.



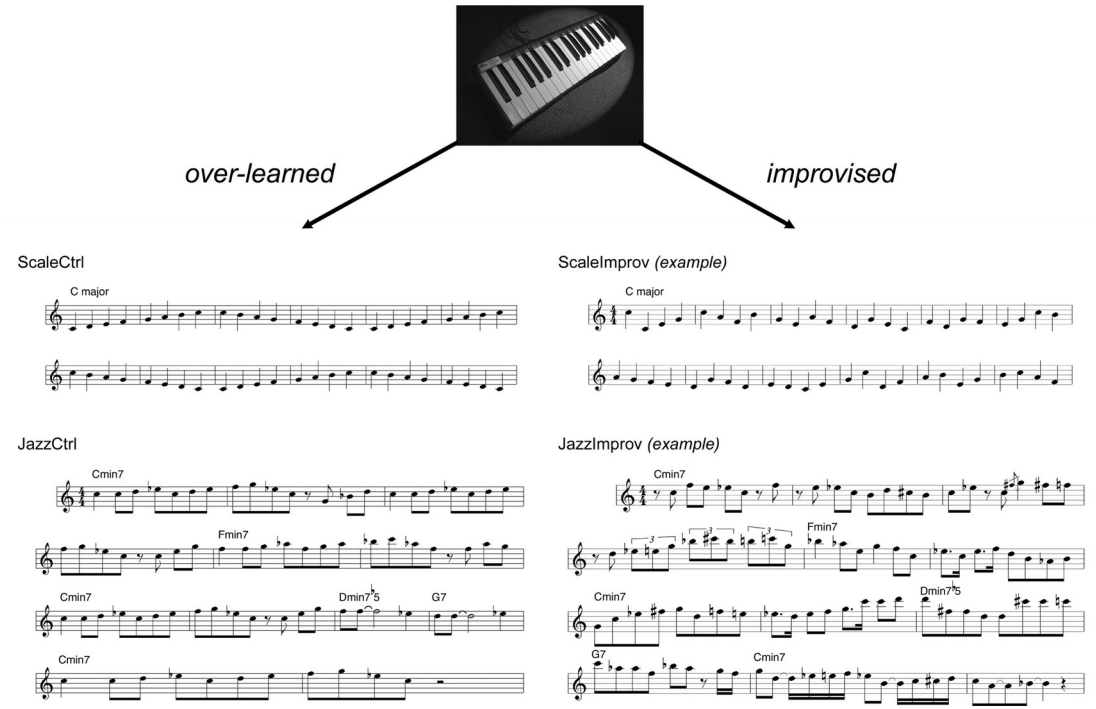
Hattori and Tomonaga (2019): Quadrupedal movement during sound presentation.

Improvisation creates choices within learned patterns

Improvisation creates a musical sequence as it is performed, using learned knowledge without replaying one fully memorized sequence. A jazz musician must select upcoming notes while monitoring timing, movement, and the sounds already produced. The process combines auditory prediction with action selection rather than operating without musical structure or prior experience.

Limb and Braun studied professional jazz pianists performing with an MRI-compatible keyboard. They compared improvisation with the production of overlearned musical material in tasks that differed in complexity. The comparison asked which activity patterns changed when musicians generated new sequences while continuing to perform within an established musical setting.

The brain measurements were blood-oxygenation signals collected during performance. They identify changes associated with the task comparison, while the musical behavior specifies what the pianist was doing. Linking the two requires comparing defined performance conditions, rather than treating any frontal activation as a direct measurement of an individual's general creativity.



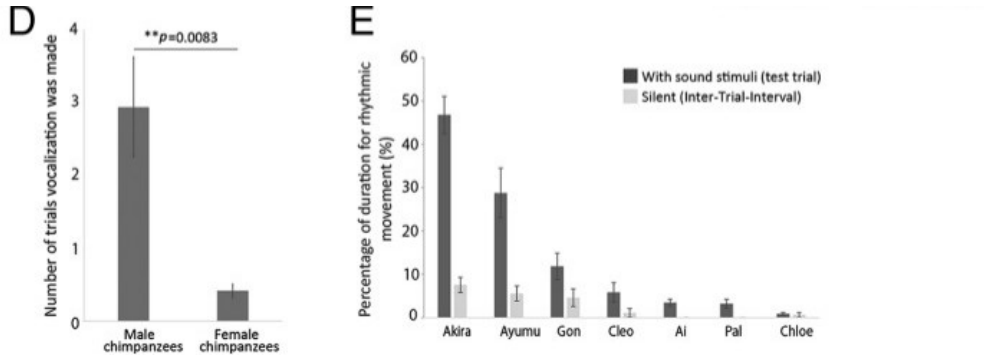
Limb and Braun (2008), Fig. 1.

Rhythmic movement can recruit several parts of the body

Rhythmic movement need not involve the entire body. The third chimpanzee movie shows hand clapping during sound presentation. This changes the measurable action from a broad shift of body position to repeated contact between the hands, supplying another way in which an acoustic sequence can accompany an organized motor pattern.

A hand-clap time can be measured more discretely than the continuous path of a swaying head. However, the interpretation still depends on its relationship to the auditory sequence. Counting claps alone would not establish whether their phase followed the sound, whether their rate changed with tempo, or whether they occurred equally often without sound.

The range of observed actions indicates that sound-induced movement can recruit several motor patterns. Which pattern appears depends on the individual and its current posture and behavior. The study records those actions rather than assuming that all bodies translate an auditory sequence into the same preferred limb or head movement.



Hattori and Tomonaga (2019), Fig. 1, selected panels.



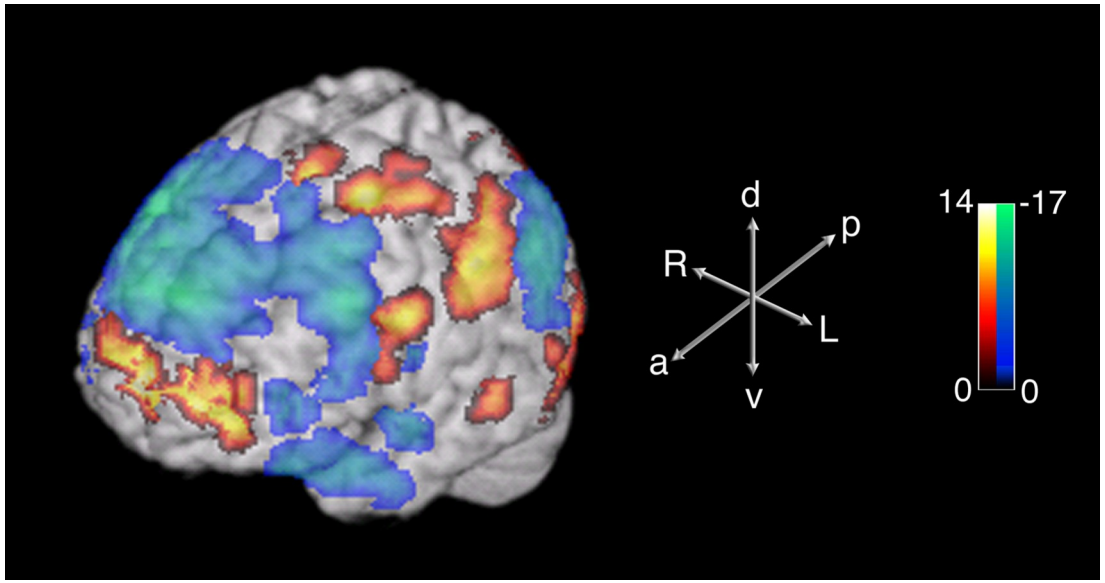
Hattori and Tomonaga (2019): Hand clapping during sound presentation.

Improvisation changes frontal activity

During jazz improvisation, Limb and Braun observed increased activity in medial prefrontal regions alongside decreases in lateral prefrontal regions relative to the overlearned comparison. The distributed pattern appeared across their two performance paradigms. Auditory and motor regions were also involved because the musicians continued hearing and producing sequences throughout the task.

The authors interpreted the frontal pattern as consistent with changes in internally generated activity and deliberate self-monitoring. That interpretation concerns processes recruited by the task, rather than evidence for a single creativity center. One frontal region can change differently from another while both remain part of the wider network supporting the performance.

Generating a new musical phrase still requires constraints. The player must preserve the current temporal framework, select actions that the hands can execute, and compare their results with the unfolding passage. Changes in monitoring and selection can alter the balance among these operations without making improvisation independent of memory, auditory feedback, or practiced motor skill.



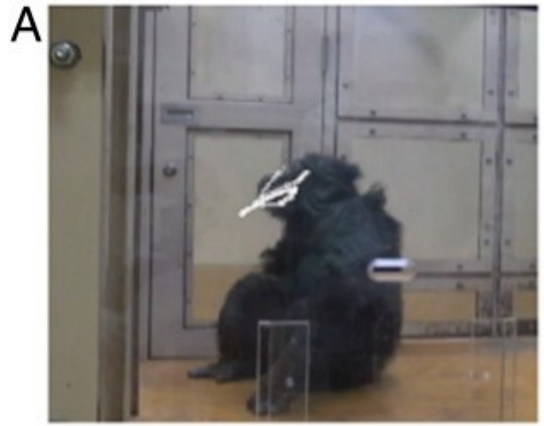
Limb and Braun (2008), Fig. 3.

Spontaneous movement differs from trained tapping

The fourth source movie shows foot tapping rather than a keyboard response or whole-body sway. In the swaying study, these movements emerged during sound presentation without requiring the animal to learn that every tap would produce a reward. This differs from an experiment in which correct alignment with an external metronome is explicitly reinforced.

Training can shape which actions an animal produces and which temporal errors it corrects. Spontaneous sound-induced motion tests another question: whether an acoustic pattern recruits repeated movement before the investigator teaches that particular timing relationship. The two approaches investigate complementary components of auditory-motor behavior rather than providing interchangeable measures of one capacity.

The original foot-tapping example preserves the study's sound and behavioral context. Its value lies in making the type of action recognizable. The quantitative comparisons of regularity, posture, and movement timing come from the full experimental analyses, which establish how these visible examples relate to the broader set of recorded responses.



Hattori and Tomonaga (2019), Fig. 2, selected panels.



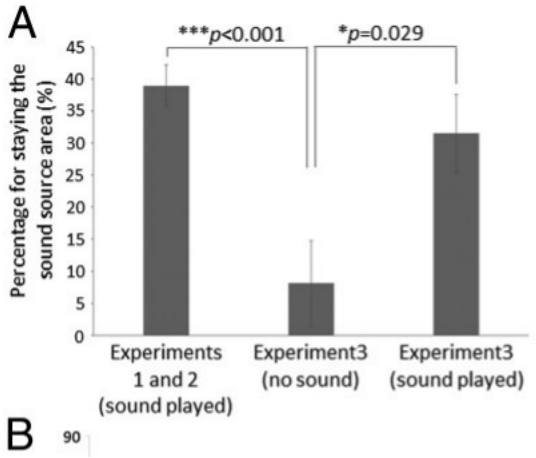
Hattori and Tomonaga (2019): Foot tapping during sound presentation.

Testing movement without rewarding each action

The fifth source movie shows rhythmic swaying accompanied by vocalization. Sound production and body movement occur together, supplying another observable example of coordinated behavior during the auditory condition. The recording does not require the assumption that the animal was reproducing a human song or reporting a particular musical emotion.

The investigators also measured whether a chimpanzee remained close to the sound source when it could choose where to stand. Greater proximity during the auditory condition supplied behavioral evidence about attraction to the sound. This outcome differs from measuring the phase of a movement, because approaching a source need not involve precise alignment with its beats.

The combination distinguishes several effects that listening can have on behavior. Sound can attract an animal to a location, initiate repeated body motion, or accompany a vocal response. A study can identify these components separately while asking which temporal properties influence them, rather than treating one visible response as proof that every component of musical perception is present.



Hattori and Tomonaga (2019), Fig. 3, selected panels.



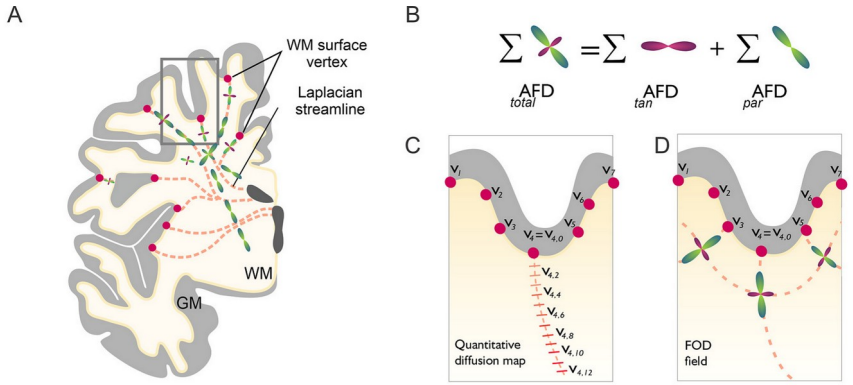
Hattori and Tomonaga (2019): Swaying and vocalization during sound presentation.

White-matter organization covaries with tapping accuracy

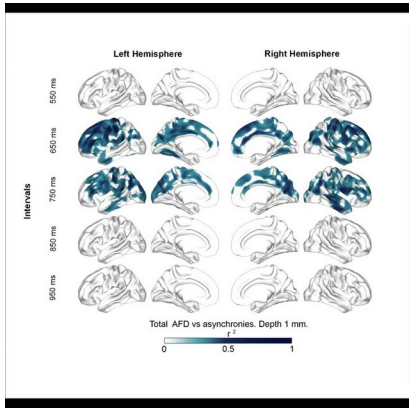
Tapping with a beat requires communication between regions that analyze sound and regions that prepare movement. **White matter** contains bundles of axons carrying signals between these regions. Short curved bundles called **U-fibers** pass beneath the cortical surface and connect nearby areas, allowing neighboring auditory and motor-related regions to exchange information during rhythmic behavior.

Researchers compared people's tapping accuracy with diffusion MRI, which measures how water movement differs across tissue directions. People with different estimates of superficial fiber density also differed in how closely their taps aligned with an auditory metronome. The clearest association occurred in the right audiomotor system at tempos near one and a half beats per second.

The rotating map displays where these statistical associations occurred across the brain's surface. Its colors represent a comparison across people, rather than axons firing during an individual tap. Diffusion measurements estimate tissue organization indirectly; they do not count synapses or demonstrate that a period of musical training produced the observed differences in white matter.



Garcia-Saldivar et al. (2024), Fig. 3, selected panels.



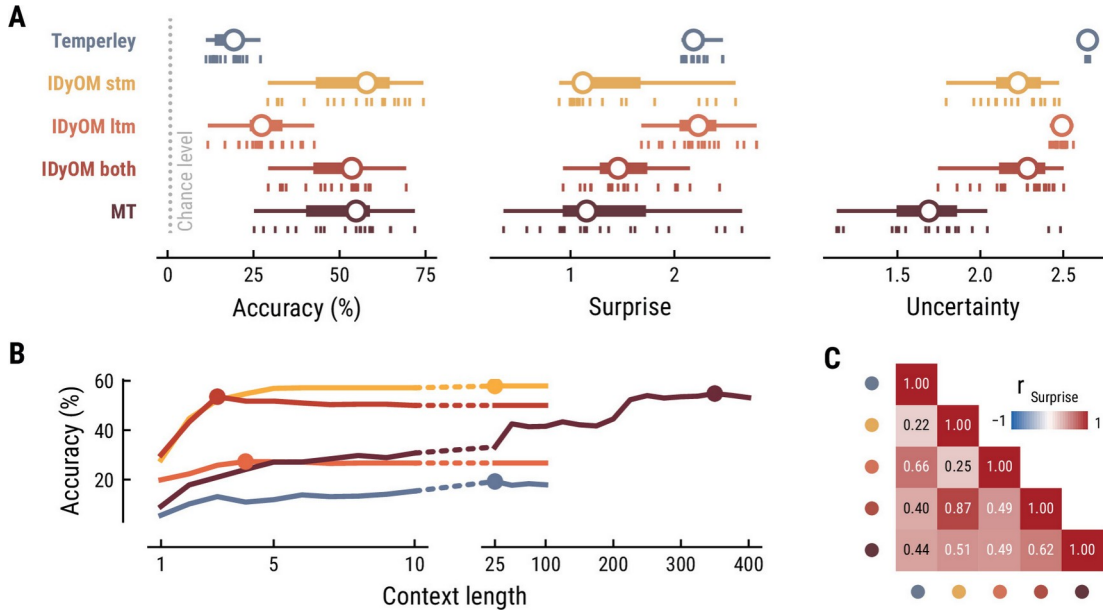
Garcia-Saldivar et al. (2024): Superficial white-matter association maps.

Music depends on memory over several timescales

A familiar melody creates expectations at several timescales. A listener may anticipate the next note from the immediately preceding notes, recognize a recurring phrase, or remember how the entire song continues. These expectations require the nervous system to retain information after a sound ends while incoming sounds continue to update the current musical pattern.

Kern and colleagues modeled musical expectations using learned regularities in note sequences and compared those predictions with human brain responses. A note's unexpectedness depends on its context, rather than its pitch alone. The same physical note can be highly predictable after one phrase and surprising after another because the preceding sequence changes what the listener expects.

Short and longer musical contexts supply different information. Recent notes constrain the immediate transition, whereas a longer passage establishes repeated motifs and broader structure. Comparing predictive models that use different amounts of preceding music helps researchers determine which temporal relationships explain neural responses beyond the sound energy and onset timing of each note.



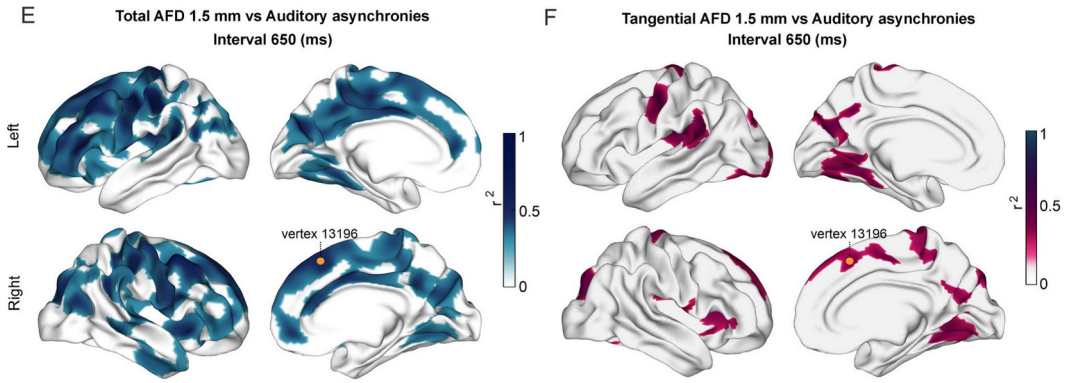
Kern et al. (2022), Fig. 2.

Measuring superficial fiber orientations

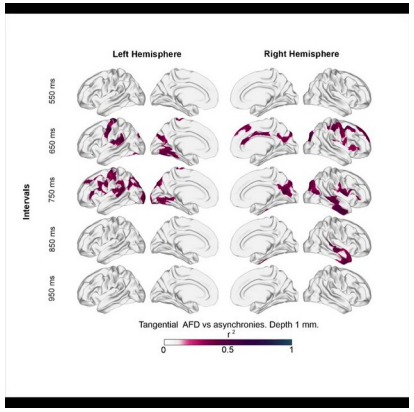
Diffusion MRI estimates fiber orientation by measuring how readily water moves in different directions. Water tends to move more easily along an organized axon bundle than across it. Researchers can use this directional pattern to estimate properties of white matter immediately beneath the cortex, including components that follow the surface and components that extend toward deeper tissue.

The source animation shows the cortical distribution of associations between tapping accuracy and one estimated orientation component of superficial white matter. Each location represents a relationship across participants. It does not show a signal traveling along a fiber, and a region changing color during the rotation does not mean that the participant changed tapping speed at that moment.

Separating orientation components allows a more specific anatomical question than measuring total white-matter volume. Nearby cortical areas exchange signals through short surface-running pathways, while other fibers enter or leave deeper bundles. Differences in these estimated components can therefore identify distinct candidate routes for coordinating auditory timing with movement, even when the overall amount of tissue looks similar.



Garcia-Saldivar et al. (2024), Fig. 3, selected panels.



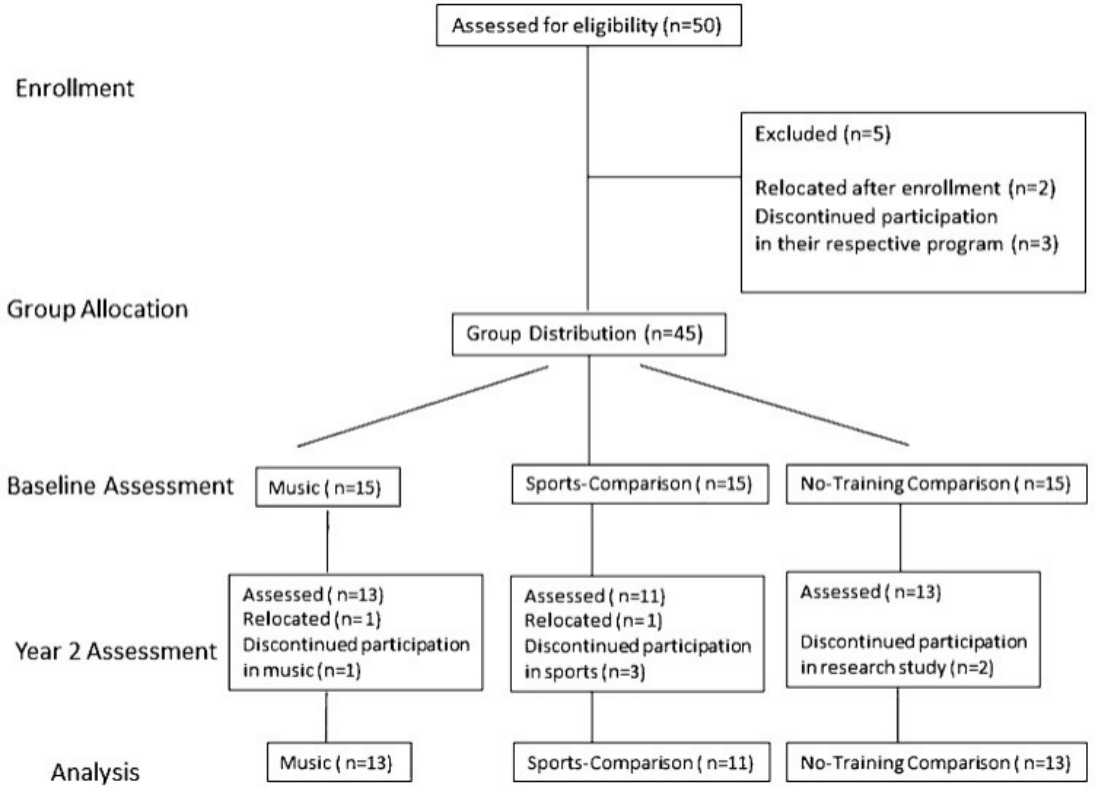
Garcia-Saldivar et al. (2024): Tangential white-matter component maps.

Following children through a music program

Learning an instrument repeatedly pairs a heard sound with a planned movement and a prediction about the next sound. Researchers at USC's **Brain and Creativity Institute** followed children beginning music instruction around six to seven years of age. Their group-based program required listening, coordinated performance, practice, and feedback from teachers and other children over several years.

The investigators also followed children taking part in sports and children without a systematic after-school training program. Measurements began before the training effects of interest had developed, when the groups did not differ on the measured baseline abilities. Following the same children over time allowed the researchers to examine developmental change within each child as well as differences between groups.

A sports comparison helps distinguish music experience from participating in a structured group activity that also requires practice and adult guidance. Families were not randomly assigned to every aspect of these programs, so later differences cannot all be attributed to music alone. The study measured specific auditory and cognitive outcomes rather than assuming that instrument lessons improve every mental ability.



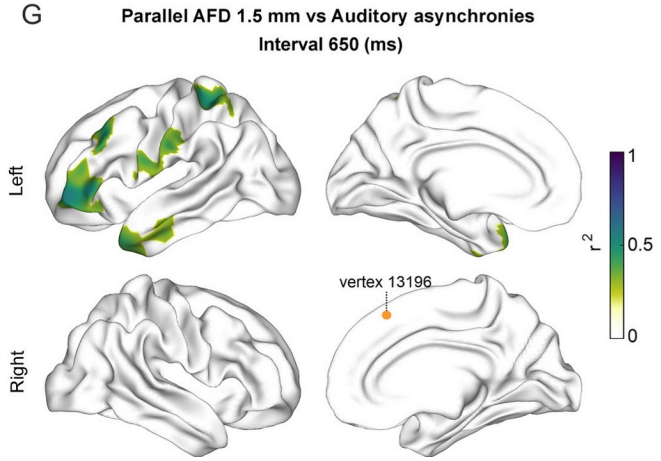
Habibi et al. (2016), Fig. 1.

Anatomical variation and training ask different questions

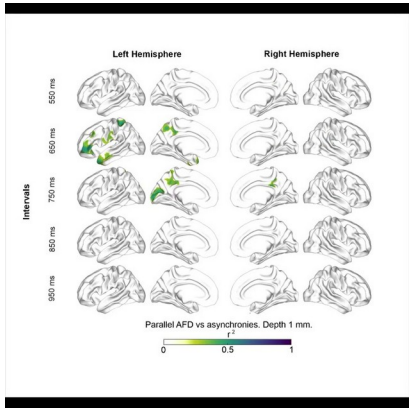
Keeping a steady rhythm involves both local cortical communication and communication between the two hemispheres. The **corpus callosum** is a large bundle of axons connecting the cerebral hemispheres. In the tapping study, its estimated fiber density and bundle diameter varied with how accurately participants aligned their movements with sounds presented at different intervals.

The relationships were organized along the corpus callosum: shorter and longer tapping intervals were associated with different anterior and posterior portions. Timing a rapid sequence and maintaining a slower interval therefore did not produce an identical anatomical pattern. The accompanying cortical animation displays another estimated superficial white-matter component, allowing its distribution to be compared with the preceding maps.

These measurements compare existing differences among people who performed the synchronization task. They do not track newly formed fibers after practice. A training experiment would require measurements before and after a defined period of experience, along with an appropriate comparison group, to determine whether changing rhythmic behavior is followed by changes in these estimated tissue properties.



Garcia-Saldivar et al. (2024), Fig. 3, selected panels.



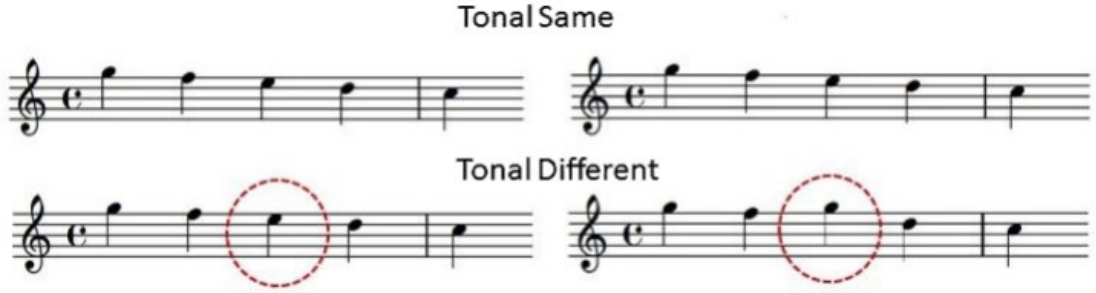
Garcia-Saldivar et al. (2024): Parallel white-matter component maps.

Music experience and auditory development

An **event-related potential**, or ERP, is a pattern of electrical activity averaged across repeated presentations of an event, such as the onset of a musical tone. Electrodes on the scalp detect the combined activity of large neuronal populations. Averaging emphasizes responses that consistently occur after the sound while reducing activity that varies unpredictably from trial to trial.

In the USC longitudinal study, children with two years of music training were better at detecting changes in a tonal environment than the comparison groups. Their tone-evoked cortical responses also showed a more mature developmental pattern. Auditory responses normally change as children grow, so the investigators compared the timing and shape of those responses across repeated assessments.

The behavioral task required children to notice a change in musical pitch relationships, while the ERP measured activity following individual sounds. These measures describe related but different processes: recognizing a tonal change requires comparing sounds across time, whereas an evoked response reveals how cortical populations react after a particular event. Neither measurement is a direct recording of a single child's musical talent.



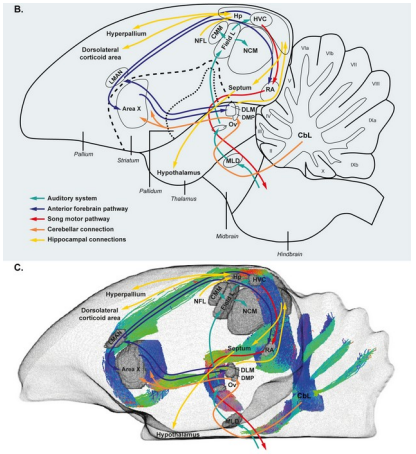
Habibi et al. (2016), Fig. 3.

Songbird brain pathways change across the seasonal cycle

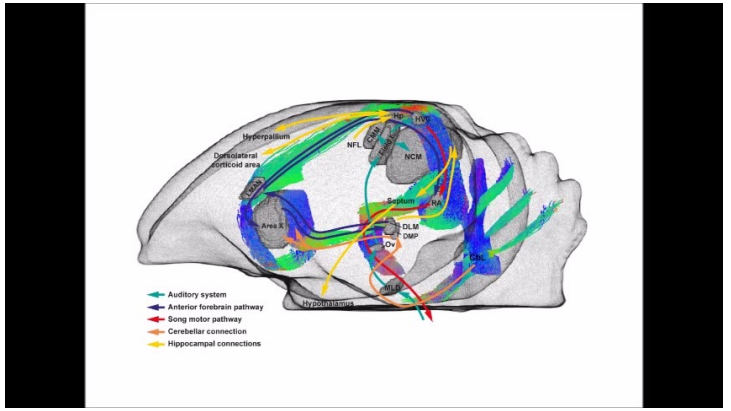
In songbirds, producing and learning vocal sequences involves specialized forebrain nuclei and the pathways between them. A **nucleus** is a cluster of neurons within the brain. Starlings also have auditory regions that analyze incoming sounds, allowing their own vocal output and other birds' songs to influence behavior throughout the annual cycle.

Researchers repeatedly imaged male and female starlings as day length changed across seasonal conditions. The source animation identifies song-related structures and reconstructed pathways in three dimensions. These anatomical views make the locations and spatial relationships of the regions visible; the animation does not display neurons firing while a bird sings or listens to another bird.

Changes were not restricted to one song nucleus or to males. The study detected seasonal tissue changes in song-control, auditory, visual, and cerebellar systems in both sexes. Some patterns of hemispheric organization differed between males and females, while the seasonal changes also extended to regions involved in coordinating movement and processing other sensory information.



Orije et al. (2021), Fig. 1, selected panels.



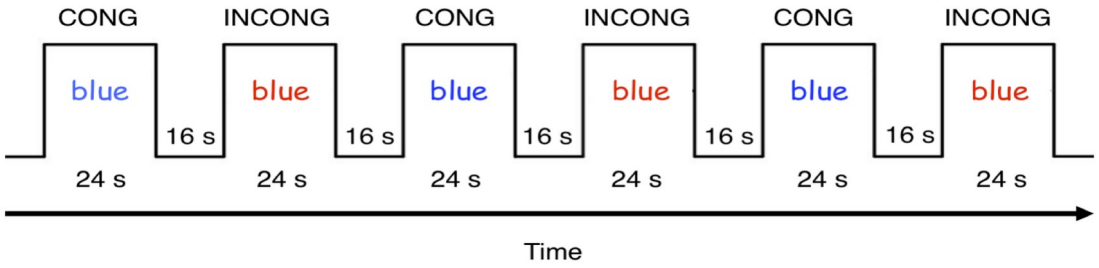
Orije et al. (2021): Starling song-related anatomy and pathways.

USC: cognitive-control responses in young musicians

The **Stroop task** creates conflict between a familiar response and the response a person has been instructed to make. If the word BLUE appears in red ink, naming the ink color requires suppressing the tendency to read the word. Frontal and related brain regions help maintain the rule and select the appropriate response when these competing signals disagree.

Sachs and colleagues examined this task in children from the USC music, sports, and comparison groups. After two years, the music group recruited several frontal and insular regions more strongly than children without systematic training during conflicting trials. These included regions involved in preparing actions, monitoring competing responses, and directing attention toward behaviorally relevant information.

The groups did not differ in their behavioral executive-function performance, and the music-versus-sports brain comparison did not show the same significant differences. A larger blood-oxygen response therefore did not mean that the music group performed the task better. It described a difference in neural recruitment during a particular conflict task, with the sports group providing a useful comparison for structured extracurricular experience.



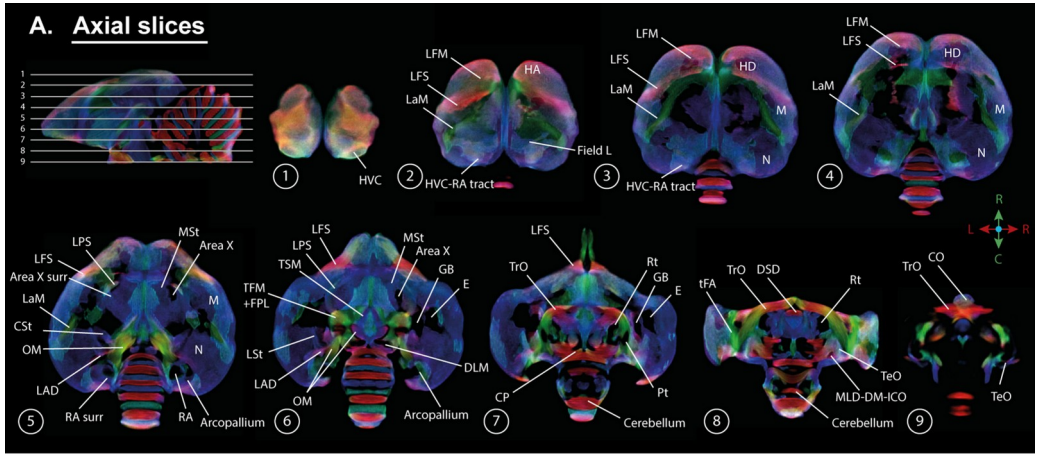
Sachs et al. (2017), Fig. 2.

A seasonal window of heightened neural change

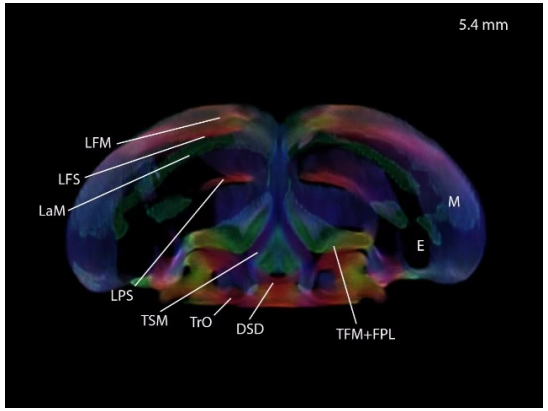
A **sensitive period** is a developmental or physiological interval during which experience or changing conditions can produce especially large changes in a neural system. In seasonal birds, these intervals can occur during the annual cycle. They need not coincide with the period when reproductive hormones are highest or when an observer hears the most conspicuous singing behavior.

The starling imaging study found prominent changes during the photosensitive period, when birds were responsive to increasing day length and gonadal hormone levels were low. Changes appeared across multiple sensory and motor pathways. The accompanying animation moves through a reconstructed population tractogram, showing the spatial arrangement of pathways rather than an animal's behavior at a particular moment.

A **tractogram** estimates trajectories from the directional diffusion of water through tissue. Seasonal differences in these measurements can reflect changes in tissue organization without identifying every cellular event that caused them. Repeated measurements in the same birds establish when the imaging properties changed across the year, while hormonal state and day length describe the physiological conditions during those changes.



Orije et al. (2021), Fig. 2, selected panels.



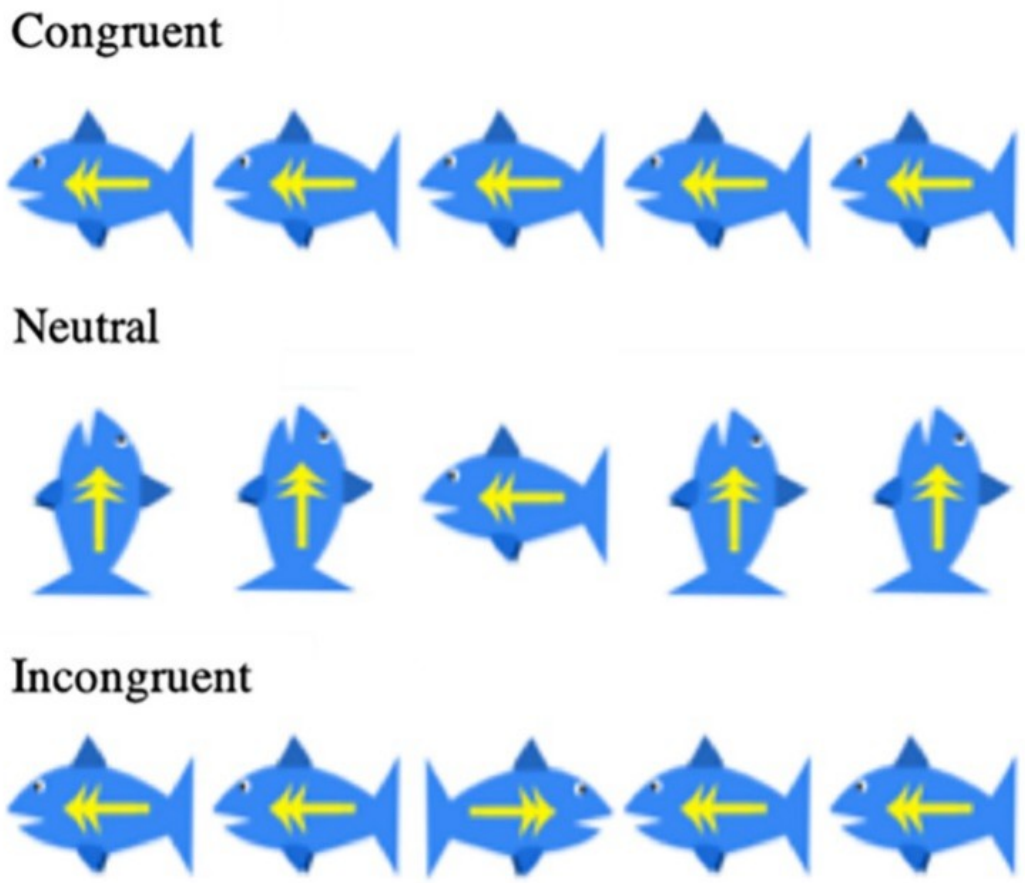
Orije et al. (2021): Starling population tractogram.

USC: following cognitive control across development

Inhibitory control allows a person to withhold a dominant response when another response better fits the current goal. A flanker task tests this ability by placing distracting symbols around a target. When surrounding symbols indicate the opposite direction, the participant must use the central target to choose a response while ignoring competing directional information.

Hennessy and colleagues continued following the USC children across several years. Beginning after three years, children in the music group more often chose a larger delayed reward over a smaller immediate reward than the comparison group. Their flanker accuracy also improved over time, while the study continued to find no group differences in behavioral performance on the color-word Stroop task.

The brain differences also changed with development. After four years, the broader pattern of music-related Stroop activation reported earlier was reduced to a difference in the right inferior frontal gyrus. Different control tasks require different combinations of attention, response selection, and waiting, so an improvement on one task does not establish an equal improvement on all forms of self-control.



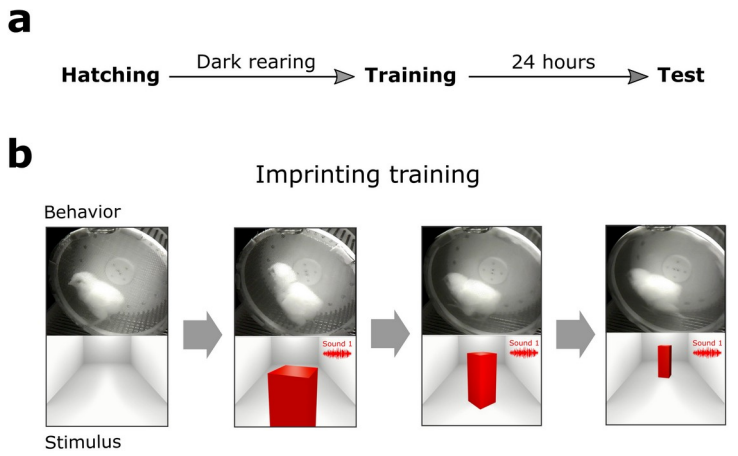
Hennessy et al. (2019), Fig. 2.

Auditory imprinting creates an early preference

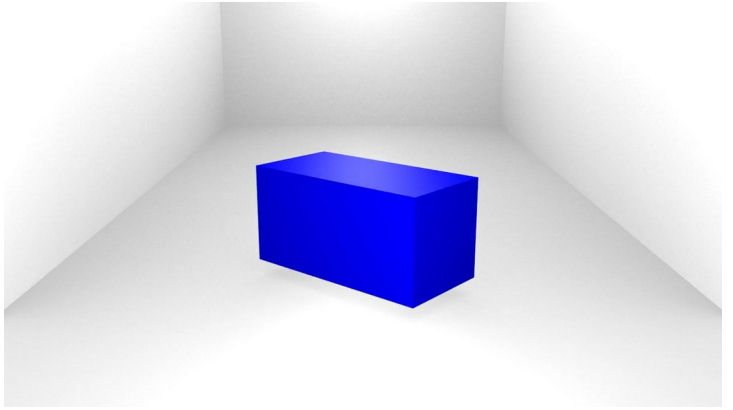
Newly hatched chicks can learn to approach a particular sound or moving object through **imprinting**, a form of early learning that changes later preferences. The sound becomes recognizable through experience during a period when the young animal is especially responsive to such learning. Researchers can then test memory by comparing approach toward familiar and unfamiliar stimuli.

Batista and colleagues trained chicks with audiovisual displays presented while the birds moved in a running wheel. During training, sound and visual information were available together, allowing the young bird to experience the paired cues repeatedly. Later tests separated auditory and visual preferences, allowing the investigators to ask whether the same cellular manipulation affected both kinds of learning equally.

An approach preference measures a learned behavioral response: the chick moves toward one stimulus more readily than another. It does not directly reveal the activity of every neuron storing the memory. By changing a protein-synthesis control mechanism before training and then measuring the preference, the researchers could test how that cellular mechanism affected the formation of auditory imprinting.



Batista et al. (2016), Fig. 1, selected panels.



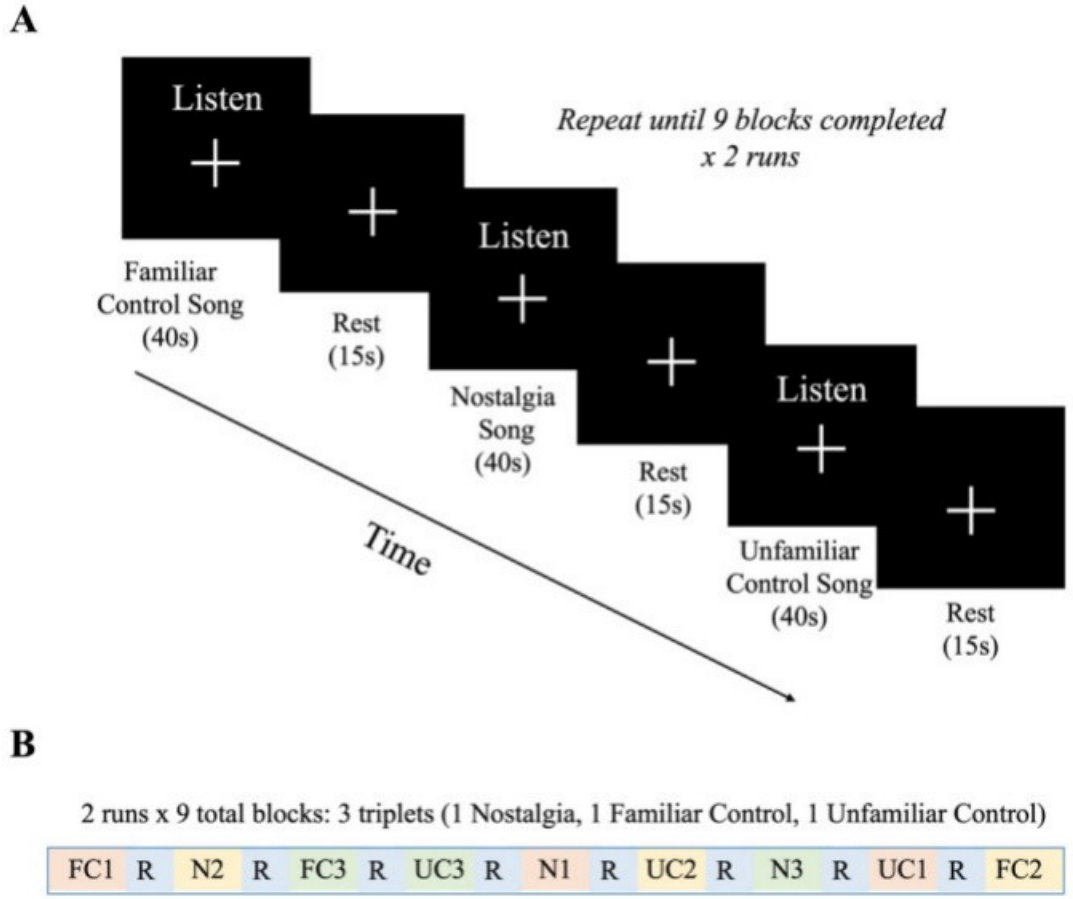
Batista et al. (2016): Original chick-imprinting training stimulus.

USC: music, nostalgia, and personal memories

A song can bring back a personal event even when its lyrics do not describe that event. **Autobiographical memory** includes memories of one's own experiences, and nostalgia often combines those memories with mixed pleasant and wistful feelings. Familiarity alone is different: a person can recognize a frequently heard song without associating it with a meaningful period of life.

In a recent Brain and Creativity Institute study, Hennessy and colleagues selected personally nostalgic songs for younger and older adults. They compared those songs with familiar but nonnostalgic songs and unfamiliar songs matched for musical features. This design helped separate the response to personal nostalgia from responses to simple familiarity or broad differences in the sound of the music.

Nostalgic listening produced greater activity in regions involved in self-related processing, emotion, reward, and memory than the comparison conditions. Activity in posteromedial regions also varied more closely with activity in the insula during nostalgic listening. This functional connectivity describes coordinated fluctuations in the imaging signals, rather than proving that one area sent a particular message directly to the other.



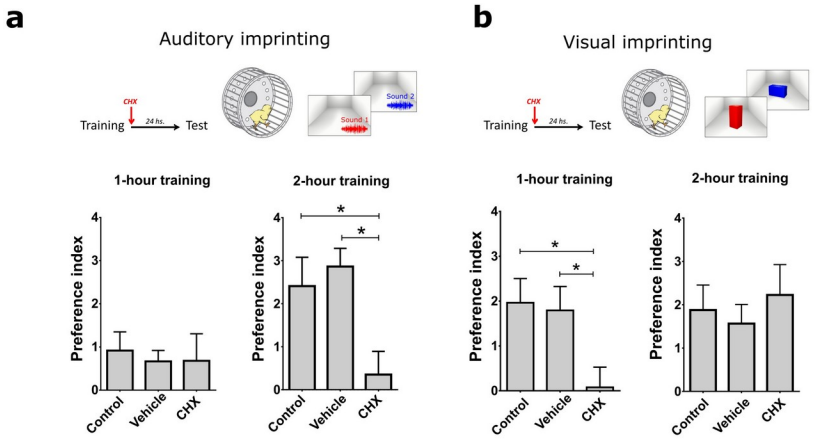
Hennessy et al. (2025), Fig. 1.

Protein synthesis helps control an auditory learning window

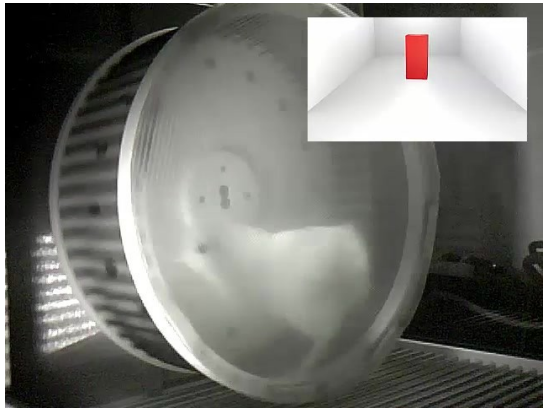
Long-lasting learning can require neurons to make proteins that alter synapses, the sites where one neuron communicates with another. **eIF2α** helps regulate the initiation of protein production. Adding a phosphate group to this protein changes that regulation, altering which proteins are made and how readily the cell can support lasting structural and functional changes.

In the chick imprinting study, increasing eIF2α phosphorylation interfered with auditory imprinting and associated neuronal plasticity. Manipulations that reduced this restrictive control improved auditory imprinting and could reopen the period when chicks learned the auditory preference. Visual imprinting did not change in the same way, indicating that the manipulation did not simply eliminate every form of approach learning.

A chick approaches a screen during the original recorded preference test. The movement is the behavioral readout; the protein manipulation occurred before the test and is evaluated across experimental conditions. Reopening auditory imprinting in this preparation does not mean that every adult brain region regains an unrestricted capacity to learn after the same molecular change.



Batista et al. (2016), Fig. 3, selected panels.



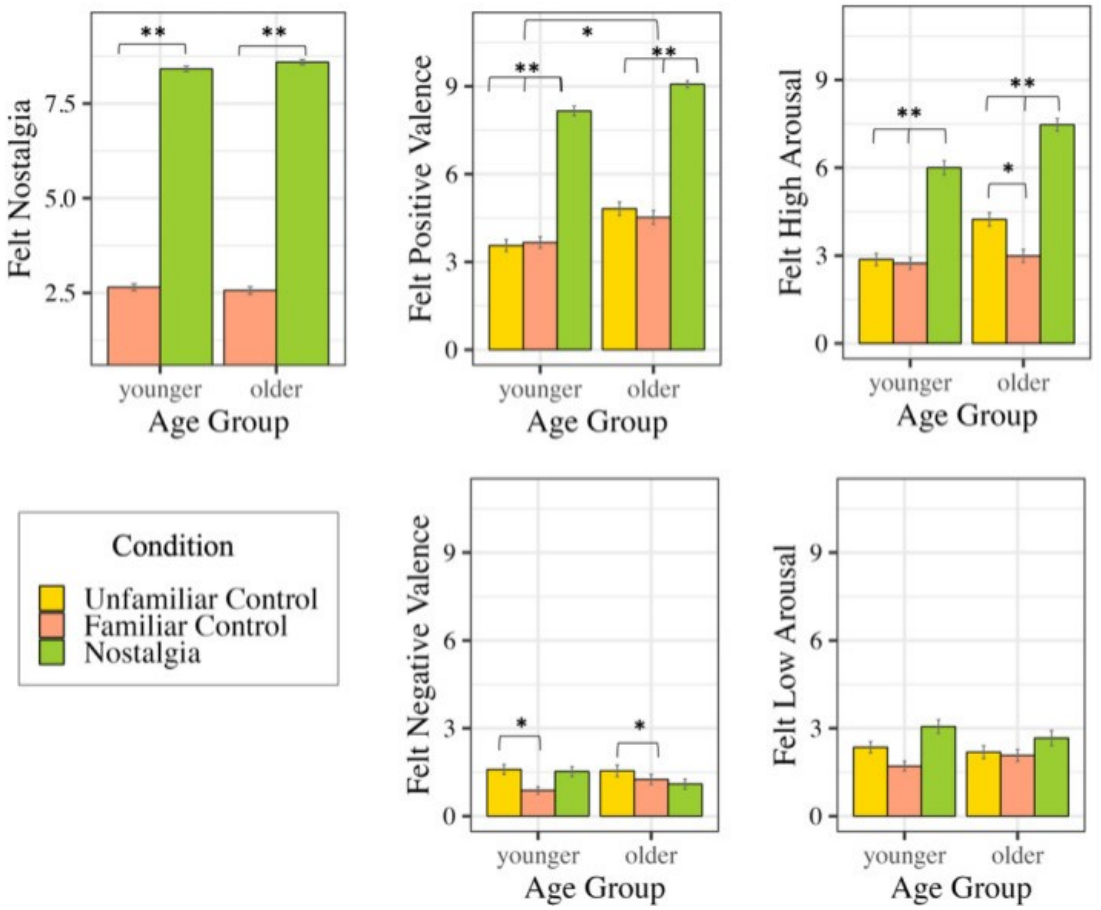
Batista et al. (2016): Chick approach during the original preference test.

Musical pleasure differs from melody recognition

Recognizing a song's structure, remembering a personal event, and enjoying the sound require overlapping but separable processes. **Amusia** refers to a difficulty processing music, which can involve impaired perception of pitch or melodic relationships. A listener's emotional response can remain partly available even when the person has difficulty identifying the musical pattern that another listener recognizes easily.

Nostalgic songs add a further component because their emotional value depends on the listener's history. In the USC study, older adults showed stronger blood-oxygen responses than younger adults in several nostalgia-related regions. Among younger adults, responses varied with trait nostalgia and cognitive measures; among older adults, responses were related to their reported emotional reactions to the music.

The participants in that study were healthy adults, and the experiment measured responses during listening. It did not test whether a music program treats dementia or restores lost autobiographical memories. Distinguishing recognition, personal recollection, and felt emotion allows researchers to measure each process separately rather than labeling every positive reaction to a familiar song as improved memory.



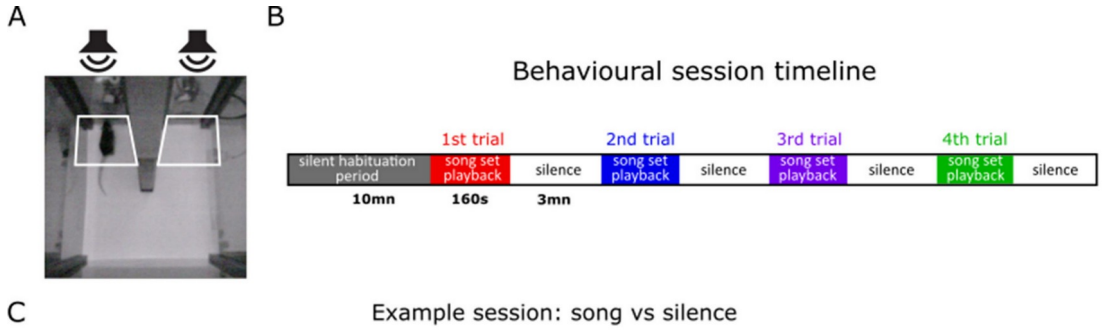
Hennessy et al. (2025), Fig. 3.

The timing of mouse calls changes their social significance

Mouse vocalizations contain sequences of brief sound elements separated by intervals of silence. These sequences can influence social approach, so their timing provides a way to study how an animal's brain converts an acoustic pattern into a behavioral decision. Female mice can approach a speaker playing male vocalizations even though the sounds are produced by a recording rather than a nearby male.

Researchers compared intact male vocalizations with versions in which selected properties were changed. Altering the regularity of the timing reduced the females' normal approach response more than several changes to syllable order or individual syllable structure. The result identifies the temporal organization of the sound sequence as a particularly useful cue in this social listening task.

In this recorded intact-song trial, the active speaker is on the left, and the mouse's position is tracked as it moves through the apparatus. The controlled evidence comes from how the animals responded across the different original and modified playbacks, including trials in which the temporal pattern was disrupted.



Perrodin et al. (2023), Fig. 2, selected panels.



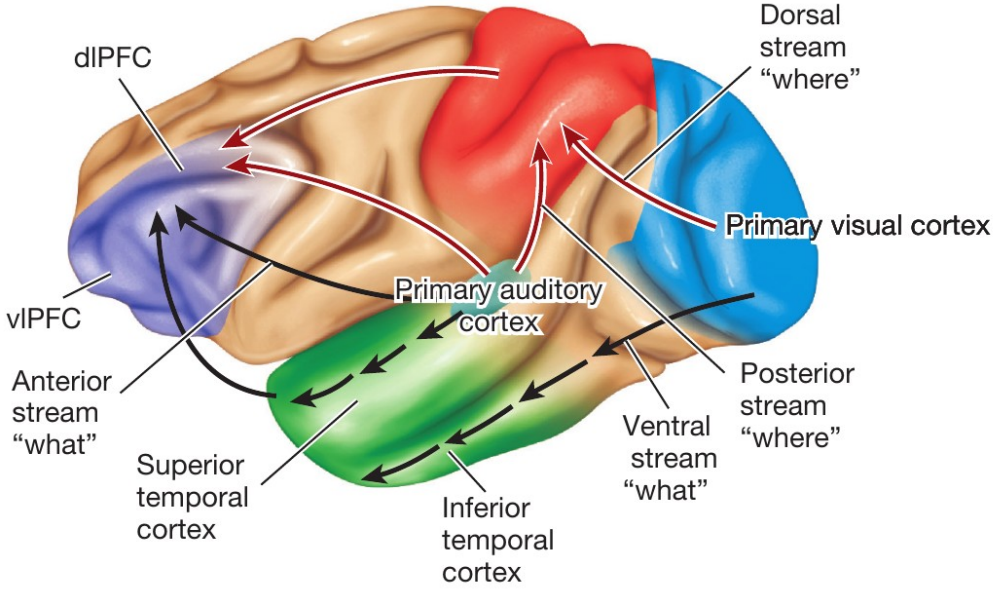
Perrodin et al. (2023): Intact male-song playback: original mouse trial 1.

From a sound pattern to an approach decision

A sound becomes behaviorally useful when its representation can influence a choice. Auditory regions identify features and patterns, while connected frontal, striatal, and motor systems help select actions appropriate to the situation. The same call can prompt approach, withdrawal, or no movement depending on the listener's experience, current state, and the presence of other relevant cues.

An **approach decision** can be measured by where an animal moves, how quickly it starts moving, or how long it remains near the sound source. These measurements answer different questions. Remaining close to a speaker measures sustained preference, while a brief head turn may reflect initial orientation even when the animal never approaches the speaker afterward.

To determine which sound feature controls the response, researchers change one property while holding others as constant as possible. Changing syllable timing while preserving the original syllable sounds tests the contribution of sequence timing. Reversing speaker positions across trials helps separate preference for the playback from an existing preference for one side of the apparatus.



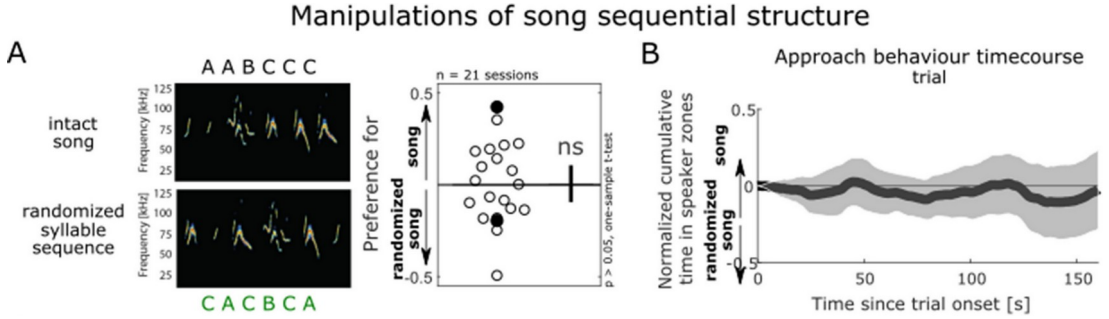
Carlson and Birkett (2022), Fig. 7.16, p. 206.

Which vocal patterns attract a listening mouse?

A sound sequence has several kinds of organization. **Syllable order** specifies which sound element follows another, while **rhythm** describes their timing and the intervals between them. Changing one does not necessarily change the other. Rearranging a sequence can preserve its overall pace, whereas shifting onset times can disrupt rhythm while retaining the original collection of syllables.

The mouse playback experiment manipulated these properties separately. Female approach was especially sensitive to irregular timing, while some alterations of syllable sequence or within-syllable acoustic structure had smaller effects. The behavior therefore depended strongly on when elements occurred, rather than requiring an exact copy of every sound element and its original place in the sequence.

This second source recording is another intact-playback trial, not a demonstration of the irregular condition. Comparing different intact trials shows the natural variation in the movement used to measure preference. The experimental timing result comes from the full set of playback comparisons; a single animal walking toward one speaker cannot by itself establish which acoustic feature caused the approach.



Perrodin et al. (2023), Fig. 3, selected panels.



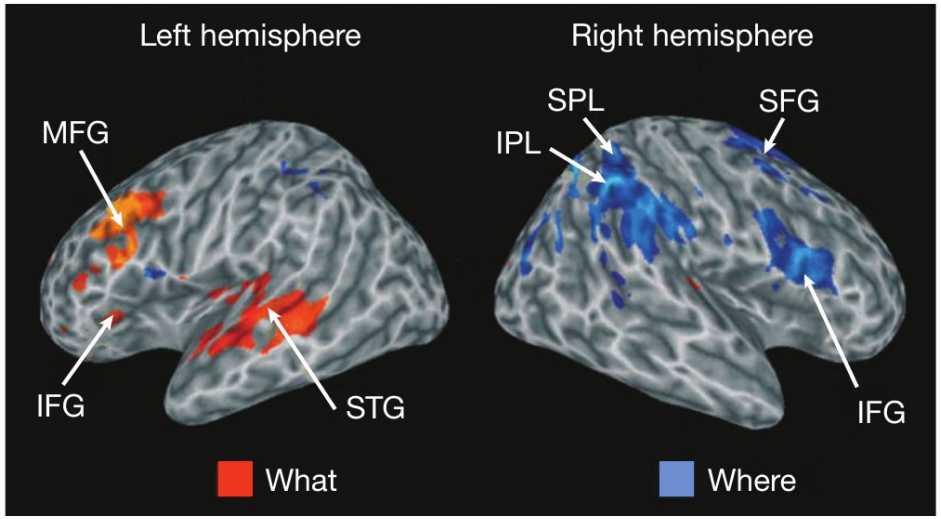
Perrodin et al. (2023): Intact male-song playback: original mouse trial 2.

When hearing a sound does not reveal its identity

Auditory agnosia is an impairment in recognizing sounds despite enough sensory detection to hear that something occurred. A person may detect a sound and judge its loudness yet struggle to identify its source. The affected operation is recognition, which requires relating a changing acoustic pattern to stored information about objects, voices, speech, or music.

Different injuries can impair different kinds of auditory recognition. Damage involving speech-processing networks can make spoken words difficult to understand, while other damage can disproportionately affect recognition of environmental sounds or musical relationships. These patterns follow from the partial specialization and interaction of cortical networks, rather than a single cortical point that recognizes every sound category.

Clinical testing separates these possibilities by asking different questions about the same or related stimuli. Detecting a tone, discriminating two pitches, identifying a familiar melody, and recognizing a spoken word require different operations. A failure to name a melody could reflect damaged melodic processing, impaired access to its identity, or difficulty producing its name, depending on the person's other preserved abilities.



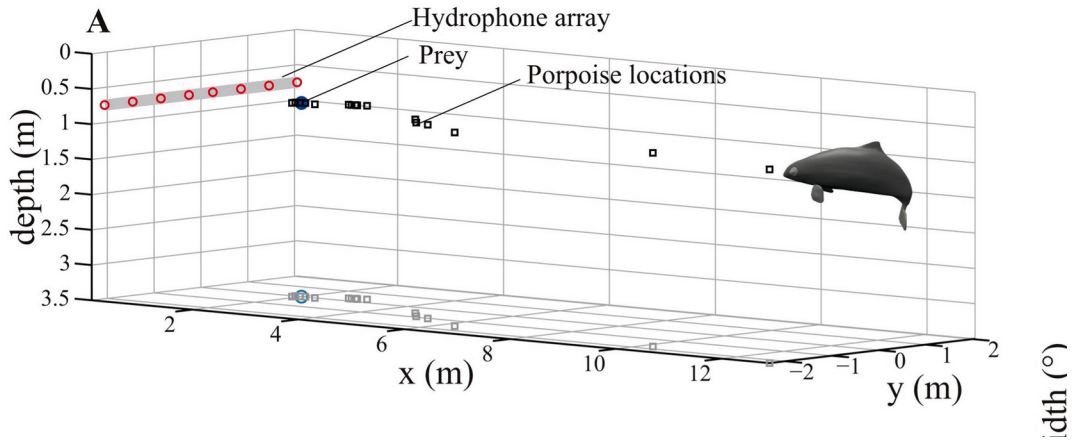
Carlson and Birkett (2022), Fig. 7.17, p. 206.

A porpoise finds prey without visual guidance

An echolocating porpoise repeatedly emits clicks and uses returning echoes while approaching a target. Each movement changes the target's distance and direction, so the animal must update its estimate rather than rely on a single initial sound. Blindfolding during a controlled task removes visual guidance and makes the animal's acoustic sampling especially clear in the recorded approach.

Wisniewska and colleagues filmed porpoises capturing fish and approaching artificial targets while hydrophones measured the outgoing signals. The original fish-capture recording combines the animal's movement with the acoustic measurements. Near the target, the porpoise switches to a rapid terminal buzz, producing frequent updates during the final part of the approach.

The study measured changes in acoustic sampling and the animal's behavior, rather than recording neurons that calculate distance. Those measurements show how the information available to the central auditory system changes as the animal closes in. The animal must coordinate its next movement with echoes from the changing scene while avoiding confusion between successive outgoing clicks and returning signals.



Wisniewska et al. (2015), Fig. 3, selected panels.



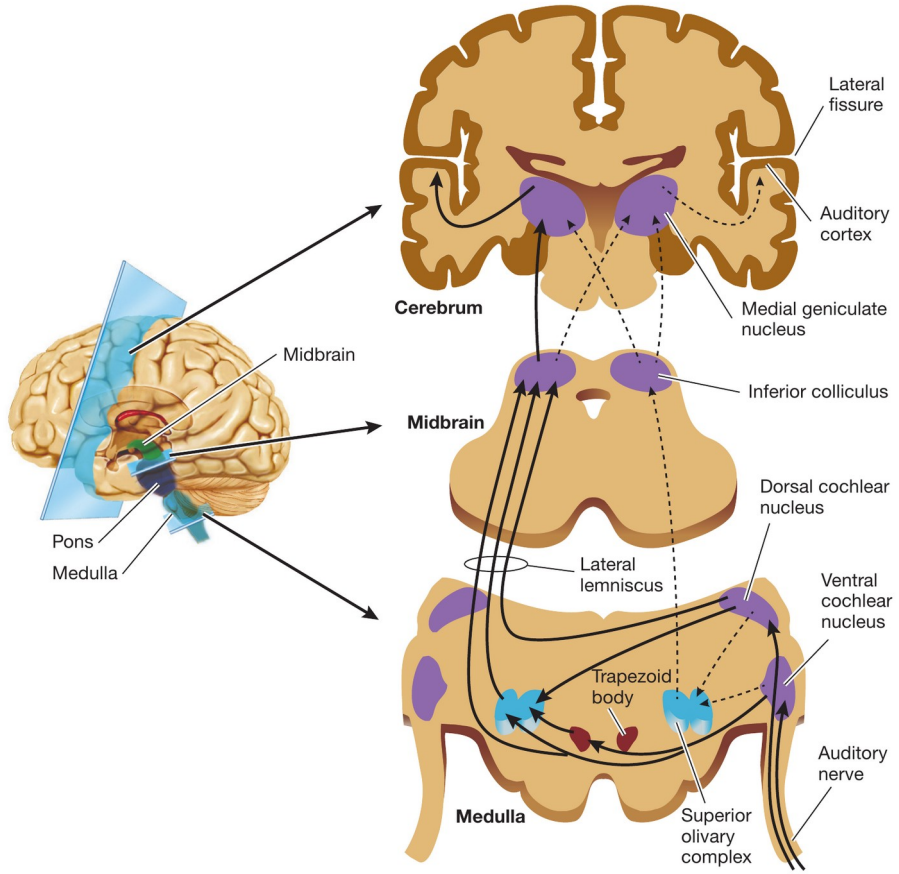
Wisniewska et al. (2015): Blindfolded porpoise capturing a fish.

Echo timing supplies information about distance to a target

An echo returns after sound travels to an object and then back to the listener. **Round-trip delay** therefore increases with distance. If the speed of sound is known, distance equals that speed multiplied by the echo delay and divided by two. The division accounts for the outward and returning portions of the sound's journey.

Central neurons can compare the timing of a self-generated signal with later returning sound. Different response properties allow populations to represent particular delays, frequencies, and changes across repeated echoes. A sequence of shorter delays indicates that a target is getting closer, while changes in the direction and structure of the echo provide additional information about the target's location and surface.

Active sensing also requires distinguishing expected consequences of the animal's own action from unrelated sounds. A motor command provides information about when a click was produced, and the arriving echoes supply information about the environment. The porpoise experiments measure the behavior and acoustics of this sampling process; they do not identify the individual central neurons implementing every comparison.



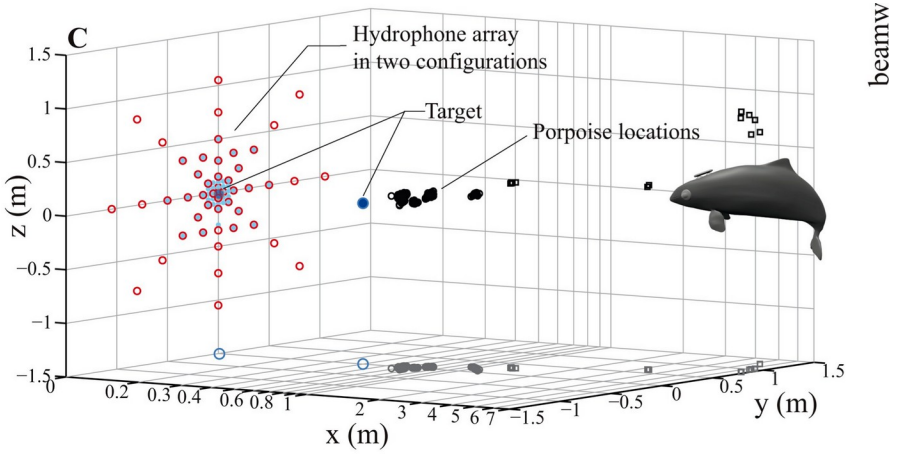
Carlson and Birkett (2022), Fig. 7.7, p. 198.

Active listening changes during target approach

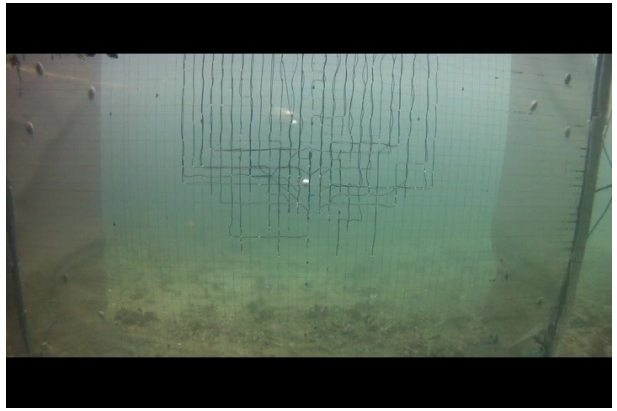
An animal approaching a target faces a changing sampling problem. At a distance, the target occupies a small region of space; close to contact, small movements can shift it rapidly relative to the animal. Frequent acoustic updates help the porpoise guide the final approach while its own movement continually changes the pattern of returning sound.

In the artificial-target experiment, a porpoise approached an aluminum target while an array of hydrophones measured the sound field. The source recording combines the approach with these measurements. As the animal entered the terminal part of the approach, the measured acoustic field widened, changing the region of space sampled during rapid close-range clicking.

These observations describe a coordinated behavioral sequence: approach, rapid sampling, and target interception. They do not demonstrate a particular motor command inside the brain. Central auditory estimates must be updated quickly enough to guide this sequence, but identifying the neurons that control each adjustment requires neural measurements or manipulations beyond the acoustic field and movement recordings shown here.



Wisniewska et al. (2015), Fig. 3, selected panels.



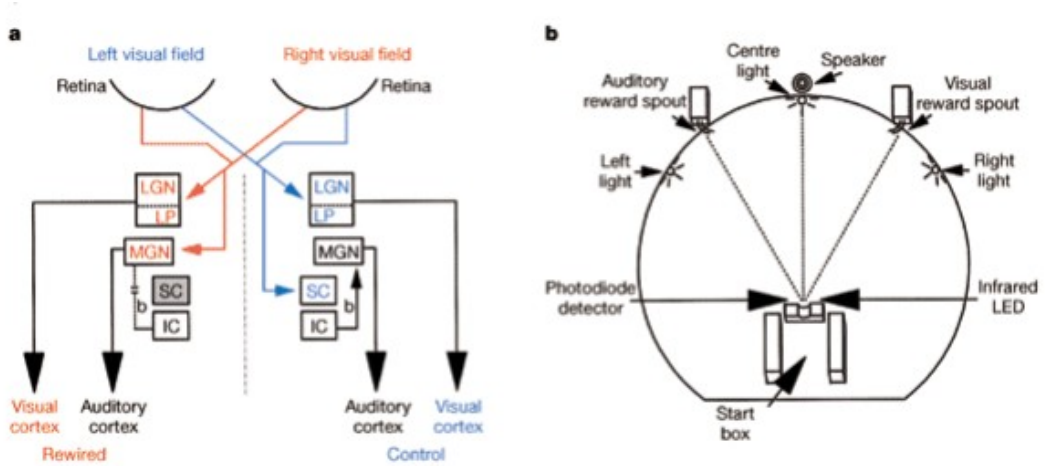
Wisniewska et al. (2015): Porpoise approaching an artificial target.

Early input shapes an auditory pathway

Early sensory input can influence what a cortical pathway comes to represent. In the rewired-ferret experiments, neonatal surgery redirected retinal input toward the auditory thalamus, the **medial geniculate nucleus**. This nucleus retained its projection to auditory cortex, creating a route by which patterned visual information reached tissue that normally develops with auditory input.

The intervention changed the available pathways early in development by reducing normal retinal targets and removing competing input to the auditory thalamus. It did not simply connect an adult eye to a fully developed auditory cortex. Neurons in the resulting pathway developed visual response properties, including selectivity for features such as the orientation of a line in the visual scene.

Behavioral tests showed that the animals could use information carried through the altered pathway for visual judgments. The result demonstrates that early input can help shape cortical responses while anatomical pathways constrain where that input travels. It does not mean that an adult hears a visible object or that every cortical region can take over any function without developmental conditions.



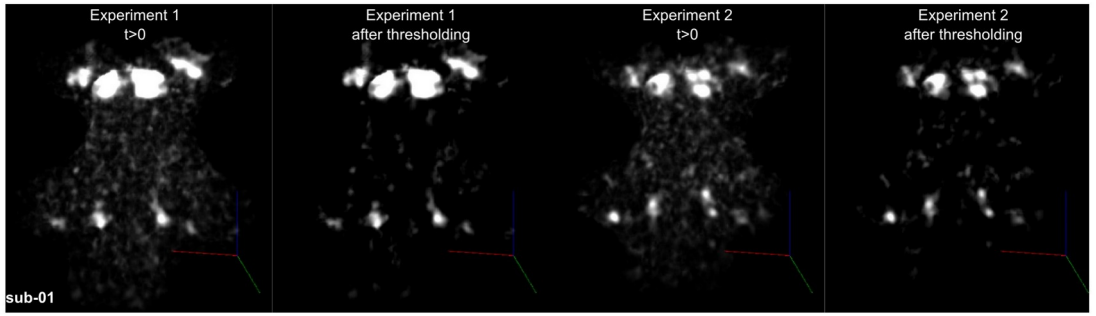
von Melchner et al. (2000), Fig. 1.

Can human auditory maps be reproduced?

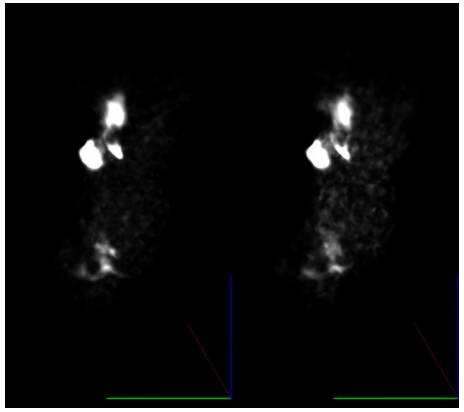
A functional brain map should be examined for whether the same procedure identifies a similar response when repeated. This is especially demanding in the auditory brainstem because several nuclei are small and close together. Slight differences in alignment or image quality can shift an apparent response relative to the anatomical structure that the researcher is trying to identify.

Sitek and colleagues compared high-field functional MRI responses across repeated natural-sound experiments. The accompanying source animation displays group-average positive statistical maps from two experiments. These colors summarize evidence for a response during the listening conditions; they do not display individual action potentials or identify what a participant consciously thought about each sound.

Agreement across repeated measurements helps distinguish a stable response location from a feature that appears in only one scan. Individual anatomical information remains necessary because averaging across participants can blur small neighboring structures. The study combined functional responses with anatomical imaging and histological information to identify the central auditory structures more precisely than any single image type could provide alone.



Sitek et al. (2019), Fig. 11.



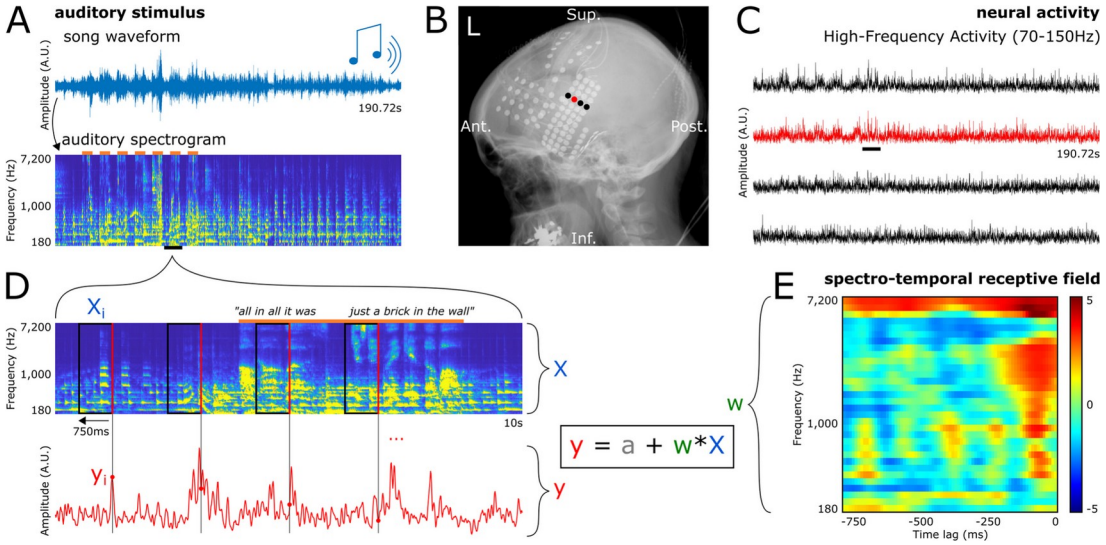
Sitek et al. (2019): Repeat auditory fMRI experiments: group maps.

Reconstructing heard music from cortical activity

Electrocorticography records electrical activity from electrodes placed on the cortical surface. Bellier and colleagues measured these signals while patients listened to Pink Floyd's Another Brick in the Wall, Part 1. They analyzed changes in high-frequency activity, a measure associated with local population processing, to determine which features of the heard music could be recovered from the recordings.

A **spectrogram** represents how much sound energy occurs at different frequencies over time. A computational model learned to predict this pattern from the cortical recordings. The researchers evaluated reconstruction on song segments held apart from model fitting, then converted the predicted spectrogram into an audible waveform. The result retained recognizable aspects of the original song, including parts of its vocals.

Superior temporal regions contributed strongly to the reconstruction. Some recording sites responded most strongly to sound onsets, others maintained responses during musical passages, and another pattern emphasized musical rhythm. The model could combine these different population responses to estimate the evolving sound. Reconstructing the heard song depended on both the information present at the electrodes and the learned mapping from activity to acoustic features.



Bellier et al. (2023), Fig. 1.

Key takeaways

Central circuits compare signals. Precisely timed excitation and inhibition transform bilateral inputs into representations of sound location and other acoustic features.

Auditory processing is distributed and recurrent. Brainstem, midbrain, thalamic, and cortical populations exchange ascending and descending signals; one region does not perform every listening operation.

Context changes what is heard. Temporal grouping, visual speech, attention, and learned expectations alter the organization and interpretation of incoming sound.

Music combines several neural operations. Pitch, timbre, rhythm, movement, prediction, reward, and autobiographical memory depend on interacting networks with partly distinct responses.

Experience changes specific responses. Music-training studies measure particular developmental outcomes; early rerouting, imprinting, and seasonal plasticity depend on defined biological conditions.

Neural measurements answer different questions. Anatomy describes pathways, activity reveals responses, and controlled manipulations test causal contributions; decoding heard music does not establish decoding private thoughts.